

MERGE ON WORKSPACES AS HOPF ALGEBRA MARKOV CHAIN

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ABSTRACT. We study the dynamical properties of a Hopf algebra Markov chain with state space the binary rooted forests with labelled leaves. This Markovian dynamical system describes the core computational process of structure formation and transformation in syntax via the Merge operation, according to Chomsky's Minimalism model of generative linguistics. The dynamics decomposes into an ergodic dynamical system with uniform stationary distribution, given by the action of Internal Merge, while the contributions of External Merge and (a minimal form of) Sideward Merge reduce to a simpler Markov chain with state space the set of partitions and with combinatorial weights. The Sideward Merge part of the dynamics prevents convergence to fully formed connected structures (trees), unless the different forms of Merge are weighted by a cost function, as predicted by linguistic theory. Results on the asymptotic behavior of the Perron-Frobenius eigenvalue and eigenvector in this weighted case, obtained in terms of an associated Perron-Frobenius problem in the tropical semiring, show that the usual cost functions (Minimal Search and Resource Restrictions) proposed in the linguistic literature do not suffice to obtain convergence to the tree structures, while an additional optimization property based on the Shannon entropy achieves the expected result for the dynamics. We also comment on the introduction of continuous parameters related to semantic embedding and other computational models, and also on some filtering of the dynamics by coloring rules that model the linguistic filtering by theta roles and phase structure, and on parametric variation and the process of parameter setting in Externalization.

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1. INTRODUCTION

In work of Diaconis, Pang, and Ram in [14] and of Pang in [27], [28], a theory of Hopf algebra Markov chains was developed. These are discrete dynamical systems associated to combinatorial Hopf algebras. These Hopf algebras (see [19]) are graded connected commutative Hopf algebras $\mathcal{H}$ with a basis given by combinatorial objects (in our case, these will be binary forests). The dynamical systems called Hopf algebra Markov chains act on the vector space of the Hopf algebra as a composition of the form

$$(1.1) \quad \mathcal{K} = \mu \circ \mathcal{L} \circ \Delta,$$

where μ is the Hopf algebra product, Δ is the Hopf algebra coproduct, and $\mathcal{L}$ is a linear endomorphism of $\mathcal{H} \otimes \mathcal{H}$. In other words, one decomposes combinatorial objects using the coproduct Δ , acts with $\mathcal{L}$ on the parts of the decomposition, and recomposes a resulting object using the product μ . In the main examples studied so far $\mathcal{L}$ is a projection. More general forms have also been studied in [14], [27], [28] where one uses powers Δ^k and μ^k in the composition (1.1), namely decompositions into multiple parts. For our purposes, we will only consider cases of the form (1.1), where $\mathcal{L}$ is not just a projection but the composition of a projection and a “grafting” operation. While in the cases studied in [14], [27], [28] the special form of $\mathcal{L}$ leads to a combinatorial description of the stationary distribution, the case we consider does not have that structure, but we will still be able to derive some general results about the stationary distribution and the dynamics, based on the Perron–Frobenius theorem and explicit counting of combinatorial objects, involving partitions and generalized multinomial coefficients.

The form of the Hopf algebra Markov chain we consider is motivated by a mathematical model of the Merge operation in linguistics, as developed in work of the first author, Chomsky, and Berwick in [22].

The Minimalist Program of syntax in generative linguistics was first introduced by Noam Chomsky in the early 1990s, [6]. Since then it underwent significant developments, culminating in the recent formulation, starting around 2013, where Chomsky introduced the “free symmetric Merge” as the fundamental structure-building operation of syntax, and isolated this universal core computational mechanism, subject to optimality constraints, [7], [8], [9]. In this model, the core Merge action accounts for all the structure-formation and key computational aspects of syntax (compositional and transformational properties), while an “externalization interface” accounts for parametric syntactic variation across languages and embodiment into specific languages, and a “syntax-semantics interface” accounts for semantic parsing. In 2023, joint work of Marcolli, Chomsky, and Berwick showed that the core computational mechanism of the free symmetric Merge admits a precise mathematical formalization in terms of Hopf algebras and Hopf algebra Markov chain dynamical systems, [22]. The work [22], as well as subsequent work [26], [24], [25], [21], [31] and further ongoing developments show that the two interfaces (externalization and syntax-semantics) also admit good mathematical modeling, though providing a full mathematical account of their functioning still requires a significant amount of further investigation.

In this paper we focus on the core computational structure of the free symmetric Merge, as formulated in [22], and we further investigate its mathematical properties as a dynamical system acting on syntactic workspaces. Namely, we study the universal dynamical properties of syntax formation and transformation, as a Hopf algebra Markov chain.

A mathematically precise model of structure formation in language is of particular current interest, because current artificial systems that simulate human language (such as large language models) are extremely wasteful in terms of both training and computational resources, and are based purely on statistical data (computing probabilities of extensions of strings of text), while completely ignoring a wealth of very detailed knowledge that we possess about the structure of language and its underlying computational mechanism. The astonishing theory-gap that one can observe in recent developments in the field of artificial intelligence makes it a urgent task to formulate the existing scientific theory of language-production, broadly known as “generative grammar”, in a precise mathematical language. While applications are not the focus of this and other theoretical papers, they are one of the natural motivations for developing our mathematical models of syntax as a computational system.

1.1. Summary of the Merge action on workspaces. We recall here some basic results and notation from [22] that we will be using throughout the paper.

- $\mathcal{SO}_0$ is a finite set. Elements $\alpha \in \mathcal{SO}_0$ are called *lexical items and syntactic features* (we will just call them “lexical data” here for brevity).
- $\mathcal{SO} = \mathfrak{T}_{\mathcal{SO}_0}$ is the countably infinite set of full binary rooted trees (nonplanar, that is, with no assigned planar embedding) with leaves decorated by elements of the set $\mathcal{SO}_0$. Elements $T \in \mathcal{SO}$ are called *syntactic objects*. We write $\mathfrak{T}_{\mathcal{SO}_0, n}$ for the set of syntactic objects $T \in \mathcal{SO}$ with $n = \#L(T)$ leaves.
- $\mathcal{SO}$ is the free nonassociative commutative magma generated by the set $\mathcal{SO}_0$

$$\mathcal{SO} = \text{Magma}_{c,na}(\mathcal{SO}_0, \mathfrak{M}) \cong \mathfrak{T}_{\mathcal{SO}_0},$$

with binary operation $\mathfrak{M}$ that acts on a pair of trees by joining them to a common root (as nonplanar trees),

$$\mathfrak{M}(T, T') := \widehat{T \quad T'} = \widehat{T' \quad T}.$$

- $\mathfrak{F}_{\mathcal{SO}_0}$ is the countably infinite set of binary rooted forests, namely forests whose connected components are nonplanar full binary rooted trees in $\mathfrak{T}_{\mathcal{SO}_0}$. Elements $F \in \mathfrak{F}_{\mathcal{SO}_0}$ are called *workspaces*. We also write $\mathfrak{F}_{\mathcal{SO}_0, n}$ for the set of workspaces $F = T_1 \sqcup \cdots \sqcup T_r \in \mathfrak{F}_{\mathcal{SO}_0}$ with $n = \#L(F) = \#L(T_1) + \cdots + \#L(T_r)$ leaves.
- An *accessible term* $T_v \subset T$ of a syntactic objects is the subtree consisting of a non-root vertex v of T and all its descendants. Thus, the set of accessible terms of T can be identified with the set of non-root vertices of T .
- The vector space $\mathcal{V}(\mathfrak{F}_{\mathcal{SO}_0})$ endowed with the product given by the disjoint union $\sqcup$ is the free commutative algebra (polynomial algebra) generated by $\mathfrak{T}_{\mathcal{SO}_0}$.
- There are different forms of coproduct on $\mathcal{V}(\mathfrak{F}_{\mathcal{SO}_0})$ with slightly different algebraic properties (see the discussion in §1.2 of [22]). All of them are of the form

$$(1.2) \quad \Delta(T) = \sum_{\underline{v}} F_{\underline{v}} \otimes T/F_{\underline{v}},$$

with $F_{\underline{v}} = T_{v_1} \sqcup \cdots \sqcup T_{v_n}$ a collection of disjoint accessible terms of T and $T/F_{\underline{v}}$ the resulting quotient. The difference rests on different ways of obtaining the quotient term $T/F_{\underline{v}}$. We will focus here on one form of the coproduct, called the “deletion coproduct” Δ^d in [22], where the quotient $T/F_{\underline{v}}$ is taken to be the maximal full binary tree obtained by edge contractions (that eliminate non-branching vertices) on the (non-full) binary tree obtained from T by cutting the edge above each non-root vertex v_i . Another form of the coproduct, called the “contraction coproduct” in [22] uses a form of the quotient $T/F_{\underline{v}}$ where each component of the forest $F_{\underline{v}}$ is contracted to its root vertex. (We will only discuss the contraction coproduct in §10.)

In addition to the different forms of the quotient $T/F_{\underline{v}}$, different forms of the coproduct may also differ according to conditions (types of admissible cuts) on the extraction of the accessible terms T_{v_i} . In the case of the deletion coproduct that we focus on in this paper, some of the algebraic properties of the coproduct change according to whether we allow arbitrary admissible cuts that extract the forest $F_{\underline{v}}$ (where admissible means that no two edges of the cut lie on the same path from the root to one of the leaves, so that $F_{\underline{v}}$ indeed consists of accessible terms), or else one requires the stricter condition that the list $\underline{v} = (v_1, \dots, v_n)$ of vertices does not contain any pair of sister-vertices (one never cuts two edges below the same vertex). We will not discuss in this paper the difference between these two choices at the level of algebraic properties, as that will be analyzed elsewhere. However, we will discuss briefly how these two different choices for admissible cuts affect the properties of the dynamical system and why, from this perspective, it is preferable to include the possibility of cutting both edges below the same vertex.

- The *Merge operation on workspaces* is a dynamical system on $\mathcal{V}(\mathfrak{F}_{\mathcal{SO}_0})$ generated by the linear operators defined on basis elements F by

$$(1.3) \quad \mathcal{K}_2(F) = \sqcup \circ (\mathfrak{B} \otimes \text{id}) \circ \Pi^{(2)} \circ \Delta \quad \text{and} \quad \mathcal{K}_1(F) = \sqcup \circ \Pi^{(1)} \circ \Delta,$$

where $\mathfrak{B}$ is the grafting operator from forests to trees $\mathfrak{B}(T \sqcup T') = \mathfrak{M}(T, T')$, and $\Pi^{(k)}$ is the linear projection on $\mathcal{V}(\mathfrak{F}_{\mathcal{SO}_0}) \otimes \mathcal{V}(\mathfrak{F}_{\mathcal{SO}_0})$ that is the identity on basis elements $F \otimes F'$ where $F = T_1 \sqcup \dots \sqcup T_k$ has k components and is zero otherwise. We can write it as a single operator

$$(1.4) \quad \tilde{\mathcal{K}}(F) = \mathcal{K}_2(F) + \mathcal{K}_1(F) = \sqcup \circ (\mathfrak{B} \otimes \text{id}) \circ \tilde{\Pi}^{(2)} \circ \Delta,$$

where $\tilde{\Pi}^{(k)} = \oplus_{j=1, \dots, k} \Pi^{(j)}$, and where, for a single tree T , we extend $\mathfrak{B}$ as $\mathfrak{B}(T) = T$, so that the last expression in (1.4) is indeed equal to $\mathcal{K}_2(F) + \mathcal{K}_1(F)$.

While (1.4) is the most natural way of writing the Merge dynamical system from the mathematical perspective, as it has exactly the form (1.1) of a Hopf algebra Markov chain, it is preferable for adherence to a more transparent linguistic interpretation, to separate out the operator $\mathcal{K}$ of (1.3) into a sum

$$\mathcal{K}_2 = \sum_{S, S'} \mathfrak{M}_{S, S'}$$

over $S, S' \in \mathcal{SO}$, with

$$(1.5) \quad \mathfrak{M}_{S, S'} = \sqcup \circ (\mathfrak{B} \otimes \text{id}) \circ \delta_{S, S'} \circ \Delta,$$

with $\delta_{S, S'}$ the Kronecker-delta operator that is the identity on terms in the range of Δ that have $S \sqcup S'$ in the left-hand-side of the coproduct (and zero otherwise). The sum over S, S' always reduces to a finite sum when applied to a workspace F . These operations $\mathfrak{M}_{S, S'}$ correspond to the linguistic External Merge ($\mathfrak{M}_{S, S'}$ with both S, S' connected components of the workspace) and Sideward Merge ($\mathfrak{M}_{S, S'}$ where at least one of S, S' is an accessible term).

The operator $\mathcal{K}_1$ of (1.3) can be similarly written in the form $\mathcal{K}_1 = \sum_S \mathfrak{M}_{S, 1}$ where 1 is the unit of the magma $\mathcal{SO}$ (the formal empty tree). These operators $\mathfrak{M}_{S, 1}$ don't have a direct linguistic interpretation on their own (one can call them *virtual Merge*), but they give rise to a very important operation, that accounts for “movement” or “transformation” in syntax, namely Internal Merge, which is realized here as a composition $\mathfrak{M}_{S, T/S} \circ \mathfrak{M}_{S, 1}$.

In order to make our formulation here as close as possible to the linguistic interpretation, we will write the Merge Markov chain $\mathcal{K}$ in terms of the operations of External Merge (EM), Internal Merge (IM) and Sideward Merge (SM), rather than in terms of EM, SM and virtual Merge (vM) as in (1.3) and (1.4), so that we ensure that each of the arrows in the Merge graph that describes the Markov chain dynamics has a linguistic interpretation. All the arguments we present can be easily adapted to the case of (1.4).

- The Merge dynamical system generated by $\tilde{\mathcal{K}}$ as in (1.4) defines a Hopf algebra Markov chain in the sense of [14], [27], [28].
- The individual Merge operations $\mathfrak{M}_{S, S'}$ and $\mathfrak{M}_{S, T/S} \circ \mathfrak{M}_{S, 1}$ can be weighted according to certain cost functions (Minimal Search, Resource Restrictions) as discussed in [22] and [26].
- In addition to the core computational mechanism of syntax described by the Merge operations $\mathfrak{M}_{S, S'}$ and $\mathfrak{M}_{S, T/S} \circ \mathfrak{M}_{S, 1}$, one can consider a system of filtering on the syntactic objects constructed by the Merge action on workspaces, which eliminates structures that are non-viable for semantic interpretation. There are two main such filters, which are analyzed in [25] and in [24], one dealing with theta-role assignments and one with the structure of phases. As shown in [25] and in [24] these filters can be modeled in terms of a

colored operad with a bud generating system in the sense of [17], and can equivalently be implemented by pruning the Merge operations according to coloring rules. From the point of view of the dynamical system we consider here, this pruning corresponds to eliminating some of the arrows from the directed graph of the Markov chain dynamics, which correspond to Merge operations that are not compatible with the coloring restrictions. We will discuss this in §15.

We take this as the basic structure defining the dynamical system that we analyze in this paper. Orbits of the action of Merge on workspaces, starting from a forest consisting only of labelled leaf vertices and no edges, correspond to linguistic derivations of sentence formation as repeated applications of Merge operations, building syntactic structures from an unstructured set of lexical items.

2. THE MERGE GRAPHS

The dynamical system defined by the action of Merge on workspaces can be described in terms of a directed graph $\mathcal{G}$ with vertices the forests $F \in \mathfrak{F}_{\mathcal{SO}_0}$ with nonempty set of edges, $E(F) \neq \emptyset$. The graph $\mathcal{G}$ has an edge from a vertex F to a vertex F' whenever there is a Merge operation with $F' = \mathfrak{M}_{S,S'}(F)$ or $F' = \mathfrak{M}_{S,T/S}(\mathfrak{M}_{S,1}(F))$.

The reason why we consider only forests F with non-trivial set of edges is because every Merge operation (even if applied to a disjoint union of vertices with no edges) produces a forest with some edges, so the forests consisting only of leaves with no edges are transient for the dynamics that never returns to them, so they do not contribute to the main properties of the dynamics like stationary distribution, cycles, etc.

Since all the Merge operations (with this form of the coproduct) preserve the number of leaves of the forest, the infinite graph $\mathcal{G}$ breaks into a disjoint union of finite graphs

$$\mathcal{G} = \bigsqcup_{n \geq 2} \mathcal{G}_n.$$

(We do not include here the case $n = 1$ as there is no dynamics there.) Moreover, the Merge operation does not alter the labeling of the leaves so each graph $\mathcal{G}_n$ is itself a disjoint union over all such labelings

$$\mathcal{G}_n = \bigsqcup_{A \in \text{Sym}^n(\mathcal{SO}_0)} \mathcal{G}_{n,A}.$$

The set of vertices of $\mathcal{G}_{n,A}$ is the finite subset $\mathfrak{F}_{A,n} \subset \mathfrak{F}_{\mathcal{SO}_0}$ of forests $F \in \mathfrak{F}_{\mathcal{SO}_0}$ with $n = \#L(F)$ for $L(F)$ the set of leaves, with set of labels $\{\alpha_\ell\}_{\ell \in L(F)} = A$, and with a non-empty set $E(F)$ of edges.

In our analysis we will restrict to a fixed $n \geq 2$ and a fixed set $A = \{\alpha_1, \dots, \alpha_n\}$ of lexical items, hence working only with one $\mathcal{G}_{n,A}$ graph at a time.

2.1. Counting forests. The first step in understanding the structure of the Merge graph $\mathcal{G}_{n,A}$ is to have a good counting formula for the number of vertices. Vertices are workspaces, namely forests whose connected components are non-planar full binary rooted trees with leaves decorated by the elements α_i of A .

In order to count vertices, namely count forests $F \in \mathfrak{F}_{A,n}$, we first split the set $\mathfrak{F}_{A,n}$ into subsets, according to how the set $L(F)$ of leaves of a forest $F = T_1 \sqcup \dots \sqcup T_r$ splits into the sets $L(T_i)$ of leaves of the tree components T_i of F .

Let $\mathcal{P}(n)$ be the set of all partitions of n as a sum $n = k_1 + \dots + k_r$ of integers with $k_i \geq 1$. We write $\varphi = \{k_1, \dots, k_r\}$ (with possible multiplicities among the k_i) for a partition $n = k_1 + \dots + k_r$.

We write $\wp_{1^n} = \{\underline{1}^n\} = \{1, \dots, 1\}$ for the partition $n = 1 + \dots + 1$ into a sum of 1's and we let

$$(2.1) \quad \mathcal{P}'(n) := \mathcal{P}(n) \setminus \{\wp_{1^n}\}.$$

be the set of all partitions $n = k_1 + \dots + k_r$ where at least one of the k_i satisfies $k_i \geq 2$.

Lemma 2.1. *The set of vertices $V(\mathcal{G}_{n,A}) = \mathfrak{F}_{A,n}$ of the directed graph $\mathcal{G}_{n,A}$ is partitioned as*

$$(2.2) \quad V(\mathcal{G}_{n,A}) = \bigsqcup_{\wp \in \mathcal{P}'(n)} V_{\wp,n,A},$$

where for $\wp = \{k_1, \dots, k_r\}$, with $n = k_1 + \dots + k_r$, the set $V_{\wp,n,A}$ consists of all the vertices F of $\mathcal{G}_{n,A}$ that are forests of the form $F = T_1 \sqcup \dots \sqcup T_r$ with $T_i \in \mathfrak{T}_{\mathcal{SO}_0, k_i}$, with nonempty set of edges.

Proof. Given a workspace $F = T_1 \sqcup \dots \sqcup T_r$ in $\mathfrak{F}_{A,n}$ with n leaves with labels $A = \{\alpha_1, \dots, \alpha_n\}$, the set of leaves $L(F)$ also decomposes as a disjoint union

$$(2.3) \quad L(F) = L(T_1) \sqcup \dots \sqcup L(T_r)$$

where $L(T_i)$ is the set of leaves of the connected component of F given by the tree T_i . If $k_i = \#L(T_i)$, we obtain a partition $\wp \in \mathcal{P}(n)$ of the form with $n = \#L(F) = k_1 + \dots + k_r$. Thus, we can split the set $\mathfrak{F}_{A,n}$ into disjoint subsets $V_{\wp,n,A}$ for $\wp \in \mathcal{P}(n)$ where $F \in V_{\wp,n,A}$ are forests where the leaves split in (2.3) according to the partition $\wp$. Forests with nonempty set of edges, $E(F) \neq \emptyset$, correspond to partitions in $\mathcal{P}'(n)$. $\square$

Partitions $\wp = \{k_1, \dots, k_r\} \in \mathcal{P}(n)$ can be represented as Young diagrams, consisting of n boxes arranged in r rows of lengths k_i (drawn in non-decreasing order). For example we have

$$\wp = \{4, 2, 2, 1\} \in \mathcal{P}(9) \Leftrightarrow D(\wp) = \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline \end{array}.$$

This partition is the image $\wp = p(F)$ for different workspaces

$$F = \begin{array}{c} \diagup \quad \diagdown \\ \bullet \quad \bullet \quad \bullet \quad \bullet \end{array} \sqcup \begin{array}{c} \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \sqcup \begin{array}{c} \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \sqcup \bullet \quad \text{or} \quad F = \begin{array}{c} \diagup \quad \diagdown \\ \bullet \quad \bullet \quad \bullet \quad \bullet \end{array} \sqcup \begin{array}{c} \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \sqcup \begin{array}{c} \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \sqcup \bullet$$

where the $\bullet$ slots at the leaves can be filled with a permutation of the elements of $A = \{\alpha_1, \dots, \alpha_9\}$. In particular, Young diagrams account for partitioning of the leaves over the different components of the workspace but not for the different tree topologies of the components. To keep track of different possible assignments of labels α_i at the leaves, we can also consider Young tableaux on the alphabet A , such as

α_{i_1}	α_{i_2}	α_{i_3}	α_{i_4}
α_{i_5}	α_{i_6}		
α_{i_7}	α_{i_8}		
α_{i_9}			

with $\alpha_{i_1}, \dots, \alpha_{i_9}$ a permutation of $A = \{\alpha_1, \dots, \alpha_9\}$.

For a partition $\wp = \{k_1, \dots, k_m\} \in \mathcal{P}(n)$, the *multinomial coefficient*

$$(2.4) \quad \mu_{\wp,n} := \binom{n}{k_1, \dots, k_r} = \frac{n!}{k_1! \dots k_m!}$$

is the size of the coset S_n/S_φ of the symmetric group S_n by the Young group of the partition,

$$(2.5) \quad S_\varphi := S_{k_1} \times \cdots \times S_{k_r}$$

with the k_i ordered as in the Young diagram $D(\varphi)$. Note that in (2.4) we list the k_i with multiplicities. When listing explicitly the multiplicities as $n = a_1 k_1 + \cdots + a_r k_r$, we write

$$(2.6) \quad \mu_{\varphi,n} := \binom{n}{\underbrace{k_1, \dots, k_1}_{a_1\text{-times}}, \dots, \underbrace{k_r, \dots, k_r}_{a_r\text{-times}}} = \frac{n!}{(k_1!)^{a_1} \cdots (k_r!)^{a_r}}.$$

The *generalized multinomial coefficient* of a partition

$$(2.7) \quad \varphi = \{\underbrace{k_1, \dots, k_1}_{a_1}, \dots, \underbrace{k_r, \dots, k_r}_{a_r}\} \quad \text{with} \quad n = \underbrace{k_1 + \cdots + k_1}_{a_1\text{-times}} + \cdots + \underbrace{k_r + \cdots + k_r}_{a_r\text{-times}}$$

is given by

$$(2.8) \quad \Upsilon_{\varphi,n} := \frac{1}{a_1! \cdots a_r!} \binom{n}{\underbrace{k_1, \dots, k_1}_{a_1\text{-times}}, \dots, \underbrace{k_r, \dots, k_r}_{a_r\text{-times}}} = \frac{1}{a_1! \cdots a_r!} \frac{n!}{(k_1!)^{a_1} \cdots (k_r!)^{a_r}}.$$

We also consider the quantities

$$(2.9) \quad \Lambda_{\varphi,n} := \frac{1}{a_1! \cdots a_r!} \binom{n}{\underbrace{k_1, \dots, k_1}_{a_1\text{-times}}, \dots, \underbrace{k_r, \dots, k_r}_{a_r\text{-times}}} \prod_{i=1}^r ((2k_i - 3)!!)^{a_i}$$

We also write (2.9) equivalently as

$$(2.10) \quad \Lambda_{\varphi,n} = \mu_{\varphi,n} \cdot \Gamma_{\varphi,n}$$

where $\mu_{\varphi,n}$ is the multinomial coefficient (2.6) and

$$(2.11) \quad \Gamma_{\varphi,n} := \frac{1}{a_1! \cdots a_r!} \prod_{i=1}^r ((2k_i - 3)!!)^{a_i}.$$

Lemma 2.2. *The number of vertices in $V(\mathcal{G}_{n,A}) = \mathfrak{F}_{A,n}$ is given by*

$$(2.12) \quad \Lambda_n := \#\mathfrak{F}_{A,n} = \sum_{\varphi=\{k_1, \dots, k_r\} \in \mathcal{P}'(n)} \Lambda_{\varphi,n},$$

with $\Lambda_{\varphi,n}$ as in (2.9) the sum over the set $\mathcal{P}'(n)$ of (2.1).

Proof. The number of non-planar full binary rooted trees with n labelled leaves is given by the odd double factorial

$$(2.13) \quad \#\mathfrak{T}_{A,n} = (2n - 3)!!.$$

Thus, the number of forests of the form $F = T_1 \sqcup \cdots \sqcup T_r$, with $T_i \in \mathfrak{T}_{A,k_i}$ and $k_1 + \cdots + k_r = n$ is given by

$$(2.14) \quad \#\{F \in \mathfrak{F}_{A,n} \mid F = T_1 \sqcup \cdots \sqcup T_r, T_i \in \mathfrak{T}_{A,k_i}\} = \Upsilon_{\varphi,n} \prod_{i=1}^r ((2k_i - 3)!!)^{a_i}$$

where the multinomial coefficients account for how the labels are split between the components and the normalization dividing by the factor $a_1! \cdots a_r!$ account for the symmetry with respect to

exchanging rows of identical length in the Young diagram of the partition. Thus, the total number of forests in $\mathfrak{F}_{A,n}$ is a sum

$$(2.15) \quad \#\mathfrak{F}_{A,n} = \Lambda_n = \sum_{\wp \in \mathcal{P}'(n)} \Lambda_{\wp,n},$$

with $\Lambda_{\wp,n}$ as in (2.9), with $\Lambda_{\wp,n} = \#V_{\wp,n,A}$. $\square$

Note, for comparison, that the counting of the sets of non-planar full binary rootes trees with n *undecorated* leaves is more complicated. It is given by the n -th Wedderburn–Etherington numbers W_n , for which there is no closed formula, but their generating function is the solution to the equation

$$\mathcal{T}(x) = x + \frac{1}{2}\mathcal{T}(x)^2 + \frac{1}{2}\mathcal{T}(x^2).$$

Correspondingly, the generating function for the binary rooted forests (with unlabelled leaves) is then

$$\mathcal{F}(x) = \exp(\mathcal{T}(x)).$$

2.2. Indegrees and outdegrees in the Merge graphs. As edges of the graphs $\mathcal{G}$ and of its components $\mathcal{G}_{n,A}$ we will consider the Merge transformations, subdividing them into External Merge, Internal Merge, and Sideward Merge arrows. In the case of Internal Merge, we will not consider the case where the accessible term extracted is one of the two trees T_1, T_2 under the root of a component tree $T = \mathfrak{M}(T_1, T_2)$ because, with the form of the coproduct we are using, the result of IM on either T_1 or T_2 will simply reproduce T and we do not include such identity maps as part of the dynamics. In the case of Sideward Merge, we will restrict to considering only the “minimal” Sideward Merge transformations (in the sense discussed in [26]), namely those that only extract and Merge atomic components (single leaves), because these are the lowest cost SM, in the cost counting of [22], [26].

We first introduce some notation for convenience. Given a workspace F , we write:

- $\pi_0(F)$ for the set of connected components of F with $\#\pi_0(F) = b_0(F)$,
- $\pi_{0,E}(F)$ for the subset of connected components of F consisting of a tree with nontrivial set of edges, and $c(F) := \#\pi_{0,E}(F)$,
- $\mathcal{C}(F) \subset L(F) \times L(F)$ for the subset of pairs (ℓ, ℓ') of leaves of F that belong to a cherry tree in F , (subtrees $T_v \subset F$ that consist of two sister leaves ℓ, ℓ' and a vertex v above them and two edges), with $d(F) = \#\mathcal{C}(F)$.
- $\mathcal{C}'(F) = \mathcal{C}(F) \cap \pi_0(F)$ for the set of cherries of F that are connected components (as opposed to accessible terms of components), with $d'(F) = \#\mathcal{C}'(F)$.
- $\tilde{c}(F) = c(F) - d'(F)$ for the number of connected components of F that have at least 3 leaves.
- $d''(F) = d(F) - d'(F)$ for the number of cherries that are accessible terms of components (rather than full components),
- $\mathcal{C}(T_i)$ for the set of cherry subtrees of a component T_i , with $d(T_i) = \#\mathcal{C}(T_i)$.

We can describe the edges in the graph $\mathcal{G}_{n,A}$ in the following way.

Proposition 2.3. *The edges of the graph $\mathcal{G}_{n,A}$ with vertex set $\mathfrak{F}_{A,n}$ are described in the following way. Consider a vertex F of $\mathcal{G}_{n,A}$ with $b_0(F) = r$ connected components, $F = T_1 \sqcup \cdots \sqcup T_r$, respectively with $k_i = \#L(T_i)$ leaves, with $\sum_i k_i = n$. Without loss of generality, assume that $T_1, \dots, T_{c(F)}$ are the components with nonempty set of edges, and among them $T_1, \dots, T_{d'(F)}$ are the components that are cherries.*

(1) *The number of outgoing External Merge (EM) arrows at the vertex F is equal to*

$$(2.16) \quad N_{\text{EM}}^{\text{out}}(F) = \begin{cases} \binom{r}{2} & r = b_0(F) \geq 2 \\ 0 & r = 1. \end{cases}$$

(2) *The number of incoming External Merge (EM) arrows at the vertex F is equal to*

$$(2.17) \quad N_{\text{EM}}^{\text{in}}(F) = c(F).$$

(3) *The number of outgoing Internal Merge (IM) arrows at the vertex $F = T_1 \sqcup \cdots \sqcup T_r$ with $k_i = \#L(T_i)$ is equal to*

$$(2.18) \quad N_{\text{IM}}^{\text{out}}(F) = \sum_{i=1}^{c(F)} (2k_i - 4).$$

(4) *The number of incoming IM arrows at F is equal to*

$$(2.19) \quad N_{\text{IM}}^{\text{in}}(F) = \sum_{i=1}^{c(F)} (2k_i - 4).$$

(5) *The number of outgoing minimal Sideward Merge arrows at F is given by*

$$(2.20) \quad N_{\text{SM}, \min}^{\text{out}}(F) = \sum_{i=d'(F)+1}^{c(F)} \binom{k_i}{2} + \sum_{i=1}^{c(F)} k_i(r - c(F)) + \sum_{i \neq j, i, j=1}^{c(F)} k_i k_j,$$

if extraction of a pair of sister leaves from a component tree with at least 3 leaves is allowed and

$$(2.21) \quad N_{\text{SM}, \min}^{\text{out}}(F) - d''(F)$$

if no extraction of both leaves below the same vertex is allowed, with $N_{\text{SM}, \min}^{\text{out}}(F)$ as in (2.20).

(6) *The number of incoming minimal Sideward Merge arrows at F is zero if $d'(F) = 0$ and otherwise*

$$(2.22) \quad N_{\text{SM}, \min}^{\text{in}}(F) = 6d'(F)(n - c(F)) + 2 \sum_{i=1}^{d'(F)} \sum_{a \neq b, a, b \neq i}^{c(F)} (2k_a - 2)(2k_b - 2) +$$

$$2 \sum_{i=1}^{d'(F)} \sum_{j \neq i, j=1}^{c(F)} (2k_j - 2)(2k_j + 2 + n - c(F))$$

when cutting sister leaves is allowed in a component with at least 3 leaves, while in the case where no cut of two leaves below the same vertex is allowed, the factor $2k_j + 2 + n - c(F)$ is replaced by $2k_j + 1 + n - c(F)$.

Proof. (1) If $F = T$ has only one connected component then there is no outgoing EM arrow as EM merges two components of the workspace. For $r \geq 2$ connected components the possible outgoing EM arrows correspond to all the possible choices of two components of F that are merged by EM, so there are $\binom{r}{2}$ arrows.

(2) Any nonplanar full binary tree T with a nonempty set of edges can be written in a unique way as $T = \mathfrak{M}(T_1, T_2)$, hence it is the image under EM of a workspace of the form $T_1 \sqcup T_2$. Suppose given a workspace $F = T_1 \sqcup \cdots \sqcup T_r$, where $T_1, \dots, T_{c(F)}$ have a non-empty set of edges, so that $T_i = \mathfrak{M}(T_{i,1}, T_{i,2})$ for $i = 1, \dots, c(F)$. We write $F = T_i \sqcup \hat{F}_i = \mathfrak{M}(T_{i,1}, T_{i,2}) \sqcup \hat{F}_i$ where

$\hat{F}_i = \sqcup_{j \neq i} T_j$. Then F is the target of $c(F)$ EM arrows with sources the workspaces $T_{i,1} \sqcup T_{i,2} \sqcup \hat{F}_i$, for $i = 1, \dots, c(F)$.

(3) Each component T_i of F that has more than two edges is the source of a number of nontrivial (non-identity) IM arrows, one for each accessible term $T_v \subset T_i$ where the non-root vertex v is not one of the two vertices immediately below the root of T_i (with the form of the coproduct we consider, IM applied to one of those two accessible terms would just be the identity). If the component tree T_i has k_i leaves, there are $2k_i - 2$ non-root vertices hence $2k_i - 4$ excluding the two below the root, so we obtain (2.18).

(4) For F to be the target of an IM arrow, one of the components $T_i = \mathfrak{M}(T_{i,1}, T_{i,2})$ of F with nontrivial set of edges must be obtained via IM from a tree of the form $T_{i,2} \triangleleft_e T_{i,1}$ or $T_{i,1} \triangleleft_{e'} T_{i,2}$ where $\triangleleft_e$ is the insertion pre-Lie operation at an edge e . We consider here as source of the arrows the workspaces where the T_i component of F is replaced by one of these edge-insertions and all the other components of F are the same. If $k_i = k_{i,1} + k_{i,2} = \#L(T_i)$ with $k_{i,1} = \#L(T_{i,1})$, $k_{i,2} = \#L(T_{i,2})$, there are $2k_{i,1} - 2$ edges in $T_{i,1}$ where the insertion $\triangleleft_e$ can take place and $2k_{i,2} - 2$ edges in $T_{i,2}$, for a total of $2k_i - 4$ possible edge insertions, that is, $2k_i - 4$ sources for IM arrows, for each component T_i of T . This gives a total number of incoming IM arrows as in (2.19).

(5) Minimal Sideward Merge operations select accessible terms or components that are single leaves and combines them into a new component. The possible cases are: two single leaves that belong to the same component that is not a cherry, two leaves belonging to two different components in $\pi_{0,E}(F)$, and one component of F consisting of a single leaf and another leaf from one of the components with non-empty set of edges. (The case of two components that are both single leaves is already counted among the EM arrows and is not a case of SM.) Thus, the number of outgoing minimal SM arrows is the sum of these three possibilities.

The first case, adding over the $\tilde{c}(F) = c(F) - d'(F)$ components that are not cherries gives

$$\sum_{i=d'(F)+1}^{c(F)} \binom{k_i}{2},$$

while in the case where one does not allow any extraction of a pair of leaves in a cherry, even if the cherry is part of a larger tree, one needs to subtract $d''(F)$.

The second case adds up to a contribution of

$$\sum_{T_i \neq T_j \in \pi_{0,E}(F)} k_i \cdot k_j.$$

The third case adds up to

$$(r - c(F)) \cdot \sum_{i=1}^{c(F)} k_i$$

with the first factor giving the choice of a single leaf component and the second factor the choice of a leaf in one of the components with non-empty set of edges. Thus the total number of outgoing minimal SM arrows gives (2.20).

(6) Since every minimal SM transformation generates a cherry component, if F has $d'(F) = 0$ there can be no incoming SM arrows. If $d'(F) \neq 0$ then each of the $d'(F)$ cherry component can be generated by an SM transformation. For a given cherry component of F , say $T_i = \{\alpha, \beta\}$, the source workspace of an SM arrow that creates T_i will have one of the following possibilities for the leaves α and β :

- (1) one of α or β is a single component and the other is inserted at an edge of a component in $\pi_{0,E}(F)$,

- (2) one of α or β is a single component and the other is joined to a single-leaf component of F to form a cherry component,
- (3) α and β are edge-inserted in two different components in $\pi_{0,E}(F)$,
- (4) one of α or β is inserted in a component in $\pi_{0,E}(F)$ and the other is joined to a single leaf,
- (5) both α and β are edge-inserted in the same component in $\pi_{0,E}(F)$ (without forming a cherry subtree, in the case where cutting the two edges below the same vertex inside a tree is not allowed)
- (6) one of α or β is joined to a single leaf to form a cherry and the other is inserted at an edge of that cherry.

The first case accounts for

$$2 \sum_{i=1}^{d'(F)} \sum_{j \neq i, j=1}^{c(F)} (2k_j - 2).$$

The second case accounts for

$$2 \cdot d'(F) \cdot (n - c(F)).$$

The third case accounts for

$$2 \sum_{i=1}^{d'(F)} \sum_{a \neq b, a, b \neq i, a, b=1}^{c(F)} (2k_a - 2)(2k_b - 2).$$

The fourth case accounts to

$$2 \cdot (n - c(F)) \sum_{i=1}^{d'(F)} \sum_{j \neq i, j=1}^{c(F)} (2k_j - 2).$$

The fifth case accounts for

$$2 \sum_{i=1}^{d'(F)} \sum_{j \neq i, j=1}^{c(F)} (2k_j - 2)(2k_j + 1)$$

where the factor $2k_j - 2$ accounts for the choice of the edge of T_j where the first insertion happens. There are then $2k_j + 1$ choices of an edges in $T_j \triangleleft_e \alpha$ where β can be inserted. In the case where we do not allow cutting of the two leaves of a cherry inside a tree component, we only have $2k_j$ choices for the second edge, so the counting is replaced by

$$2 \sum_{i=1}^{d'(F)} \sum_{j \neq i, j=1}^{c(F)} (2k_j - 2)2k_j.$$

The sixth case accounts for

$$4 \cdot d'(F) \cdot (n - c(F)).$$

This results in (2.22). □

As we will further discuss, one of the reasons why it is preferable, from the point of view of the properties of the Merge dynamical system, to include the admissible cuts that remove both edges below a vertex, is that the counting of the in/out-degrees for the SM arrows becomes then dependent only on data of the partition $\wp = p(F)$ rather than on the specific topology of the forest F , while distinguishing between admissible cuts that sever both edges below the same vertex or not depends on specific information about F and not just on the partition $\wp$. As we will discuss in §4, the fact that the in/out degrees only depend on $\wp$ greatly simplifies the dynamics.

2.3. Sparsity. Proposition 2.3 also shows the *sparsity* property of these graphs. Namely, as n grows in size, the graph $\mathcal{G}_{n,A}$ becomes increasingly sparse at a very fast rate. The density of a directed graph is defined as

$$(2.23) \quad D(\mathcal{G}) = \frac{\#E(G)}{\#V(G) \cdot (\#V(G) - 1)},$$

namely the ratio to the maximal number of edges. A family of graphs $\mathcal{G}_n$ is growing increasingly sparse if

$$\lim_{n \rightarrow \infty} D(\mathcal{G}_n) = 0.$$

Corollary 2.4. *The density $D(\mathcal{G}_{n,A})$ satisfies*

$$(2.24) \quad D(\mathcal{G}_{n,A}) \leq \frac{P(n)}{\Lambda_n},$$

with $P(n)$ a polynomial, at most cubic in n , and with Λ_n as in (2.12).

Proof. The number of edges $\#E(\mathcal{G}_{n,A})$ can be obtained from the number of incoming and outgoing edges computed in Proposition 2.3 since, with $\deg$ denoting the valence of vertices,

$$\begin{aligned} \#E(\mathcal{G}_{n,A}) &= \frac{1}{2} \sum_{F \in V(\mathcal{G}_{n,A})} \deg(F) \\ &= \frac{1}{2} \sum_{F \in V(\mathcal{G}_{n,A})} (N_{\text{EM}}^{\text{out}}(F) + N_{\text{EM}}^{\text{in}}(F) + N_{\text{IM}}^{\text{out}}(F) + N_{\text{IM}}^{\text{in}}(F) + N_{\text{SM min}}^{\text{out}}(F) + N_{\text{SM min}}^{\text{in}}(F)). \end{aligned}$$

We can bound these terms by

$$N_{\text{EM}}^{\text{out}}(F) \leq \binom{n}{2} = O(n^2), \quad N_{\text{EM}}^{\text{in}}(F) \leq n$$

$$N_{\text{IM}}^{\text{out}}(F) \leq 2n(n-2) = O(n^2), \quad N_{\text{IM}}^{\text{in}}(F) \leq 2n(n-1) = O(n^2)$$

$$N_{\text{SM min}}^{\text{out}}(F) \leq 2n(n-1) = O(n^2), \quad N_{\text{SM min}}^{\text{in}}(F) \leq 2n^2(3n+1) = O(n^3).$$

We then obtain an overall polynomial growth estimate times the number of vertices

$$\#E(\mathcal{G}_{n,A}) \leq P(n) \cdot \#V(\mathcal{G}_{n,A}).$$

Thus, the density satisfies

$$D(\mathcal{G}_{n,A}) \leq \frac{P(n)}{\#V(\mathcal{G}_{n,A})}.$$

The number of vertices $\#V(\mathcal{G}_{n,A})$ is the number of forests $F \in \mathfrak{F}_{A,n}$. This can be computed as in Lemma 2.2. $\square$

2.4. Strong connectedness. Recall that a directed graph G is strongly connected if there is a directed path of edges between any pair of vertices in G . The graph G is periodic if all the cycle-lengths in the graph have a common gcd greater than 1 and is aperiodic if the gcd of the cycle-lengths is equal to 1.

Proposition 2.5. *The graphs $\mathcal{G}_{n,A}$ are strongly connected and aperiodic for every choice of (n, A) .*

There are valid reasons to consider both possibilities: the usual claim in the linguistics literature that the Merge operator itself is Markovian is rigorously formalized by the HAMC approach described here, while the random walk approach can be absorbed into the fact that a weighting of the Merge operations is needed in any case, to describe cost functions and optimality properties, as we will discuss more in detail in §11, and the rescaling used in the first approach can also be combined into an overall cost function (though with a less transparent interpretation in terms of the linguistic model).

However, there is an additional property that makes the HAMC approach preferable, namely the HAMC dynamical system satisfies an entropy optimization property, as we discuss in §3.3. In fact, as we will discuss further in §13, other forms of optimization with respect to the Shannon entropy will play an important role in the dynamical properties of the Merge action.

For these reasons we will discuss both approaches here, but we will only focus on the HAMC formulation.

3.1. Random walk on the Merge graph. In the case of an undirected graph G , the random walk has a simple behavior. It is the Markov chain with transition probabilities, for $v, v' \in V(G)$,

$$(3.1) \quad \mathbb{P}(v, v') = \begin{cases} \frac{1}{\deg(v)} & \exists e \in E(G) : \partial e = \{v, v'\} \\ 0 & \text{otherwise.} \end{cases}$$

The stationary distribution

$$(3.2) \quad \pi(v') = \sum_{v \in V(G)} \pi(v) \mathbb{P}(v, v')$$

is then just given by

$$\pi(v) = \frac{\deg(v)}{2\#E(G)},$$

where we again use the fact that $\sum_{v \in V(G)} \deg(v) = 2\#E(G)$.

However, the random walk on a directed graph is more subtle. It is given by the Markov chain with transition probabilities, for $v, v' \in V(G)$,

$$(3.3) \quad \mathbb{P}(v, v') = \begin{cases} \frac{1}{\deg^{\text{out}}(v)} & \exists e \in E(G) : v = s(e), v' = t(e) \\ 0 & \text{otherwise.} \end{cases}$$

where now only the outward directions are chosen at each vertex. In this case one does not have a simple form of the stationary distribution in terms of valencies as in the undirected case, since we would now have $\sum_v \deg^{\text{out}}(v) \mathbb{P}(v, v') = \deg^{\text{in}}(v')$. If the graph G is strongly connected, however, it is possible to obtain the stationary distribution through the Perron–Frobenius theorem.

The convergence to the stationary distribution then depends on the aperiodicity of the graph. Periodicity with the gcd of the cycle-lengths equal to $k > 1$ is equivalent to the existence of a morphism $G \rightarrow C_k$, where C_k is a cyclically directed polygon graph of length k , and is in turn equivalent to the spectrum of the transition matrix containing k -th roots of unity, and in turn implies that one does not have convergence by iteration to the stationary distribution, but an oscillatory behavior. On the other hand, in the case of aperiodicity, the Perron-Frobenius theorem ensures that all the other eigenvalues have smaller absolute value so the random walk always converges to a stationary distribution given by the Perron-Frobenius eigenvector (3.2).

In the case of the Merge graph, this approach considers the random walk on the directed graphs $\mathcal{G}_{n,A}$ described above. We know from Proposition 2.5 that the Merge graphs $\mathcal{G}_{n,A}$ are both strongly connected and aperiodic. The original Merge operator is the linear operator on $\mathcal{V}(\mathfrak{F}_{S\mathcal{O}_0})$ represented in matrix form $\mathcal{K} = (\mathcal{K}_{F,F'})$ for the standard basis $\{F \in \mathfrak{F}_{S\mathcal{O}_0}\}$, on the invariant subspaces $\mathcal{V}(\mathfrak{F}_{A,n})$, by the adjacency matrix of the directed graph $\mathcal{G}_{n,A}$. The random

walk operator is then the linear operator on the same vector space, represented in the same basis by rescaling the entries $\mathcal{K}_{F,F'} \mapsto \deg^{out}(F)^{-1} \mathcal{K}_{F,F'}$ with

$$\deg^{out}(F) = N_{EM}^{out}(F) + N_{IM}^{out}(F) + N_{SM\min}^{out}(F).$$

We use the notation $\mathcal{RW}$ and

$$(3.4) \quad \mathcal{RW}_{F,F'} := \deg^{out}(F)^{-1} \mathcal{K}_{F,F'}$$

to denote this random walk operator on the Merge graph, writing $\mathcal{RW}^{(A,n)}$ and $\mathcal{RW}_{F,F'}^{(A,n)}$ when we want to explicitly keep track of the data (A, n) of the graph $\mathcal{G}_{n,A}$.

From the point of view of the linguistic model that motivates our construction, the problem with the use of the random walk operator $\mathcal{RW}^{(A,n)}$ is that one does not have a natural theoretical justification for the rescaling of the entries of $\mathcal{K} = (\mathcal{K}_{F,F'})$ (the linguistic Merge operations) by the inverses $\deg^{out}(F)^{-1}$ the out-degrees: from the perspective of theoretical linguistics it is the Merge action itself that is Markovian. Thus, a better formulation is provided by the following approach.

3.2. Hopf algebra Markov chain. The other approach we mentioned above does not normalize the entries of the $\mathcal{K}_{F,F'}$ matrix to make it stochastic, rather it finds the appropriate change of basis in which the original linear operator is represented by a stochastic matrix. This can be done, as discussed in [14], [27]. Since the graphs $\mathcal{G}_{n,A}$ are strongly connected, we can apply the Perron-Frobenius theorem that ensures the existence of right Perron-Frobenius eigenvector $\eta(F) = \eta_{n,A}(F) > 0$,

$$(3.5) \quad \sum_{F'} \mathcal{K}_{F,F'}^{(A,n)} \eta(F') = \lambda \eta(F),$$

where $\lambda = \lambda_{n,A}$ is the Perron-Frobenius eigenvalue of the adjacency matrix $\mathcal{K}^{(A,n)}$ of the graph $\mathcal{G}_{n,A}$. We can then rescale the basis elements by $F \mapsto \eta(F)^{-1} F$. In this new basis $\{\eta(F)^{-1} F\}$ the same Merge operator is represented by the matrix $\hat{\mathcal{K}}_{F,F'}^{(A,n)}$ of the form

$$(3.6) \quad \hat{\mathcal{K}}_{F,F'}^{(A,n)} = \lambda^{-1} \frac{\eta(F')}{\eta(F)} \mathcal{K}_{F,F'}^{(A,n)}.$$

The matrix $\hat{\mathcal{K}}^{(A,n)} = (\hat{\mathcal{K}}_{F,F'}^{(A,n)})$ obtained as in (3.6) is by construction stochastic, hence it defines the transition matrix of a Markov chain. This is the rigorous formulation of the ‘‘Merge is Markovian’’ property, with (3.6) the actual associated Hopf algebra Markov chain.

One can then study the stationary distribution (3.2) with $\mathbb{P}(F, F') = \hat{\mathcal{K}}_{F,F'}^{(A,n)}$, and other properties of this Markov chain that describe the dynamics of Merge on workspaces. Proposition 2.5 already provides some important information about the Merge action on workspaces.

Proposition 3.1. *For a strongly connected graph $\mathcal{G}$ with adjacency matrix $\mathcal{K}$, with Perron-Frobenius eigenvalue λ and (right) eigenvector η , let $\hat{\mathcal{K}}$ be the associated Markov chain defined as in (3.6). The stationary distribution π of $\hat{\mathcal{K}}$,*

$$(3.7) \quad \sum_x \pi(x) \hat{\mathcal{K}}_{x,y} = \pi(y),$$

is given by

$$(3.8) \quad \pi(x) = \psi(x) \eta(x),$$

where ψ, η are the left and right Perron-Frobenius eigenvectors of $\mathcal{K}$.

Proof. Since the graph $\mathcal{G}$ is strongly connected, the Markov chain $\hat{\mathcal{K}}$ is irreducible hence its (normalized) stationary distribution is unique. Thus, it suffices to see that $\psi(x)\eta(x)$ satisfies (3.7). We have

$$\sum_x \psi(x)\eta(x)\lambda^{-1}\frac{\eta(y)}{\eta(x)}\mathcal{K}_{x,y} = \sum_x \psi(x)\mathcal{K}_{x,y}\lambda^{-1}\eta(y) = \psi(y)\eta(y).$$

□

We will prove in Proposition 9.1 a more refined version of this simple observation, adapted to a projection between Markov chains.

Remark 3.2. A Markov chain with transition matrix $\mathcal{K}$ is *ergodic* if the stationary distribution is unique and any given initial distribution on the set of vertices of the underlying graph G converges under iterations of $\mathcal{K}$ to the stationary distribution.

Ergodicity, as formulated in Remark 3.2 above, follows from strong connectedness and aperiodicity of G , hence Proposition 2.5 has the following direct consequence.

Proposition 3.3. *The Merge Hopf algebra Markov chain $\hat{\mathcal{K}}^{(A,n)}$ is ergodic, and so is the Merge random walk $\mathcal{RW}^{(A,n)}$.*

This result carries interesting theoretical information about the Minimalist model of syntax. It shows that syntax formation and transformation (what the Merge operator models) is an ergodic dynamical system. This means that the dynamics has a unique stationary distribution that captures the long-term behavior of the dynamics (the long-term probability of being in one of the possible states, which are the workspaces, the syntactic structures). Ergodic Markov chains are those for which the long-term behavior is *predictable*: ergodicity, for dynamical systems, is usually viewed as the property that the time average is the same as the space average. In the case of Markov chains, this can be more precisely expressed as in Remark 3.2, as the property that the long-term probability of being in a given state does not depend on the choice of the initial state. The values $\pi(F)$ of the stationary distribution also compute the expected return times of the dynamics, which are given by $1/\pi(F)$.

Note here, however, that these nice properties of the dynamics hold when we include Sideward Merge as part of the dynamics (because the strong connectedness property of the graphs $\mathcal{G}_{n,A}$ require the SM arrows). We will discuss more in detail in §6 what changes when the SM arrows are absent, and when the different types of arrows, EM, IM, and SM are weighted differently by a suitable cost function.

3.3. Hopf algebra Markov chain as maximal entropy random walk. We also point out here an important property that the HAMC formulation of Merge, as in (3.6) satisfies, that the random walk operator $\mathcal{RW}$ does not have: maximal entropy optimization.

Suppose given a directed graph $\mathcal{G}$ with adjacency matrix $\mathcal{K}_{\mathcal{G}}$. Suppose that $\mathcal{S}$ is a Markov chain on the same underlying directed graph $\mathcal{G}$, namely a weighted version of the adjacency matrix $\mathcal{K}_{\mathcal{G}}$ that is a stochastic matrix.

The entropy of a Markov chain with transition probabilities $\mathcal{S}(x, y)$ and with stationary distribution $\pi(x)$ is given by

$$(3.9) \quad \text{Sh}(\mathcal{S}) = - \sum_x \pi(x) \sum_y \mathcal{S}(x, y) \log \mathcal{S}(x, y).$$

This is also called the *entropy rate* of $\mathcal{S}$.

The entropy rate $\text{Sh}(\mathcal{S})$ is equal to the asymptotic behavior (in the length of the path becoming large) of the average entropy of the probability distribution in the space of paths. Namely, suppose

given a directed path $\gamma = e_0 e_1 \dots e_{\ell-1}$ in the graph $\mathcal{G}$, with source and target vertices of the edges $s(e_i) = x_i$ and $t(e_i) = x_{i+1}$. The probability distribution on paths of length ℓ is given by

$$(3.10) \quad \mathbb{P}_\ell(\gamma) = \pi(x_0) \mathcal{S}(x_0, x_1) \cdots \mathcal{S}(x_{\ell-1}, x_\ell).$$

The entropy rate (3.9) is the limit

$$(3.11) \quad \text{Sh}(\mathcal{S}) = \lim_{\ell \rightarrow \infty} \frac{\text{Sh}(\mathbb{P}_\ell)}{\ell},$$

where $\text{Sh}(\mathbb{P}_\ell)$ is the Shannon entropy of the probability (3.10).

When $\mathcal{S}$ varies over all Markov chains with the same underlying directed graph $\mathcal{G}$, it is known (see [29], [30]) that the maximal possible entropy rate $\text{Sh}(\mathcal{S})$ is the topological entropy of the directed graph $\mathcal{G}$, which is given by

$$(3.12) \quad S_{\text{top}}(\mathcal{G}) := \log \lambda_{\mathcal{G}},$$

where $\lambda_{\mathcal{G}}$ is the Perron-Frobenius eigenvalue of the adjacency matrix $\mathcal{K}_{\mathcal{G}}$.

The following fact is well known (see for instance [5], [11], [29], [30]) but we recall it here for the reader's convenience.

Proposition 3.4. *Suppose given a strongly connected directed graph $\mathcal{G}$. Let $\mathcal{K}_{\mathcal{G}}$ be the adjacency matrix of $\mathcal{G}$, with Perron-Frobenius eigenvalue λ and with η and ψ the right and left Perron-Frobenius eigenvectors, normalized so that*

$$(3.13) \quad \sum_{x,y} (\psi(x)\eta(y))^2 = 1.$$

The maximal entropy random walk (MERW) is the Markov chain

$$(3.14) \quad \mathcal{S}_{\text{MERW}}(x, y) = \hat{\mathcal{K}}_{\mathcal{G}}(x, y) = \frac{1}{\lambda} \frac{\eta(y)}{\eta(x)} \mathcal{K}_{\mathcal{G}}(x, y).$$

The $\mathcal{S}_{\text{MERW}}$ satisfies the following properties.

- (1) *The probability distribution (3.10), with $\mathcal{S} = \mathcal{S}_{\text{MERW}}$, is uniform on all paths with the same length and endpoints.*
- (2) *The entropy rate is maximal,*

$$\text{Sh}(\mathcal{S}_{\text{MERW}}) = S_{\text{top}}(\mathcal{G}).$$

Proof. One defines the MERW Markov chain as (3.14), and one can see that it determines a uniform distribution on the space of paths with fixed endpoints and length by observing that by (3.10) we have

$$(3.15) \quad \mathbb{P}_{\text{MERW},\ell}(\gamma) = \frac{1}{\lambda^\ell} \pi(x_0) \frac{\eta(x_\ell)}{\eta(x_0)} \mathcal{K}_{\mathcal{G}}(x_0, x_1) \cdots \mathcal{K}_{\mathcal{G}}(x_{\ell-1}, x_\ell) = \frac{\psi(x_0)\eta(x_\ell)}{\lambda^\ell},$$

where in the second equality we used Proposition 3.1 and the fact that, for a directed edge e_i in $\mathcal{G}$ with $s(e_i) = x_i$ and $t(e_i) = x_{i+1}$, we have $\mathcal{K}_{\mathcal{G}}(x_i, x_{i+1}) = 1$. To see that MERW also maximizes the entropy rate, we use the limit (3.11). We have

$$\text{Sh}(\mathbb{P}_{\text{MERW},\ell}) = - \sum_{\gamma} \mathbb{P}_{\text{MERW},\ell}(\gamma) \log \mathbb{P}_{\text{MERW},\ell}(\gamma),$$

with the sum over all paths γ of length ℓ . This is

$$\text{Sh}(\mathbb{P}_{\text{MERW},\ell}) = - \sum_{x_0, x_\ell} \mathcal{K}_{\mathcal{G}}^\ell(x_0, x_\ell) \frac{\psi(x_0)\eta(x_\ell)}{\lambda^\ell} \log \left(\frac{\psi(x_0)\eta(x_\ell)}{\lambda^\ell} \right)$$

$$= - \sum_{x_0, x_\ell} \frac{\mathcal{K}_\mathcal{G}^\ell(x_0, x_\ell)}{\lambda^\ell} \psi(x_0) \eta(x_\ell) \log(\psi(x_0) \eta(x_\ell)) + \sum_{x_0, x_\ell} \frac{\mathcal{K}_\mathcal{G}^\ell(x_0, x_\ell)}{\lambda^\ell} \psi(x_0) \eta(x_\ell) \log(\lambda^\ell),$$

where $\mathcal{K}_\mathcal{G}^\ell(x, y) = \#\{\gamma \mid s(\gamma) = x, t(\gamma) = y, \text{length}(\gamma) = \ell\}$. Thus, we have

$$\text{Sh}(\mathcal{S}_{\text{MERW}}) = \lim_{\ell \rightarrow \infty} \frac{\text{Sh}(\mathbb{P}_{\text{MERW}, \ell})}{\ell} = \lim_{\ell \rightarrow \infty} \sum_{x, y} \frac{\mathcal{K}_\mathcal{G}^\ell(x, y)}{\lambda^\ell} \psi(x) \eta(y) \log(\lambda).$$

The Perron-Frobenius theorem shows that, when $\ell \rightarrow \infty$

$$\frac{\mathcal{K}_\mathcal{G}^\ell(x, y)}{\lambda^\ell} \sim \psi(x) \eta(y),$$

hence the above limit and the normalization (3.13) give

$$\text{Sh}(\mathcal{S}_{\text{MERW}}) = \sum_{x, y} (\psi(x) \eta(y))^2 \log(\lambda) = \log(\lambda) = S_{\text{top}}(\mathcal{G}).$$

□

Note how the transition probabilities of MERW are non-local (unlike those of the random walk $\mathcal{RW}$ that only depend on the (out)degrees at vertices) and depend on the whole graph topology through the Perron-Frobenius eigenvalue and eigenvector of the adjacency matrix.

3.4. Free energy optimization. We also recall, for later use, a generalization of Proposition 3.4, where instead of the adjacency matrix one uses a weighted version and MERW induces a Boltzmann distribution on paths and correspondingly optimizes the free energy instead of the entropy rate (see [30]).

Let $\mathcal{G}$ be a strongly connected directed graph and let $\mathcal{K}_\mathcal{G}^{(\omega, \beta)}$ be a weighted adjacency matrix, namely

$$(3.16) \quad \mathcal{K}_\mathcal{G}^{(\omega, \beta)}(x, y) = e^{-\beta \omega(x, y)} \mathcal{K}_\mathcal{G}(x, y),$$

where $\omega(x, y) \in \mathbb{R}$ are real weights and $\mathcal{K}_\mathcal{G}$ is the adjacency matrix of $\mathcal{G}$. We define the energy function $\mathbb{E} : E(\mathcal{G}) \rightarrow \mathbb{R}_+$ on edges of $\mathcal{G}$ as

$$\mathbb{E}(e) = \omega(x, y), \quad \text{for } x = s(e) \text{ and } y = t(e).$$

Even though the weights are not necessarily positive, the energy is bounded below since the graph is finite, hence it is non-negative after a global shift. Let $\mathcal{S}$ be a Markov chain with underlying directed graph $\mathcal{G}$. The *free energy* $\mathbb{F}(\mathcal{S})$ of the Markov chain is defined as

$$(3.17) \quad \mathbb{F}(\mathcal{S}) = \bar{\mathbb{E}}(\mathcal{S}) - \beta^{-1} \text{Sh}(\mathcal{S})$$

where $\bar{\mathbb{E}}(\mathcal{S})$ is the expected energy

$$(3.18) \quad \bar{\mathbb{E}}(\mathcal{S}) := \sum_x \pi^\mathcal{S}(x) \sum_y \mathcal{S}(x, y) \omega(x, y),$$

with $\pi^\mathcal{S}$ the stationary distribution of $\mathcal{S}$. One looks for Markov chains that *minimizes* $\mathbb{F}(\mathcal{S})$. Note that in the case with no weights, $\omega = 0$, this is equivalent to maximizing the entropy $\text{Sh}(\mathcal{S})$ so one recovers the previous case. Sometimes one rephrases this question equivalently in terms of a *maximization* of the free energy $\tilde{\mathbb{F}}$ written as

$$(3.19) \quad \tilde{\mathbb{F}}(\mathcal{S}) = \text{Sh}(\mathcal{S}) - \beta \bar{\mathbb{E}}(\mathcal{S})$$

as in the formulation in [30], while minimization of (3.17) that we follow here is more suitable for direct comparison with the thermodynamics semirings used in [21].

Definition 3.5. The Boltzmann distribution on the set of edges of $\mathcal{G}$ is given by

$$(3.20) \quad \mathbb{P}_{(\omega,\beta)}(e) := \frac{\psi^{(\omega,\beta)}(s(e))\eta^{(\omega,\beta)}(t(e))e^{-\beta\mathbb{E}(e)}}{Z_\omega(\beta)} \quad \text{with } Z_\omega(\beta) = \sum_e \psi^{(\omega,\beta)}(s(e))\eta^{(\omega,\beta)}(t(e)) e^{-\beta\mathbb{E}(e)},$$

with $\psi^{(\omega,\beta)}$ and $\eta^{(\omega,\beta)}$ the left and right Perron-Frobenius eigenvectors of $\mathcal{K}_\mathcal{G}^{(\omega,\beta)}$, with Perron-Frobenius eigenvalue $\lambda_{(\omega,\beta)}$. We use the normalization

$$(3.21) \quad \sum_x \psi^{(\omega,\beta)}(x)\eta^{(\omega,\beta)}(x) = 1.$$

Lemma 3.6. The partition function of the Boltzmann distribution (3.20) satisfies

$$(3.22) \quad Z_\omega(\beta) = \lambda_{\omega,\beta}$$

Proof. We have

$$\begin{aligned} \sum_e \psi^{(\omega,\beta)}(s(e))\eta^{(\omega,\beta)}(t(e)) e^{-\beta\mathbb{E}(e)} &= \sum_{x,y} \psi^{(\omega,\beta)}(x)\eta^{(\omega,\beta)}(y)e^{-\beta\omega(x,y)}\mathcal{K}_\mathcal{G}(x,y) = \\ &= \lambda_{\omega,\beta} \sum_x \psi^{(\omega,\beta)}(x)\eta^{(\omega,\beta)}(x) = \lambda_{\omega,\beta} \end{aligned}$$

with the normalization (3.21). □

We denote by $\hat{\mathcal{K}}_\mathcal{G}^{(\omega,\beta)}$ the Markov chain associated to the weighted adjacency matrix by

$$(3.23) \quad \hat{\mathcal{K}}_\mathcal{G}^{(\omega,\beta)}(x,y) := \frac{1}{\lambda_{(\omega,\beta)}} \frac{\eta^{(\omega,\beta)}(y)}{\eta^{(\omega,\beta)}(x)} \mathcal{K}_\mathcal{G}^{(\omega,\beta)}(x,y).$$

This has the following property.

Lemma 3.7. For the Markov chain $\hat{\mathcal{K}}_\mathcal{G}^{(\omega,\beta)}$, the expected energy satisfies

$$(3.24) \quad \mathbb{F}(\hat{\mathcal{K}}_\mathcal{G}^{(\omega,\beta)}) = \bar{\mathbb{E}}_{\mathbb{P}_{(\omega,\beta)}} - \beta^{-1} \text{Sh}(\mathbb{P}_{(\omega,\beta)}),$$

with $\mathbb{P}_{(\omega,\beta)}$ the Boltzmann distribution (3.20) and

$$\bar{\mathbb{E}}_{\mathbb{P}_{(\omega,\beta)}} = \sum_e \mathbb{P}_{(\omega,\beta)}(e) \mathbb{E}(e).$$

Proof. We have

$$\begin{aligned} \mathbb{F}(\hat{\mathcal{K}}_\mathcal{G}^{(\omega,\beta)}) &= \sum_x \pi^{(\omega,\beta)}(x) \sum_y \hat{\mathcal{K}}_\mathcal{G}^{(\omega,\beta)}(x,y) (\omega(x,y) + \beta^{-1} \log(\hat{\mathcal{K}}_\mathcal{G}^{(\omega,\beta)}(x,y))) \\ &= \sum_x \pi^{(\omega,\beta)}(x) \sum_y \hat{\mathcal{K}}_\mathcal{G}^{(\omega,\beta)}(x,y) (\omega(x,y) + \beta^{-1} \log(e^{-\beta\omega(x,y)}) + \beta^{-1} \log(\hat{\mathcal{K}}_\mathcal{G}(x,y))) \\ &= \beta^{-1} \sum_x \pi^{(\omega,\beta)}(x) \sum_y e^{-\beta\omega(x,y)} \hat{\mathcal{K}}_\mathcal{G}(x,y) \log(\hat{\mathcal{K}}_\mathcal{G}(x,y)). \end{aligned}$$

By Lemma 3.6 and Proposition 3.1 applied to $\hat{\mathcal{K}}_\mathcal{G}^{(\omega,\beta)}$, we also have

$$\bar{\mathbb{E}}_{\mathbb{P}_{(\omega,\beta)}} = \sum_{x,y} \frac{\psi^{(\omega,\beta)}(x)\eta^{(\omega,\beta)}(y)}{\lambda_{(\omega,\beta)}} e^{-\beta\omega(x,y)} \mathcal{K}_\mathcal{G}(x,y) \omega(x,y) = \sum_x \pi^{(\omega,\beta)}(x) \sum_y e^{-\beta\omega(x,y)} \hat{\mathcal{K}}_\mathcal{G}(x,y) \omega(x,y)$$

and

$$\begin{aligned}
-\beta^{-1} \text{Sh}(\mathbb{P}_{(\omega, \beta)}) &= \beta^{-1} \sum_x \pi^{(\omega, \beta)}(x) \sum_y \hat{\mathcal{K}}_{\mathcal{G}}^{(\omega, \beta)}(x, y) \log(\hat{\mathcal{K}}_{\mathcal{G}}^{(\omega, \beta)}(x, y)) \\
&= \beta^{-1} \sum_x \pi^{(\omega, \beta)}(x) \sum_y \hat{\mathcal{K}}_{\mathcal{G}}^{(\omega, \beta)}(x, y) \log(e^{-\beta \omega(x, y)}) + \beta^{-1} \sum_x \pi^{(\omega, \beta)}(x) \sum_y e^{-\beta \omega(x, y)} \hat{\mathcal{K}}_{\mathcal{G}}(x, y) \log(\hat{\mathcal{K}}_{\mathcal{G}}(x, y)) \\
&= - \sum_x \pi^{(\omega, \beta)}(x) \sum_y e^{-\beta \omega(x, y)} \hat{\mathcal{K}}_{\mathcal{G}}(x, y) \omega(x, y) + \beta^{-1} \sum_x \pi^{(\omega, \beta)}(x) \sum_y e^{-\beta \omega(x, y)} \hat{\mathcal{K}}_{\mathcal{G}}(x, y) \log(\hat{\mathcal{K}}_{\mathcal{G}}(x, y)),
\end{aligned}$$

so that

$$\bar{\mathbb{E}}_{\mathbb{P}_{(\omega, \beta)}} - \beta^{-1} \text{Sh}(\mathbb{P}_{(\omega, \beta)}) = \beta^{-1} \sum_x \pi^{(\omega, \beta)}(x) \sum_y e^{-\beta \omega(x, y)} \hat{\mathcal{K}}_{\mathcal{G}}(x, y) \log(\hat{\mathcal{K}}_{\mathcal{G}}(x, y)).$$

□

We also have the following properties of the Boltzmann distribution.

Lemma 3.8. *The Boltzmann distribution $\mathbb{P}_{(\omega, \beta)}$ of (3.20) satisfies*

$$(3.25) \quad \bar{\mathbb{E}}_{\mathbb{P}_{(\omega, \beta)}} = -\frac{d}{d\beta} \log Z_{\omega}(\beta),$$

$$(3.26) \quad \text{Sh}(\mathbb{P}_{(\omega, \beta)}) = \log Z_{\omega}(\beta) - \beta \frac{d}{d\beta} \log Z_{\omega}(\beta) + \text{Sh}(\pi^{(\omega, \beta)}),$$

$$(3.27) \quad \mathbb{F}_{\mathbb{P}_{(\omega, \beta)}} = \bar{\mathbb{E}}_{\mathbb{P}_{(\omega, \beta)}} - \beta^{-1} \text{Sh}(\mathbb{P}_{(\omega, \beta)}) = -\beta^{-1} (\text{Sh}(\pi^{(\omega, \beta)}) + \log Z_{\omega}(\beta)).$$

Proof. We obtain (3.25) as the identity

$$\begin{aligned}
\bar{\mathbb{E}}_{\mathbb{P}_{(\omega, \beta)}} &= \sum_e \mathbb{P}_{(\omega, \beta)}(e) \mathbb{E}(e) = \frac{\sum_e \psi^{(\omega, \beta)}(s(e)) \eta^{(\omega, \beta)}(t(e)) e^{-\beta \mathbb{E}(e)} \mathbb{E}(e)}{Z_{\omega}(\beta)} \\
&= \frac{-\frac{d}{d\beta} \sum_e \psi^{(\omega, \beta)}(s(e)) \eta^{(\omega, \beta)}(t(e)) e^{-\beta \mathbb{E}(e)}}{Z_{\omega}(\beta)} = -\frac{d}{d\beta} \log Z_{\omega}(\beta).
\end{aligned}$$

We also have

$$\begin{aligned}
\text{Sh}(\mathbb{P}_{(\omega, \beta)}) &= - \sum_e \frac{\psi^{(\omega, \beta)}(s(e)) \eta^{(\omega, \beta)}(t(e)) e^{-\beta \omega(e)}}{Z_{\omega}(\beta)} \log\left(\frac{\psi^{(\omega, \beta)}(s(e)) \eta^{(\omega, \beta)}(t(e)) e^{-\beta \omega(e)}}{Z_{\omega}(\beta)}\right) \\
&= \beta \sum_e \frac{\psi^{(\omega, \beta)}(s(e)) \eta^{(\omega, \beta)}(t(e)) e^{-\beta \omega(e)}}{Z_{\omega}(\beta)} \omega(e) \\
&\quad + \sum_e \frac{\psi^{(\omega, \beta)}(s(e)) \eta^{(\omega, \beta)}(t(e)) e^{-\beta \omega(e)}}{Z_{\omega}(\beta)} \log Z_{\omega}(\beta) \\
&\quad - \sum_e \frac{\psi^{(\omega, \beta)}(s(e)) \eta^{(\omega, \beta)}(t(e)) e^{-\beta \omega(e)}}{Z_{\omega}(\beta)} \log(\psi^{(\omega, \beta)}(s(e)) \eta^{(\omega, \beta)}(t(e))) \\
&= \beta \bar{\mathbb{E}}_{\mathbb{P}_{(\omega, \beta)}} + \log Z_{\omega}(\beta) - \sum_{x, y} \frac{\psi^{(\omega, \beta)}(x) \eta^{(\omega, \beta)}(y) \mathcal{K}_{\mathcal{G}}^{(\omega, \beta)}(x, y)}{Z_{\omega}(\beta)} \log(\psi^{(\omega, \beta)}(x) \eta^{(\omega, \beta)}(y)) \\
&= \log Z_{\omega}(\beta) - \beta \frac{d}{d\beta} \log Z_{\omega}(\beta) - \sum_{x, y} \frac{\psi^{(\omega, \beta)}(x) \eta^{(\omega, \beta)}(y) \mathcal{K}_{\mathcal{G}}^{(\omega, \beta)}(x, y)}{\lambda_{(\omega, \beta)}} \log(\psi^{(\omega, \beta)}(x))
\end{aligned}$$

$$\begin{aligned}
& - \sum_{x,y} \frac{\psi^{(\omega,\beta)}(x) \eta^{(\omega,\beta)}(y) \mathcal{K}_{\mathcal{G}}^{(\omega,\beta)}(x,y)}{\lambda_{(\omega,\beta)}} \log(\eta^{(\omega,\beta)}(y)) = \log Z_{\omega}(\beta) - \beta \frac{d}{d\beta} \log Z_{\omega}(\beta) \\
& - \sum_x \psi^{(\omega,\beta)}(x) \eta^{(\omega,\beta)}(x) \log(\psi^{(\omega,\beta)}(x)) - \sum_y \psi^{(\omega,\beta)}(y) \eta^{(\omega,\beta)}(y) \log(\eta^{(\omega,\beta)}(y)) \\
& = \log Z_{\omega}(\beta) - \beta \frac{d}{d\beta} \log Z_{\omega}(\beta) - \sum_x \psi^{(\omega,\beta)}(x) \eta^{(\omega,\beta)}(x) \log(\psi^{(\omega,\beta)}(x) \eta^{(\omega,\beta)}(x)),
\end{aligned}$$

which gives (3.26), from which (3.27) then follows. $\square$

We also have the following generalization of Proposition 3.4.

Proposition 3.9. *Suppose given a strongly connected directed graph $\mathcal{G}$ with weighted adjacency matrix $\mathcal{K}_{\mathcal{G}}^{(\omega,\beta)}$ as above, and with Perron-Frobenius eigenvalue $\lambda_{(\omega,\beta)}$ and left and right eigenvectors $\psi^{(\omega,\beta)}$ and $\eta^{(\omega,\beta)}$. Then the weighted MERW on $\mathcal{G}$ is the Markov chain (3.23),*

$$(3.28) \quad \mathcal{S}_{\text{MERW}}^{(\omega,\beta)}(x,y) = \hat{\mathcal{K}}_{\mathcal{G}}^{(\omega,\beta)}(x,y) = \frac{1}{\lambda_{(\omega,\beta)}} \frac{\eta^{(\omega,\beta)}(y)}{\eta^{(\omega,\beta)}(x)} \mathcal{K}_{\mathcal{G}}^{(\omega,\beta)}(x,y).$$

The weighted MERW has the following properties.

- (1) The probability on the space of paths induced by $\hat{\mathcal{K}}_{\mathcal{G}}^{(\omega,\beta)}$ is the Boltzmann distribution

$$(3.29) \quad \mathbb{P}_{(\omega,\beta),\ell}(\gamma) = \frac{1}{Z_{\omega,\ell}(\beta)} \psi^{(\omega,\beta)}(x_0) \eta^{(\omega,\beta)}(x_{\ell}) \exp(-\beta \mathbb{E}(\gamma)),$$

for a path $\gamma = e_0 e_1 \dots e_{\ell}$ with $s(e_i) = x_i$ and $t(e_i) = x_{i+1}$, with energy

$$\mathbb{E}(\gamma) := \sum_{i=0}^{\ell} \omega(x_i, x_{i+1}),$$

with the partition function

$$Z_{\omega,\ell}(\beta) = \sum_{\gamma} \psi^{(\omega,\beta)}(x_0) \eta^{(\omega,\beta)}(x_{\ell}) \exp(-\beta \mathbb{E}(\gamma)) = \lambda_{(\omega,\beta)}^{\ell},$$

with the sum taken over paths γ of length ℓ .

- (2) The free energy $\mathbb{F}_{\mathbb{P}_{(\omega,\beta),\ell}}$ of the Boltzmann distribution on paths of length ℓ satisfies

$$(3.30) \quad \lim_{\ell \rightarrow \infty} \frac{\mathbb{F}_{\mathbb{P}_{(\omega,\beta),\ell}}}{\ell} = \mathbb{F}(\mathcal{S}_{\text{MERW}}^{(\omega,\beta)}) + \beta^{-1} \text{Sh}(\pi^{(\omega,\beta)}),$$

where

$$(3.31) \quad \mathbb{F}_{\mathbb{P}_{(\omega,\beta),\ell}} := -\beta^{-1} \text{Sh}(\mathbb{P}_{(\beta,\omega),\ell}) + \sum_{\gamma} \mathbb{P}_{(\beta,\omega),\ell}(\gamma) \mathbb{E}(\gamma),$$

and the limit (3.30) gives the optimal value

$$(3.32) \quad \lim_{\ell \rightarrow \infty} \frac{\mathbb{F}_{\mathbb{P}_{(\omega,\beta),\ell}}}{\ell} = -\beta^{-1} \log \lambda_{(\omega,\beta)}.$$

Proof. The argument follows the same steps as in Proposition 3.4, but replacing the adjacency matrix $\mathcal{K}_{\mathcal{G}}$ with the weighted version $\mathcal{K}_{\mathcal{G}}^{(\omega,\beta)}(x,y) = \exp(-\beta \omega(x,y)) \mathcal{K}_{\mathcal{G}}(x,y)$. The distribution induced by $\hat{\mathcal{K}}_{\mathcal{G}}^{(\omega,\beta)}$ on paths is given by

$$\mathbb{P}_{(\omega,\beta),\ell}(\gamma) = \pi^{(\omega,\beta)}(x_0) \hat{\mathcal{K}}_{\mathcal{G}}^{(\omega,\beta)}(x_0, x_1) \cdots \hat{\mathcal{K}}_{\mathcal{G}}^{(\omega,\beta)}(x_{\ell-1}, x_{\ell})$$

$$= \frac{1}{\lambda_{(\omega,\beta)}^\ell} \psi^{(\omega,\beta)}(x_0) \eta^{(\omega,\beta)}(x_\ell) \mathcal{K}_{\mathcal{G}}^{(\omega,\beta)}(x_0, x_1) \cdots \mathcal{K}_{\mathcal{G}}^{(\omega,\beta)}(x_{\ell-1}, x_\ell) = \frac{1}{\lambda_{(\omega,\beta)}^\ell} \psi^{(\omega,\beta)}(x_0) \eta^{(\omega,\beta)}(x_\ell) e^{-\beta \mathbb{E}(\gamma)}.$$

To show that this agrees with (3.29) it suffices to show that $Z_{\omega,\ell}(\beta) = \lambda_{(\omega,\beta)}^\ell$. This follows as in Lemma 3.6 with

$$Z_{\omega,\ell}(\beta) = \sum_{x,y} \psi^{(\omega,\beta)}(x) \eta^{(\omega,\beta)}(y) (\mathcal{K}_{\mathcal{G}}^{(\omega,\beta)})^\ell(x, y) = \lambda_{(\omega,\beta)}^\ell \sum_x \psi^{(\omega,\beta)}(x) \eta^{(\omega,\beta)}(x) = \lambda_{(\omega,\beta)}^\ell.$$

The free energy of the distribution $\mathbb{P}_{(\omega,\beta),\ell}$ on the set of paths of length ℓ is given by

$$\mathbb{F}_{\mathbb{P}_{(\omega,\beta),\ell}} = -\beta^{-1} \text{Sh}(\mathbb{P}_{(\omega,\beta),\ell}) + \sum_{\gamma} \mathbb{P}_{(\omega,\beta),\ell}(\gamma) \mathbb{E}(\gamma)$$

where

$$-\beta^{-1} \text{Sh}(\mathbb{P}_{(\omega,\beta),\ell}) = \beta^{-1} \sum_{x,y} \psi^{(\omega,\beta)}(x) \eta^{(\omega,\beta)}(y) \frac{(\mathcal{K}_{\mathcal{G}}^{(\omega,\beta)})^\ell(x, y)}{\lambda_{(\omega,\beta)}^\ell} \log\left(\frac{\psi^{(\omega,\beta)}(x) \eta^{(\omega,\beta)}(y) e^{-\beta \mathbb{E}(\gamma)}}{\lambda_{(\omega,\beta)}^\ell}\right)$$

using as in Proposition 3.4 the fact that along the edges e_i of the path γ each $\mathcal{K}_{\mathcal{G}}(x_i, x_{i+1}) = 1$, so that we obtain

$$-\beta^{-1} \text{Sh}(\mathbb{P}_{(\omega,\beta),\ell}) = \beta^{-1} \sum_{x,y} \psi^{(\omega,\beta)}(x) \eta^{(\omega,\beta)}(y) \frac{(\mathcal{K}_{\mathcal{G}}^{(\omega,\beta)})^\ell(x, y)}{\lambda_{(\omega,\beta)}^\ell} (-\beta \mathbb{E}(\gamma) + \log\left(\frac{\psi^{(\omega,\beta)}(x) \eta^{(\omega,\beta)}(y)}{\lambda_{(\omega,\beta)}^\ell}\right)),$$

while

$$\sum_{\gamma} \mathbb{P}_{(\omega,\beta),\ell}(\gamma) \mathbb{E}(\gamma) = \sum_{x,y} \psi^{(\omega,\beta)}(x) \eta^{(\omega,\beta)}(y) \frac{(\mathcal{K}_{\mathcal{G}}^{(\omega,\beta)})^\ell(x, y)}{\lambda_{(\omega,\beta)}^\ell} \mathbb{E}(\gamma).$$

This gives

$$\begin{aligned} \mathbb{F}_{\mathbb{P}_{(\omega,\beta),\ell}} &= \beta^{-1} \sum_{x,y} \psi^{(\omega,\beta)}(x) \eta^{(\omega,\beta)}(y) \frac{(\mathcal{K}_{\mathcal{G}}^{(\omega,\beta)})^\ell(x, y)}{\lambda_{(\omega,\beta)}^\ell} \log\left(\frac{\psi^{(\omega,\beta)}(x) \eta^{(\omega,\beta)}(y)}{\lambda_{(\omega,\beta)}^\ell}\right). \\ &= \beta^{-1} \left(\sum_x \psi^{(\omega,\beta)}(x) \eta^{(\omega,\beta)}(x) \log(\psi^{(\omega,\beta)}(x)) + \sum_y \psi^{(\omega,\beta)}(y) \eta^{(\omega,\beta)}(y) \log(\eta^{(\omega,\beta)}(y)) - \beta^{-1} \log(\lambda_{(\omega,\beta)}^\ell) \right) \\ &= -\beta^{-1} (\text{Sh}(\pi^{(\omega,\beta)}) + \ell \log \lambda_{(\omega,\beta)}), \end{aligned}$$

so that the limit of $\mathbb{F}_{\mathbb{P}_{(\omega,\beta),\ell}}/\ell$ gives the optimal value $\min_{\mathcal{S}} \mathbb{F}(\mathcal{S}) = -\beta^{-1} \log(\lambda_{(\omega,\beta)})$. $\square$

3.5. Linguistic implications of entropy optimization. In the case of the Merge dynamics on $\mathcal{G}_{n,A}$, we consider the matrix representation $\hat{\mathcal{K}}^{(A,n)}$ of the Merge operations in the appropriate basis in which it is a Hopf algebra Markov chain, as explained above. As we have seen in this section, the form $\hat{\mathcal{K}}^{(A,n)}$ of the Merge dynamics is exactly the MERW on the Merge graph $\mathcal{G}_{n,A}$.

Thus, this shows a particularly interesting property of the action of Merge on workspaces, modeled as a Hopf algebra Markov chain: Merge is *information maximizing*, in the sense that it maximizes the entropy rate $\text{Sh}(\mathcal{S})$ over all the possible Markov chains on the same directed graph.

As we showed in this section, the Hopf algebra Markov chain $\hat{\mathcal{K}}^{(A,n)}$ induces the *uniform distribution* on the set of directed paths in the graph $\mathcal{G}_{n,A}$. As in Proposition 3.4, we know that the probability distribution on paths γ of length ℓ between workspaces F and F' is equal to

$$\frac{\psi(F) \eta(F')}{\lambda^\ell}.$$

What this means is that, for example, if we start a Merge derivation with given lexical material (the initial workspace F) and we obtain a resulting workspace F' (say, a fully formed sentence

$F' = T$ consisting of a single syntactic objects), all viable derivations of the same length (directed path of edges in the graph $\mathcal{G}_{n,A}$ from F to F') have the same probability in the Markov chain. This is a stronger Markovian property than just the usual observation that “Merge is Markovian”. It means that the history of the derivation does not matter and only the starting and final point and the length determine the likelihood of that derivation, but not in any way the specific history.

Indeed, the fact that such a stronger Markovian property should hold for Merge was already observed by Chomsky in [8], p.20, where he talks about “derivations strictly Markovian in a strong sense, beyond the normal Markovian property”. This property is described as the fact that in the process of structure formation in syntax the current state of the derivation does not contain the history of the derivation. We interpret this here as the property that the probability of a derivation (a path) is only dependent on its result and not on the path itself (the history).

We will discuss the effects of Minimal Yield and other optimality constraints in §11, but we want to first point out here that the uniform distribution on paths, that the dynamics $\hat{\mathcal{K}}^{(A,n)}$ satisfies, already ensures that such a stricter Markovian property holds, according to which the history of the derivation has no effect on the dynamics (except for its length, which is inevitable, since minimal length derivations ought to be favored over longer derivations, for optimality and for convergence).

There is another related optimization property involving the Shannon entropy, that will provide an important additional piece of information when evaluating the cost functions that are usually considered in linguistics (Minimal Search and Resource Restrictions) and their effect on the dynamics. We will show that it is only when one also incorporates an information optimizing property that the cost functions have the desired effect of dampening the Sideward Merge contributions to the dynamics and ensure the convergence of the structure formation process of syntax to the most connected workspaces (single trees) via External Merge. We will return to discuss this in §13.

The optimization of free energy instead of entropy that we discussed in this section will be useful when we discuss the Merge dynamics with the optimization of cost functions, where the entropy-maximizing Markov chain $\hat{\mathcal{K}}^{(A,n)}$ induces a free-energy minimizing Markov chain.

4. DECOMPOSING THE HOPF ALGEBRA MARKOV CHAIN

We proceed to analyze the dynamics of the Merge Hopf algebra Markov chain by separately analyzing different parts of the Merge graph and different Merge operations. In the linguistic model, External Merge is responsible for structure formation (including theta role assignments, to which we will return in §15), while Internal Merge is responsible for transformation via movement. Sideward Merge is invoked to justify some special linguistic phenomena (head-to-head movement, for example) as discussed in [26]. Its existence and necessity as part of the Minimalism is controversial. We will discuss in detail its role in the dynamics of Merge and what changes if only EM and IM are considered, §6.

The way we proceed in analyzing different parts of the dynamics is by separating out the set $V(\mathcal{G}_{n,A})$ of vertices F of the graph $\mathcal{G}_{n,A}$ into subsets $V_{\wp,n,A}$ labelled by a partition $\wp \in \mathcal{P}'(n)$ of the integer n , with $\mathcal{P}'(n)$ the subset of partitions as in (2.1). We then consider, for each $\wp \in \mathcal{P}'(n)$, the subgraphs $\mathcal{G}_{n,A,\wp}$ on the set of vertices $V_{\wp,n,A}$ with only the edges given by IM arrows. We study, separately, the dynamics within each graph $\mathcal{G}_{n,A,\wp}$. We also consider a graph $\mathcal{G}_{\mathcal{P}'(n)}$ that has as vertices the partitions $\wp \in \mathcal{P}'(n)$ and a directed edge $\wp \rightarrow \wp'$ whenever there are directed edges in $\mathcal{G}_{n,A}$ between some vertex in $V_{\wp,n,A}$ and some vertex in $V_{\wp',n,A}$. The dynamics on $\mathcal{G}_{\mathcal{P}'(n)}$ will provide the remaining information needed to understand how the dynamical systems on the graphs $\mathcal{G}_{n,A,\wp}$ are intertwined in the full Merge dynamics over the graph $\mathcal{G}_{n,A}$. We will see that

Proposition 4.1. *Let $\mathfrak{T}_{\mathcal{SO}_0, k}$ denote the set of nonplanar full binary rooted trees with k leaves decorated by elements of $\mathcal{SO}_0$. Let $\mathcal{P}(n)$ be the set of all partitions and $\mathcal{P}'(n) = \mathcal{P}(n) \setminus \{1, 1, \dots, 1\}$ as in (2.1). Consider the decomposition (2.2) of the set of vertices $V(\mathcal{G}_{n,A}) = \sqcup_{\wp \in \mathcal{P}'(n)} V_{\wp, n, A}$. Let $\mathcal{G}_{n,A,\wp}$ denote the subgraph of $\mathcal{G}_{n,A}$ with vertex set $V_{\wp, n, A}$ and only the edges consisting of IM arrows. The EM arrows map vertices of $\mathcal{G}_{n,A,\wp}$ to vertices of $\mathcal{G}_{n,A,\wp'}$ where $\wp'$ is a partition of the form $\wp' = \{k_i + k_j\} \cup \{k_\ell\}_{\ell \neq i, j}$, for some $i \neq j \in \{1, \dots, r\}$. If the partition $\wp$ contains a 2 and a 1, then there are some minimal SM arrows that map the set $V_{\wp, n, A}$ to itself. All other minimal SM arrows map vertices of $\mathcal{G}_{n,A,\wp}$ to vertices of $\mathcal{G}_{n,A,\wp'}$ where $\wp' \neq \wp$ is a partition of the form $\wp' = \{2, k_i - 1, k_j - 1\} \cup \{k_\ell\}_{\ell \neq i, j}$ or of the form $\wp' = \{2, k_i - 2\} \cup \{k_\ell\}_{\ell \neq i}$, for some $i \neq j \in \{1, \dots, r\}$ or $i \in \{1, \dots, r\}$, respectively. Thus, in all of these cases $\mathcal{G}_{n,A,\wp}$ is the induced subgraph on the set of vertices $V_{\wp, n, A}$.*

$$\begin{array}{c} \alpha_i \quad \alpha_j \\ \diagup \quad \diagdown \\ \alpha_i \quad \alpha_j \sqcup \alpha_k \mapsto \alpha_i \quad \alpha_k \sqcup \alpha_j \\ \\ \alpha_i \quad \alpha_j \\ \diagup \quad \diagdown \\ \alpha_i \quad \alpha_j \sqcup \alpha_k \mapsto \alpha_j \quad \alpha_k \sqcup \alpha_i. \end{array}$$

This decomposition in particular means that, if we want to study the Internal Merge dynamics, we can consider a single graph $\mathcal{G}_{n,A,\varphi}$ at a time.

5. THE INTERNAL MERGE DYNAMICS

We discuss here the long term behavior of the Internal Merge dynamics and convergence to a stationary distribution.

Definition 5.1. Let $\mathcal{G}_{n,A}^{\text{IM}}$ be the graph with vertex set $\mathfrak{F}_{A,n}$ and with edges the IM arrows (namely the graph $\mathcal{G}_{n,A}$ with the EM and SM arrows removed). We write $\mathcal{G}_{n,A,\varphi}^{\text{IM}}$ for the induced subgraph of $\mathcal{G}_{n,A}^{\text{IM}}$ with vertex set $V_{\varphi,n,A}$.

5.1. Connected components. We first consider the Internal Merge dynamics on workspaces consisting of a single tree.

Proposition 5.2. The graph $\mathcal{G}_{n,A,\{n\}}^{\text{IM}}$ with $\varphi = \{n\}$, with vertices the workspaces $F = T \in \mathfrak{F}_{A,n}$ consisting of a single tree, is connected.

Proof. The graph has $(2n-3)!!$ vertices corresponding to all the possible tree topologies of T (as a non-planar tree with labelled leaves). We need to show that all of them are in the same connected component. A full binary rooted tree T is a *comb tree* iff it contains only one cherry. In that case, if IM extracts any one of the leaves, it produces another comb tree with a permutation of the leaves labels, and in fact all permutations of the labels occur in this way. So all the comb trees are in the same connected component. Thus, to show that $\mathcal{G}_{n,A,\{n\}}^{\text{IM}}$ is connected, it suffices to show that, for any T with n -leaves labelled by the elements of A , that is not already a comb tree, there is a path of IM arrows, each extracting a single leaf, that connects T to some comb tree on n -leaves, with some labeling by the elements of A . We can do this inductively, starting with $n = 4$ (for $n = 3$ there are only comb trees), where one IM on one of the leaves of the non-comb topology produces a comb tree. Suppose that for $m < n$ leaves the statement holds. If T is not a comb tree then there is more than one cherry. Apply an IM that extracts one of the two leaves, call it α , in one of the cherries. The resulting tree has that leaf α attached directly to the root, with the vertex v that was the root of T as its sister vertex, which is now the root vertex of $T/d\alpha$. The tree $T/d\alpha$ has $n-1$ leaves, so there is a path of IM arrows extracting one leaf at a time that connect $T/d\alpha$ to a comb. But then performing the same path of IM operations on the same sequence of leaves, on the tree $\mathfrak{M}(\alpha, T/d\alpha)$ also produces a comb tree, because the comb sequence of leaves extracted and merged at the root continue a comb sequence already starting with α . $\square$

Note that this argument proves connectedness but not strong connectedness. In fact the graph $\mathcal{G}_{n,A,\{n\}}^{\text{IM}}$ is also strongly connected: this will follow from the more general argument given in Proposition 5.5.

Proposition 5.3. For a partition $\varphi \in \mathcal{P}'(n)$ of the form $\varphi = \{k_1, \dots, k_1, \dots, k_r, \dots, k_r\}$ with $n = a_1 k_1 + \dots + a_r k_r$, the graph $\mathcal{G}_{n,A,\varphi}^{\text{IM}}$ consists of separate connected components. The number of connected components is equal to the generalized multinomial coefficient of (2.8),

$$b_0(\mathcal{G}_{n,A,\varphi}^{\text{IM}}) = \Upsilon_{\varphi,n} = \frac{1}{a_1! \cdots a_r!} \binom{n}{\underbrace{k_1, \dots, k_1}_{a_1\text{-times}}, \dots, \underbrace{k_r, \dots, k_r}_{a_r\text{-times}}}.$$

We denote by C_φ the set of connected components of the graph $\mathcal{G}_{n,A,\varphi}^{\text{IM}}$ and we write $\mathcal{G}_{n,A,\varphi,\sigma}^{\text{IM}}$ for the component $\sigma \in C_\varphi$. The set C_φ can be obtained as a quotient of the conjugacy class Conj_φ in the symmetric group S_n of cycle type φ , with the relation

$$\Upsilon_{\varphi,n} \cdot \prod_{i=1}^r ((k_i - 1)!)^{a_i} = \#\text{Conj}_\varphi = \frac{n!}{\prod_{i=1}^r a_i! k_i^{a_i}}.$$

Each connected component $\mathcal{G}_{n,A,\varphi,\sigma}^{\text{IM}}$ contains a number of vertices equal to

$$\prod_{i=1}^r ((2k_i - 3)!!)^{a_i}.$$

Proof. Internal Merge acting on a workspace $F = T_1 \sqcup \dots \sqcup T_r$ acts separately on one connected component T_i of the workspace at a time, so that the action on workspaces is completely determined by the action on trees. In particular, the action preserves the splitting $L(F) = \sqcup_i L(T_i)$ by preserving each set $L(T_i) = L(\mathfrak{M}(T_v, T_i/T_v))$. This means that a forest $F \in V_{\varphi,n,A}$ will always be mapped by IM arrows to other forests in $V_{\varphi,n,A}$ with the same φ . The partition of the set of labels over the subsets $L(T_i)$ of leaves is also preserved by IM (unlike SM and EM): suppose a given $F \in V_{\varphi,n,A}$ for $\varphi = \{k_1, \dots, k_r\}$ has an assigned partition of the labels set $A = \{\alpha_1, \dots, \alpha_n\}$ into subsets of sizes k_i assigned to the sets of leaves $L(T_i)$ of the components T_i of F . Then Internal Merge will preserve this partition of the labels and cannot mix leaf labels assigned to different components of the workspace. Thus, these partitions of the set A are invariant along each directed path of Internal Merge arrows on the set $V_{\varphi,n,A}$. This means that each such orbit, determined by a label assignment, is a different connected component of the graph $\mathcal{G}_{n,A,\varphi}$.

If the partition φ is of the form $n = k_1 + \dots + k_r$ with all the $k_i \leq 2$, there are no Internal Merge arrows (we exclude the IM arrows that extract one of the accessible terms at the vertices immediately below the root since those are just the identity, so both the single leaves and the cherry trees do not have IM dynamics). So in this case each vertex of $\mathcal{G}_{n,A,\varphi}^{\text{IM}}$ is a connected component.

When the partition φ has at least one of the k_i satisfies $k_i \geq 3$, there are nontrivial IM arrows in $\mathcal{G}_{n,A,\varphi}^{\text{IM}}$. For any partition of the labels set A over the components of the workspaces, the $\prod_i (2k_i - 3)!!$ different tree topologies are in the same connected component, by Proposition 5.2.

Thus, the connected components of $\mathcal{G}_{n,A,\varphi}^{\text{IM}}$ are in bijective correspondence with the partitions of the set A of labels over components of the workspace, so that there are $\Upsilon_{\varphi,n}$ components.

Consider permutations $\sigma \in S_n$ that have the cycle structure determined by the partition φ , namely a number a_i of k_i -cycles. The cycle type φ determines a conjugacy class Conj_{φ} in the symmetric group S_n . When counting elements in the conjugacy class Conj_{φ} of cycle type φ , a factor equal to the multinomial coefficient $\mu_{\varphi,n}$ as in (2.4) counts the choices of elements that to go in each cycle. Each of these choices generates $k_i!/k_i = (k_i - 1)!$ assignments of labels counted modulo cyclic permutations. This counting is taken up to permutations of the a_i cycles of size k_i , resulting in $\#\text{Conj}_{\varphi} = \Upsilon_{\varphi,n} \cdot \prod_i ((k_i - 1)!)^{a_i} = \#C_{\varphi} \cdot \prod_i ((k_i - 1)!)^{a_i}$.

In each component $\mathcal{G}_{n,A,\varphi,\sigma}^{\text{IM}}$, one has a choice of all the possible tree topologies of the trees T_i in the workspace $F = T_1 \sqcup \dots \sqcup T_r$, which gives a product of $((2k_i - 3)!!)^{a_i}$ for each component T_i , which determines the number of vertices in each connected component. $\square$

5.2. Strong connectedness of components. We can then prove a stronger result about the Internal Merge dynamics on each connected component $\mathcal{G}_{n,A,\varphi,\sigma}^{\text{IM}}$, $\sigma \in C_{\varphi}$.

Proposition 5.4. *Let $\varphi \in \mathcal{P}'(n)$ be a partition $\varphi = \{k_1, \dots, k_r\}$ with at least one of the $k_i \geq 3$, so that there are non-trivial IM arrows. Then each connected component $\mathcal{G}_{n,A,\varphi,\sigma}^{\text{IM}}$ of the graph $\mathcal{G}_{n,A,\varphi}^{\text{IM}}$ is strongly connected and aperiodic, hence the Internal Merge Hopf algebra Markov chain $\hat{\mathcal{K}}^{(A,n,\varphi,\sigma)}$ is ergodic on each connected component.*

Proof. As we mentioned in the case of Proposition 3.3, ergodicity follows from strong connectedness and aperiodicity. To show that the Hopf algebra Markov chain $\hat{\mathcal{K}}^{(A,n,\varphi,\sigma)}$ of Internal Merge is ergodic we need a refinement of Proposition 2.5 to show that not only the graphs $\mathcal{G}_{n,A}$ are strongly

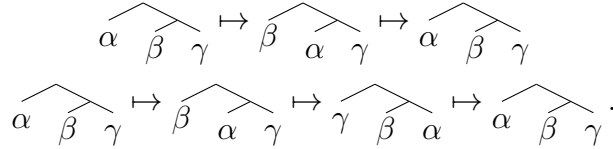
connected and aperiodic but also each connected component $\mathcal{G}_{n,A,\wp,\sigma}^{\text{IM}}$ of the subgraphs $\mathcal{G}_{n,A,\wp}^{\text{IM}}$ has the same properties.

For a given component $\mathcal{G}_{n,A,\wp,\sigma}^{\text{IM}}$, we look at the restriction $\hat{\mathcal{K}}^{(A,n,\wp,\sigma)}$ of the dynamics of $\hat{\mathcal{K}}^{(A,n,\wp)}$ to that component. By Proposition 2.3 we know that there are, at each $F \in V_{\wp,n,A,\sigma}$ as many outgoing as incoming IM edges,

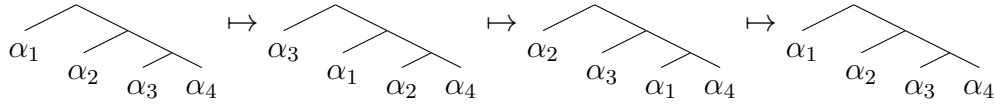
$$(5.1) \quad N_{\text{IM}}^{\text{out}}(F) = \sum_{i=1}^{c(F)} (2k_i - 4) = N_{\text{IM}}^{\text{in}}(F),$$

where $F = T_1 \sqcup \dots \sqcup T_r$ with $\wp$ the partition $n = k_1 + \dots + k_r$ for $k_i = \#L(T_i)$, and where we assume for simplicity of notation that $T_1, \dots, T_{c(F)}$ are the components with nonempty set of edges, with $c(F) \leq r$. The numbers k_i as well as $c(F) = \#\{i \mid k_i > 1\}$ are determined by the partition $\wp$ so they are the same for all vertices $F \in V_{\wp,n,A}$ for fixed $\wp$.

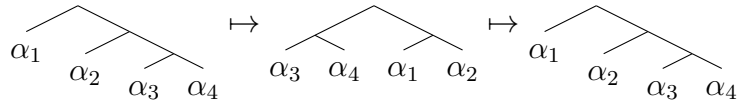
The strong connectedness property then follows from the fact that each vertex of $\mathcal{G}_{n,A,\wp,\sigma}$ has $\deg^{\text{out}}(F) = \deg^{\text{in}}(F)$, which implies that the directed graph $\mathcal{G}_{n,A,\wp,\sigma}$ is Eulerian, namely there is a directed cycle that visits each edge of $\mathcal{G}_{n,A,\wp,\sigma}$ exactly once. This then implies strong connectedness as, starting at any vertex F one can reach any other vertex F' along the Eulerian circuit. We can see that $\mathcal{G}_{n,A,\wp,\sigma}$ must also be aperiodic by refining the argument given in Proposition 2.5 that identifies explicit cycles of Internal Merge operations. It suffices to consider the case of $\wp = \{n\}$ namely the workspaces $F = T$ consisting of a single connected component $T \in \mathfrak{T}_{\mathcal{SO}_0,n}$. The graph $\mathcal{G}_{n,A,\wp=\{n\}}$ has a single connected component. If we show that these graphs are aperiodic, then all the other graphs $\mathcal{G}_{n,A,\wp,\sigma}$ will also be aperiodic because for every $k_i \geq 3$ in the partition $\wp = \{k_1, \dots, k_r\}$ they will contain copies of the cycles we exhibit in $\mathcal{G}_{n,A,\{k_i\}}$. We consider separately the cases $n = 3$ and $n = 4$ leaves and the general case $n \geq 5$ leaves. We show that in all of these cases we always have an IM cycle of order 2 and an IM cycle of order 3 so that the gcd of the cycle-lengths is equal to 1. For $n = 3$, by extracting one of the other leaf of the cherry, we obtain cycles



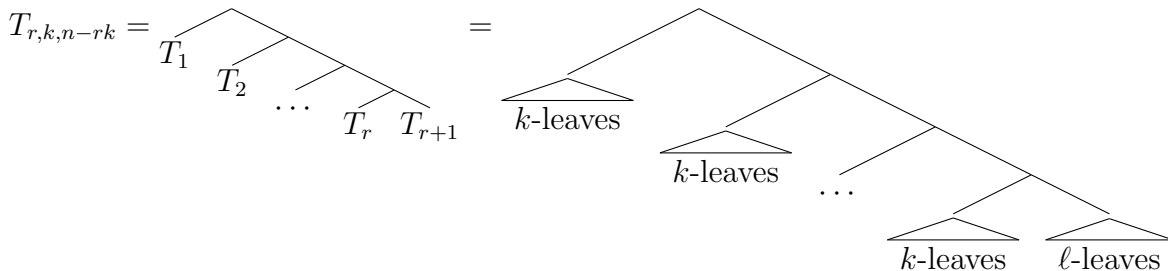
In the case with $n = 4$ we similarly have a cycle of length 3



and also a cycle of length 2 (by first extracting the $\{\alpha_3, \alpha_4\}$ cherry and then the single leaf α_1)



Thus $\mathcal{G}_{2,A\{2\}}$ and $\mathcal{G}_{3,A\{3\}}$ are aperiodic. Now suppose that $n \geq 5$. Consider trees of the form



with subtrees T_i for $i = 1, \dots, r$ with k -leaves with identical tree topologies, and an additional subtree T_{r+1} with ℓ leaves, for $n = rk + \ell$ and $\ell \geq 1$, $r \geq 2$ and $k \geq 1$. The trees considered in Proposition 2.5 are the special case $T_{n-1,1,1}$. Then the iteration of IM operations that extract the accessible terms T_r then T_{r-1} and so on until T_1 gives a cycle of length r . For $n \geq 5$ these cycles always include a cycle of length $r = 2$ and a cycle of length 3 so again the gcd of the cycle lengths is equal to one, so aperiodicity always holds. $\square$

5.3. Uniform distribution. The same property (5.1) of the in/out-degrees of the Internal Merge dynamics also give us the form of the stationary distribution on the $\mathcal{G}_{n,A,\varphi,\sigma}^{\text{IM}}$ graphs.

Proposition 5.5. *For partitions with at least one $k_i \geq 3$, the connected component $\mathcal{G}_{n,A,\varphi,\sigma}^{\text{IM}}$ of the graph $\mathcal{G}_{n,A,\varphi}^{\text{IM}}$ are strongly connected and for any $\sigma \in C_\varphi$, the part $\hat{\mathcal{K}}^{(A,n,\varphi,\sigma)}$ of the Hopf algebra Markov chain given by the action of Internal Merge on $\mathcal{G}_{n,A,\varphi,\sigma}^{\text{IM}}$ has the uniform distribution as stationary distribution. When all $k_i \leq 2$ there are no IM arrows and each vertex is a component.*

Proof. As shown in Propositions 2.3 and 5.4, at each vertex $F \in V_{\varphi,n,A,\sigma}$ there are as many outgoing as incoming IM edges, as in (5.1), $\deg^{\text{out}}(F) = \deg^{\text{in}}(F)$. Note that the value in (5.1) is constant for all vertices $F \in V_{\varphi,n,A}$, and depends only on the partition φ ,

$$N_{\text{IM}}^{\text{out}}(F) = N_{\text{IM}}^{\text{in}}(F) =: d_\varphi.$$

Let $\mathcal{K}^{(A,n,\varphi,\sigma)} = (\mathcal{K}_{F,F'}^{(A,n,\varphi,\sigma)})$ denote the adjacency matrix of the graph $\mathcal{G}_{n,A,\varphi,\sigma}^{\text{IM}}$.

For a directed graph where all the vertices have the same in and out degree, all of them equal to d_φ , the Perron-Frobenius theorem, applied to the adjacency matrix, shows that the Perron-Frobenius eigenvalue satisfies

$$\min_F \sum_{F'} \mathcal{K}_{F,F'}^{(A,n,\varphi,\sigma)} \leq \lambda \leq \max_F \sum_{F'} \mathcal{K}_{F,F'}^{(A,n,\varphi,\sigma)},$$

where $\sum_{F'} \mathcal{K}_{F,F'}^{(A,n,\varphi,\sigma)} = \deg^{\text{out}}(F)$, so if $\deg^{\text{out}}(F) = d_\varphi$ for all $F \in V_{\varphi,n,A,\sigma}$, we have that $\lambda = d_\varphi$ is the Perron-Frobenius eigenvalue and correspondingly the uniform distribution $\eta(F) = 1$ is the Perron-Frobenius eigenvalue

$$\sum_{F'} \mathcal{K}_{F,F'}^{(A,n,\varphi,\sigma)} \eta(F') = d_\varphi \eta(F).$$

Thus, the part of the Hopf algebra Markov chain given by Internal Merge simply has transition matrix

$$\hat{\mathcal{K}}_{F,F'}^{(A,n,\varphi,\sigma)} = d_\varphi^{-1} \cdot \mathcal{K}_{F,F'}^{(A,n,\varphi)},$$

with no rescaling of the basis, since $\eta(F) = 1$ for all $F \in V_{\varphi,n,A}$. This Markov chain has stationary distribution

$$\sum_F \pi(F) \hat{\mathcal{K}}_{F,F'}^{(A,n,\varphi,\sigma)} = \pi(F')$$

given by the uniform distribution $\pi(F) = 1$ (normalized to $\pi(F) = 1/\#V_{n,A,\varphi}$), since

$$d_\varphi^{-1} \sum_F \mathcal{K}_{F,F'}^{(A,n,\varphi,\sigma)} = \frac{\deg^-(F')}{d_\varphi} = 1,$$

since $\deg^+(F) = \deg^-(F) = d_\varphi$ for all $F \in V_{\varphi,n,A,\sigma}$. $\square$

One can rephrase Propositions 5.5 and 5.4 in the following way.

Corollary 5.6. *The uniform distribution is the unique stationary distribution of the Internal Merge Hopf algebra Markov chains $\hat{\mathcal{K}}^{(A,n,\varphi,\sigma)}$, on each connected component $\mathcal{G}_{n,A,\varphi,\sigma}$ of $\mathcal{G}_{n,A,\varphi}$, and any initial distribution on the set of workspaces $V_{\varphi,n,A,\sigma}$ converges under iterations of $\hat{\mathcal{K}}^{(A,n,\varphi,\sigma)}$ to the uniform distribution.*

This shows that Internal Merge by itself is an ergodic dynamical system with uniform stationary distribution. This fact singles out Internal Merge as having especially nice dynamical properties.

6. INTERNAL AND EXTERNAL MERGE DYNAMICS

In this and the next section we outline the opposite roles that the minimal Sideward Merge and the External Merge arrows plays in the dynamics by showing what happens if one either removes all the SM arrows, while keeping IM and EM arrows or (in the next section) if one removes the EM arrows and keeps IM and SM arrows. In both cases the strong connectedness property breaks down and the vertices of the graph split into transient and recurrent states, with the stationary distribution supported only on the recurrent states. The nature of the recurrent states is complementary in the two cases: predictably, the dynamics moves to the most connected structure when no SM is present and to the least connected ones when no EM is present.

The dynamical system involving only External and Internal Merge is the one that is usually considered in the linguistics literature, where it is argued (see [7], [8], [9]) that the Sideward Merge part of the dynamics is excluded on the basis of optimization with respect to certain cost functions, that we will discuss in §11. However, from the mathematical perspective, as well as in comparison with similar settings in physics models, one can argue (see [26]) that cost optimization is not a hard structural constraint that eliminates SM, but rather a soft constraint that makes it more unlikely, or contributing less significantly, the more it deviates from optimality. Using this viewpoint, it was shown in [26] that certain linguistic phenomena can be explained by a minimally optimality violating form of SM, which is exactly the minimal SM we are considering in the graphs $\mathcal{G}_{n,A}$ here.

In this section, we present the Merge dynamics in the way it would be according to the original proposal of [7], [8], [9], where the SM arrows of the graph $\mathcal{G}_{n,A}$ are completely eliminated. We show that, while in this case the dynamics does indeed converge through EM to the connected structures (workspaces consisting of a single tree) with the remaining action of IM on trees as the only recurrent part of the dynamics, the strong connectedness property is broken, eliminating the desirable dynamical property or ergodicity and the independence of the dynamics on the initial position, as well as the entropy optimizing property we discussed earlier that also relies on strong connectedness. We will return in §11 and §13 to discuss how to use the type of cost functions proposed in [7], [8], [9] and in [22] and the entropy optimization property to correct the dynamics by incorporating cost-weights that suppress the undesirable effects of Sideward Merge without losing the desirable strong connectedness and ergodicity and entropy optimization properties of the dynamics.

We consider here the subgraph $\mathcal{G}_{n,A}^{\text{IM-EM}} \subset \mathcal{G}_{n,A}$ with the same set of vertices $V(\mathcal{G}_{n,A}^{\text{IM-EM}}) = V(\mathcal{G}_{n,A}) = \mathfrak{F}_{A,n}$, that only includes as edges the IM and the EM arrows but not the SM arrows of $\mathcal{G}_{n,A}$. It is then clear that the strong connectedness property fails for this subgraph $\mathcal{G}_{n,A}^{\text{IM-EM}}$. We refer to the resulting Markov chain as $\hat{\mathcal{K}}^{\text{IM-EM},(A,n)}$.

6.1. Reducibility and transient states. A transient state in a Markov chain is a state that has a nonzero probability of never being returned to.

Lemma 6.1. *Every vertex $F \in V(\mathcal{G}_{n,A}^{\text{IM-EM}})$ given by a workspace with more than one connected component (that is, with image $\wp = p(F) \in \mathcal{P}'(n)$ with $\wp \neq \{n\}$) is a transient state of $\hat{\mathcal{K}}^{\text{IM-EM},(A,n)}$.*

Proof. Since EM always decreases the number of connected components of the workspace by one, it is clear that a state $F = T_1 \sqcup \dots \sqcup T_r \in \mathfrak{F}_{A,n}$ with $r > 1$, namely a vertex in $\mathcal{G}_{n,A,\wp=\{k_1,\dots,k_r\}}^{\text{IM-EM}}$ has a non-zero probability of leaving the graph $\mathcal{G}_{n,A,\wp=\{k_1,\dots,k_r\}}^{\text{IM-EM}}$ via one of the EM arrows and entering one of the graphs with $\mathcal{G}_{n,A,\wp'=\{k_i+k_j,k_1,\dots,\hat{k}_i,\dots,\hat{k}_j,\dots,k_r\}}^{\text{IM-EM}}$, where the notation $\hat{k}_i$ means that the entry k_i is removed. Since without the SM arrows every arrow in the graph $\mathcal{G}_{n,A}^{\text{IM-EM}}$ either maintains the same number of connected components of the workspace (IM arrows) or decreases it by one (EM arrows), there is no way to return from any vertex of $\mathcal{G}_{n,A,\wp'=\{k_i+k_j,k_1,\dots,\hat{k}_i,\dots,\hat{k}_j,\dots,k_r\}}^{\text{IM-EM}}$ to a vertex of $\mathcal{G}_{n,A,\wp=\{k_1,\dots,k_r\}}^{\text{IM-EM}}$, hence the vertices of $\mathcal{G}_{n,A,\wp=\{k_1,\dots,k_r\}}^{\text{IM-EM}}$ are transient. In the case of vertices of $\mathcal{G}_{n,A,\wp=\{n\}}^{\text{IM-EM}}$ there are no outgoing EM arrows so the vertex of this subgraph remain (via IM arrows) within the same $\mathcal{G}_{n,A,\wp=\{n\}}^{\text{IM-EM}}$ and these are the only recurrent states of the Markov chain $\hat{\mathcal{K}}^{\text{IM-EM},(A,n)}$. $\square$

Corollary 6.2. *The Hopf algebra Markov chain $\hat{\mathcal{K}}^{\text{IM-EM},(A,n)}$ is reducible. Equivalently, the graph $\mathcal{G}_{n,A}^{\text{IM-EM}}$ is not strongly connected.*

Proof. A finite irreducible Markov chain cannot have any transient state, hence Lemma 6.1 implies that $\hat{\mathcal{K}}^{\text{IM-EM},(A,n)}$ is reducible. $\square$

6.2. Communication classes. A reducible Markov chain can be decomposed into *communication classes*: this is the same as the decomposition of the underlying directed graph into *strongly connected components*.

Given a directed graph G , one says that two vertices $v, v' \in V(G)$ are strongly connected, or communicating, if either $v = v'$ or if $v \neq v'$ and there is a directed path of edges from v to v' and a directed path of edges from v' to v . Strong connectivity so defined is an equivalence relation on $V(G)$ and the equivalence classes are the strongly connected components of G (or communication classes).

The *condensation graph* $\bar{G}$ is the directed graph obtained from G by contracting each strongly connected component of G to a single vertex. The graph $\bar{G}$ is an *acyclic directed graph*. In general $\bar{G}$ so obtained is a multigraph, with possibly multiple parallel edges between the same pair of vertices. We can further identify any set of parallel edges in $\bar{G}$ to a weighted single edge.

A *closed communication class* is a strongly connected component with no arrows pointing to other strongly connected components, namely such that its strong connectedness equivalence class is a sink of the condensation directed acyclic graph.

Consider again the graphs $\mathcal{G}_{n,A,\wp,\sigma}^{\text{IM}}$, with $\sigma \in C_\wp$, that are the connected components of the IM graph $\mathcal{G}_{n,A}^{\text{IM}}$.

The following description of communication classes for $\hat{\mathcal{K}}^{\text{IM-EM},(A,n)}$ follows directly from the description of the IM dynamics on the graphs $\mathcal{G}_{n,A,\wp,\sigma}^{\text{IM}}$ and the EM arrows.

Proposition 6.3. *The graphs $\mathcal{G}_{n,A,\wp,\sigma}^{\text{IM}}$, for $\wp \in \mathcal{P}'(n)$ and $\sigma \in C_\wp$, are the strongly connected components of the graph $\mathcal{G}_{n,A}^{\text{IM-EM}}$ (communication classes of the Markov chain $\hat{\mathcal{K}}^{\text{IM-EM},(A,n)}$). The condensation graph $\bar{\mathcal{G}}_{n,A}^{\text{IM-EM}}$ is the graph $\tilde{\mathcal{G}}_{\mathcal{P}'(n)}^{\text{EM}}$ with set of vertices*

$$V(\tilde{\mathcal{G}}_{\mathcal{P}'(n)}^{\text{EM}}) = \{(\wp, \sigma) \mid \wp \in \mathcal{P}'(n), \sigma \in C_\wp\}$$

and one directed edge from $(\wp, \sigma)$ to $(\wp', \sigma')$ for

$$\wp = \{k_1, \dots, k_r\} \quad \text{and} \quad \wp' = \{k_i + k_j, k_1, \dots, \hat{k}_i, \dots, \hat{k}_j, \dots, k_r\}$$

(with the notation as in Lemma 6.1) for some $k_i, k_j \in \wp$, and with σ' the image of σ under the induced map $C_\wp \rightarrow C_{\wp'}$. The graph $\mathcal{G}_{n,A,\wp}^{\text{IM}}$ with $\wp = \{n\}$, where vertices are workspaces consisting of a single tree with n leaves, is the only closed communication class.

The induced map $C_\wp \rightarrow C_{\wp'}$ is simply determined by the fact that each assignment of labels of A to the boxes of $D(\wp)$ determines a corresponding assignment to the boxes of $D(\wp')$.

6.3. Stationary distribution on the closed communication class. This implies that, when we do not consider any Sideward Merge as part of the dynamics, the limiting distribution for $\hat{\mathcal{K}}^{\text{IM-EM},(A,n)}$ is only supported on the connected structures. This describes Merge as a process of structure formation where the dynamics evolves via External Merge toward connected structures and away from workspaces with multiple connected components, and once connected structures are reached the only dynamics left consists of the transformations via movement implemented by Internal Merge. This indeed agrees with the usual description of the Merge dynamics given by Chomsky in [7], [8], [9], in terms of just EM and IM.

Proposition 6.4. *The stationary distribution for the reducible Markov chain $\hat{\mathcal{K}}^{\text{IM-EM},(A,n)}$ is the uniform distribution supported on the vertices of the graph $\mathcal{G}_{n,A,\wp=\{n\}}^{\text{IM}}$ (workspaces consisting of a single tree with n leaves).*

Proof. If a stationary distribution π exists, it has the property that

$$(6.1) \quad \pi(F) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N (\hat{\mathcal{K}}^{\text{IM-EM},(A,n)})_{F,F'}^N,$$

for any initial state F' in the same communication class. In the reducible case, $\pi(F) = 0$ whenever F is a transient state of $\hat{\mathcal{K}}^{\text{IM-EM},(A,n)}$. Thus, π can only be supported on the recurrent states, that in this case are the vertices of the subgraph $\mathcal{G}_{n,A,\wp=\{n\}}^{\text{IM}}$. Indeed, the stationary distribution for a reducible Markov chain is always supported on the closed communication classes. In this case there is only one such class, consisting of the vertices of $\mathcal{G}_{n,A,\wp=\{n\}}^{\text{IM}}$. Moreover, on those vertices, π should agree with the stationary distribution of the Internal Merge dynamics on the graph $\mathcal{G}_{n,A,\wp=\{n\}}^{\text{IM}}$ because choosing both F, F' in (6.1) to be vertices in $\mathcal{G}_{n,A,\wp=\{n\}}^{\text{IM}}$ leads to all the nontrivial entries in the $(\hat{\mathcal{K}}^{\text{IM-EM},(A,n)})_{F,F'}^N$ being IM arrows so the limit has to be the stationary distribution of the IM dynamics on $\mathcal{G}_{n,A,\wp=\{n\}}^{\text{IM}}$, which is the uniform distribution. $\square$

6.4. The IM–EM structure formation model: linguistic aspects. The dynamics described in this section and summarized in the result of Proposition 6.4 gives, from the linguistic perspective, a very satisfactory view of structure formation in syntax, obtained via free symmetric Merge, which shows convergence to the connected structures (fully formed sentences) and Internal Merge dynamics acting on the connected structures in an ergodic way, without privileging any particular subset of available structures (uniform stationary distribution). It indeed reflects what was described in Chomsky's model of [7] and [8], where Sideward Merge is excluded on the ground of its failure of optimality.

However, as we pointed out at the beginning of this section, this has the problem that elimination of the Sideward Merge arrows looks artificial, in the sense that these Merge operations are clearly part of the natural Hopf algebra Markov chain definition of Merge, as compositions

$$\mathfrak{M}_{S,S'} = \sqcup \circ (\mathcal{B} \otimes \text{id}) \circ \delta_{S,S'} \circ \Delta,$$

where non-primitive terms of the coproduct are used, so there is no *algebraic* argument for their elimination, and cost counting only makes these Merge operations subdominant with respect to EM and IM, but still present.

As discussed in [22], one could replace the coproduct Δ with its primitive part Δ^P , that only forms partitions of workspaces without extracting accessible terms, and then this same formula would provide only External Merge without Sideward Merge. On the other hand this would not be consistent with maintaining the extraction of accessible terms in Internal Merge and still being able to see Internal Merge and External Merge as cases of the same operation (with an $\mathfrak{M}_{S,1}$ operation involved in IM). Thus, just removing the SM arrows from the Hopf algebra Markov chain $\hat{\mathcal{K}}$ of Merge is not directly justified by the underlying algebraic structure.

In Chomsky's formulation of Minimalism [7], [8], the selection of External Merge and Internal Merge over Sideward Merge is formulated as optimality (minimization of some natural cost function), where EM and IM are optimal while SM is not. There are different types of optimality that are discussed, in this context, in the linguistics literature on Merge: Minimal Search and forms of Resource Restriction like Minimal Yield. These are analyzed in terms of the mathematical model in [22]. A detailed discussion of these different cost functions and Sideward Merge is also further expanded in [26], where it is shown more precisely what roles Minimal Search and Resource Restrictions have in constraining possible forms of Sideward Merge.

The way to deal with different weights by cost functions in our mathematical setting is to replace the adjacency matrix of the Merge graph $\mathcal{G}_{n,A}$ with a weighted version, where the weights incorporate the cost functions, in a way similar to the case we analyzed in §3.4, and then argue that this weighted form of the Merge dynamics better approximates the EM–IM dynamics described in this section, with the effects of SM dampened by the cost function. While this idea seems natural and is easily confirmed in the simple example of $n = 3$ leaves computed in [26], the general case of arbitrary n is more subtle. We will return to discuss cost functions in §11 and the effects on the Merge dynamics in §11 and §13.

7. INTERNAL AND SIDEWARD MERGE DYNAMICS

We now consider, for comparison, what would happen if we only had IM and SM arrows, without EM. This is of course a purely hypothetical scenario that does not correspond to any viable linguistic model, since it clearly would not lead to structure formation in any way. It is useful, however, to discuss this briefly, in preparation for the direct comparison of EM and SM that we introduce in the following section.

The first observation we can make regarding the Sideward Merge part of the Merge dynamical system is that it plays a role similar to External Merge but in the “opposite direction” namely moving the dynamics toward more disconnected structures. Thus, if we keep the Internal Merge part of the dynamics unchanged, and we consider only the SM arrows without the EM arrows, we obtain an overall picture similar to what we have seen with IM and EM, but in the reverse direction. We assume, as in the full Merge dynamics $\hat{\mathcal{K}}$ introduced in §2, that only the minimal Sideward Merge arrows are included (those that only extract two single leaves).

Proposition 7.1. *Let $\mathcal{G}_{n,A}^{\text{IM-SM}} \subset \mathcal{G}_{n,A}$ be the subgraph on the same set of vertices, with edges given only by the IM and the minimal SM arrows (without any EM arrows). The associated Markov chain $\hat{\mathcal{K}}^{\text{IM-SM},(A,n)}$ is reducible. The strongly connected components (communication classes) are given by the following cases.*

- For partitions $\wp \in \mathcal{P}'(n)$ containing no pair $k_i = 2, k_j = 1$, the strongly connected components are graphs $\mathcal{G}_{n,A,\wp,\sigma}^{\text{IM}}$, with $\sigma \in C_\wp$;

- For partitions $\varphi \in \mathcal{P}'(n)$ containing some $\{2, 1\}$ pairs, the strongly connected components are graphs $\mathcal{G}_{n,A,\varphi,\tilde{\sigma}}^{\text{IM-SM}}$ with $\tilde{\sigma} \in C_{\tilde{\varphi}}$ for $\tilde{\varphi}$ the partition (in some $\mathcal{P}'(m)$ with $m < n$) obtained from φ by removing all the $\{2, 1\}$ pairs.

The only closed communication class is given by the induced subgraph on $\mathcal{G}_{n,A,\varphi=\{2,1,\dots,1\}}^{\text{IM-SM}}$. This graph has no IM arrows and only minimal SM arrows. The stationary distribution is the uniform distribution supported on the vertices of the subgraph $\mathcal{G}_{n,A,\varphi=\{2,1,\dots,1\}}^{\text{IM-SM}}$. The condensation graph $\bar{\mathcal{G}}_{n,A}^{\text{IM-SM}}$ is the graph $\tilde{\mathcal{G}}_{\mathcal{P}'(n)}^{\text{SM}}$ with vertex set

$$V(\tilde{\mathcal{G}}_{\mathcal{P}'(n)}^{\text{SM}}) = \{(\varphi, \sigma) \mid \varphi \in \mathcal{P}'_{2,1}(n)^c, \sigma \in C_{\varphi}\} \cup \{(\varphi, \sigma) \mid \varphi \in \mathcal{P}'_{2,1}(n), \sigma \in C_{\tilde{\varphi}}\}$$

where $\mathcal{P}'_{2,1}(n)$ is the set of partitions with $\{2, 1\}$ pairs and $\mathcal{P}'_{2,1}(n)^c$ its complement, and SM arrows $(\varphi, \sigma) \rightarrow (\varphi', \sigma')$ for $\varphi = \{k_1, \dots, k_r\}$ and $\varphi' = \{2, k_1, \dots, k_i - 2, \dots, k_r\}$ or $\varphi' = \{2, k_1, \dots, k_i - 1, \dots, k_j - 1, \dots, k_r\}$ with σ' the image of σ under the induced map $C_{\varphi} \rightarrow C_{\varphi'}$. This condensation graph has source vertices the pairs (φ, σ) with partitions $\varphi \in \mathcal{P}'(n)$ where all $k_i \neq 2$.

Proof. Any workspace $F \in \mathcal{G}_{n,A,\varphi}^{\text{IM-SM}}$ with $\varphi \neq \{2, 1, \dots, 1\}$ is a transient state. Indeed, any workspace F that contains either more than one cherry component or at least one component with $k_i \geq 3$ leaves has at least one minimal SM arrow with target F' in a different $\mathcal{G}_{n,A,\varphi'}^{\text{IM-SM}}$. In the case with some $k_i \geq 3$, there are some SM arrows where the partition φ' replaces a k_i with a pair $\{2, k_i - 2\}$, while in the case with more than one cherry, φ' replaces a pair $\{k_i = 2, k_j = 2\}$ with a triple $\{2, 1, 1\}$, increasing the number of connected components, and there are no SM (nor IM) arrows that can go in the reverse direction, decreasing the number of components. On the other hand in the induced subgraph $\mathcal{G}_{n,A,\varphi=\{2,1,\dots,1\}}^{\text{IM-SM}}$ there are no IM arrows and the only arrows in $\mathcal{G}_{n,A}^{\text{IM-SM}}$ that start at a vertex of $\mathcal{G}_{n,A,\varphi=\{2,1,\dots,1\}}^{\text{IM-SM}}$ are SM arrows of the form

$$(7.1) \quad \alpha_i \frown \alpha_j \sqcup \alpha_k \bigsqcup_{\ell \neq i,j,k} \alpha_{\ell} \mapsto \alpha_i \frown \alpha_k \sqcup \alpha_j \bigsqcup_{\ell \neq i,j,k} \alpha_{\ell}.$$

These arrows preserve $\mathcal{G}_{n,A,\varphi=\{2,1,\dots,1\}}^{\text{IM-SM}}$ so this subgraph consists of recurrent states.

For φ with no 2's and 1's among the k_i 's (these have $k_i \geq 3$) the subgraph $\mathcal{G}_{n,A,\varphi,\sigma}^{\text{IM-SM}}$ of $\mathcal{G}_{n,A,\varphi}^{\text{IM-SM}}$ on the same vertices as $\mathcal{G}_{n,A,\varphi,\sigma}^{\text{IM}}$ have no minimal SM arrows mapping the vertex set to itself, so these graphs agree with the IM graphs $\mathcal{G}_{n,A,\varphi,\sigma}^{\text{IM}}$ that are strongly connected and aperiodic.

When the partition contains some 2's and some 1's, on the other hand, $\mathcal{G}_{n,A,\varphi}^{\text{IM-SM}}$ contains minimal SM edges that map a subgraph $\mathcal{G}_{n,A,\varphi,\sigma}^{\text{IM-SM}}$ to some other $\mathcal{G}_{n,A,\varphi,\sigma'}^{\text{IM-SM}}$ through permuting some of the leaf labels as in (7.1). Thus, some different $\mathcal{G}_{n,A,\varphi,\sigma}^{\text{IM-SM}}$ become part of the same strongly connected component in this case, and the resulting strong components depend only on the assignments of labels on the rows of the diagram $D(\tilde{\varphi})$. In fact all the labels in cherries and the single leaves components of these workspaces can be permuted by repeated application of arrows like (7.1). The condensation graph $\bar{\mathcal{G}}_{n,A}^{\text{IM-SM}}$ has vertices the pairs (φ, σ) that list the strongly connected components of $\mathcal{G}_{n,A}^{\text{IM-SM}}$. Among these, the ones that are sources are those that have no incoming SM arrows, hence where the workspace has no cherry component. $\square$

As we observed above, for a partition φ that contains at least a pair $k_i = 2, k_j = 1$, there is, for each such pair, an SM transformation mapping $\mathcal{G}_{n,A,\varphi}$ to itself, that acts on the components $T_i = \mathfrak{M}(\alpha_a, \alpha_b)$ and $T_j = \alpha_c$ as

$$\alpha_a \frown \alpha_b \sqcup \alpha_c \mapsto \alpha_a \frown \alpha_c \sqcup \alpha_b$$

and as the identity on the other components of the workspace $F = T_1 \sqcup \dots \sqcup T_r$. These are the only kinds of SM arrows that preserve φ . We denote them here as $\text{SM}_{2,1}$. For any $F \in V(\mathcal{G}_{n,A,\varphi})$

we have

$$(7.2) \quad N_{\text{SM}_{2,1}}^{\text{out}}(F) = N_{\text{SM}_{2,1}}^{\text{in}}(F) = \#\{\text{pairs } \{2, 1\} \text{ in } \wp\}.$$

8. EXTERNAL AND SIDEWARD MERGE DYNAMICS VIA YOUNG DIAGRAMS

Our analysis so far has shown that, in order to understand the properties of the full Merge dynamics on $\mathcal{G}_{n,A}$, which includes all the EM, IM, and minimal SM arrows, we need to understand how the SM and EM arrows interact with each other and drive the behavior of the dynamics. As before, we restrict to only the minimal SM arrows, since these are enough SM arrows to make the Hopf algebra Markov chain ergodic and the results of [26] also show that these suffice to account for any linguistic phenomena for which SM may be a preferable explanation.

As we have previously discussed in §2.2 (and also in [26]), one can also choose to only allow those minimal SM arrows that do not cut both edges of a cherry. This choice is natural because it gives somewhat better algebraic properties for the deletion coproduct Δ^d (as will be discussed elsewhere). On the other hand, we will see that including these SM arrows makes the dynamical properties of the Merge Hopf algebra Markov chain nicer. Thus, given that this is the main focus of our investigation, we will always make here the assumption that we include among the minimal SM arrows also those that extract the two leaves of a cherry. We can limit these to the case where the cherry is a subtree of a larger tree, as when the cherry is a component, the resulting SM would be just the identity so we can ignore it.

Since we already know the behavior of the IM dynamics, the first step in further investigating the properties of the Hopf algebra Markov chain $\hat{\mathcal{K}}$ with IM, EM, and minimal SM arrows is to study graphs $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ with only EM and minimal SM arrows. Note that the graph $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ (unlike the $\tilde{\mathcal{G}}_{\mathcal{P}'(n)}^{\text{EM}}$ and $\tilde{\mathcal{G}}_{\mathcal{P}'(n)}^{\text{SM}}$ discussed above) is no longer the condensation graph of $\mathcal{G}_{n,A}$ because $\mathcal{G}_{n,A}$ is itself strongly connected so it has only one strongly connected component and no condensation graph.

We analyze the EM-SM dynamics in steps. First we introduce the graph $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ that has vertex set $\mathcal{P}'(n)$ and arrows that describe the effect of EM and minimal SM on partitions of leaves in the workspace. This graph does not include the tree topologies and does not account for assignments of labels in $A = \{\alpha_1, \dots, \alpha_n\}$ at the leaves.

As a second step, we introduce a way to keep track of the labels α_i at the leaves, by replacing the Young diagrams describing the partitions in $\mathcal{P}'(n)$ with Young tableaux on the alphabet A . Finally, we discuss how to account for the different tree topologies via appropriate multiplicities.

We will discuss in §9 how the full Merge dynamics on $\mathcal{G}_{n,A}$ relates to the dynamics on $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ that we discuss in this section, through an intermediate graph $\mathcal{G}_{\mathcal{P}'(n)}$ that also incorporates the IM part of the dynamics.

8.1. The Young diagrams graph. Young diagrams are usually organized as a lattice either with respect to the dominance order on $\mathcal{P}(n)$ or by inclusion of diagrams (also called covering relation) on $\mathcal{P} = \sqcup_n \mathcal{P}(n)$ (Young lattice). These order relations and the resulting graphs can be interpreted as discrete dynamical systems, which have applications to models like the Sand Pile Model (a dynamical system exhibiting self-organized criticality).

We consider here a different kind of discrete dynamical system on Young diagrams (not coming from a lattice structure), which is a Markov chain $\hat{\mathcal{K}}_{\mathcal{P}'(n)}$ induced on the set of partitions $\mathcal{P}'(n)$ by the Merge action $\hat{\mathcal{K}}^{(A,n)}$ on $\mathcal{G}_{n,A}$ through a projection map, resulting in a graph $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$, which we consider here with only EM and SM arrows, and we extend in §9 by additional IM self-loops.

Definition 8.1. Let $p : \mathfrak{F}_{A,n} \rightarrow \mathcal{P}'(n)$ denote the projection map that assigns to a forest $F = T_1 \sqcup \dots \sqcup T_r$ in $\mathfrak{F}_{A,n}$ the partition $\wp = p(F)$, $n = k_1 + \dots + k_r$, defined by the partitioning of the n

leaves into the different components of the workspace,

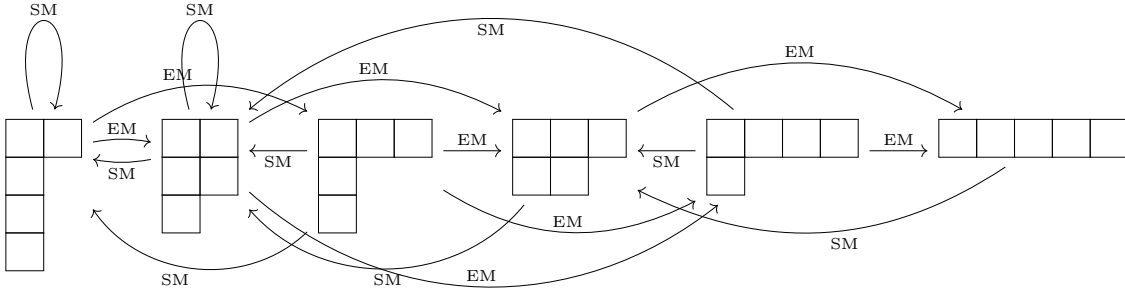
$$L(F) = L(T_1) \sqcup \cdots \sqcup L(T_r)$$

into the $k_i = L(T_i)$ leaves of each component T_i .

We consider here a graph that has partitions as vertices and edges corresponding to the effect on partitions of the EM and SM arrows on workspaces.

Definition 8.2. Let $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ denote the graph that has vertex set given by the set $\mathcal{P}'(n)$ of partitions of n , excluding the partition $\wp_1 = \{1, 1, \dots, 1\}$, representing the partitions $\wp$ as their Young diagrams $D(\wp)$. The edges are induced by the EM and SM arrows of $\mathcal{G}_{n,A}$: there is an EM or SM edge $\wp \rightarrow \wp'$ in $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ whenever there is an EM or SM arrow $F \rightarrow F'$ for some $F \in p^{-1}(\wp)$ and some $F' \in p^{-1}(\wp')$ in the graph $\mathcal{G}_{n,A}$, under the projection $p : \mathfrak{F}_{A,n} \rightarrow \mathcal{P}'(n)$. The EM arrows in $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ combine together two of the rows of the diagram into a single row, while the SM arrows extract either two boxes from the same row or two boxes from different rows and form with them a new row of length 2.

For example, in the case $n = 5$ the graph $\mathcal{G}_{\mathcal{P}'(5)}^{\text{EM-SM}}$ takes the form



We now investigate some general properties of the graph $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ and the Markov chain $\hat{\mathcal{K}}_{\mathcal{P}'(n)}^{\text{EM-SM}}$.

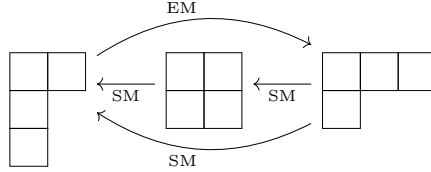
8.2. Strong connectivity. Strong connectivity in fact follows from strong connectivity of the $\mathcal{G}_{n,A}$ graph via projection, but we spell out a direct proof anyway (which reflects the proof for $\mathcal{G}_{n,A}$ given in [26]), because it will be useful later.

Lemma 8.3. *The graph $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ for all $n \geq 4$, is strongly connected and aperiodic. Thus, the Markov chain $\hat{\mathcal{K}}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ is ergodic: it has a unique stationary distribution (non-uniform, in general). Any initial distribution converges to the stationary distribution under iterations of $\hat{\mathcal{K}}_{\mathcal{P}'(n)}^{\text{EM-SM}}$.*

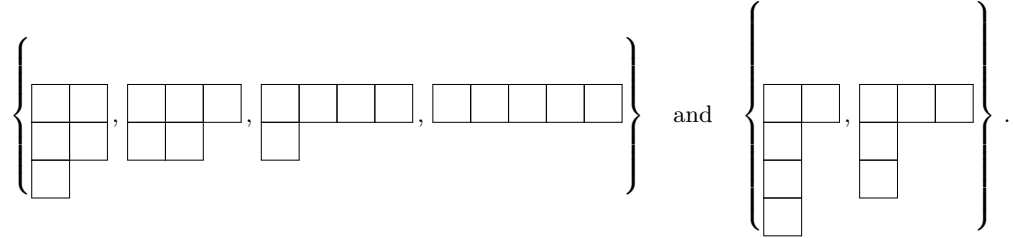
Proof. To show strong connectivity it suffices to show that, for every $\wp, \wp' \in V(\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}) = \mathcal{P}'(n)$, there is a directed path of minimal SM arrows from $\wp$ to the partition $\{2, 1, \dots, 1\}$ and that there is a directed path of EM arrows from $\{2, 1, \dots, 1\}$ to any other $\wp'$. For the first one, suppose that $\wp = \{k_1, \dots, k_r\}$. For any $k_i \geq 3$ there is a sequence of SM arrows that extract two boxes from a row of length k_i and gives $\{2, k_1, \dots, k_i - 2, \dots, k_r\}$, until each such k_i is reduced to 2 or 1. Applying this sequence of SM transformations results in a partition that has only 2's and 1's. Any such partition can be reduced to the partition with a single 2 and all 1's by minimal SM arrows by repeatedly extracting one box from two rows of length 2 resulting in one new row of length 2 and two rows of length 1 (and all the other rows unchanged), this reduces by one the number of rows of length 2 and increases by two the number of rows of length 1, until one reaches $\{2, 1, \dots, 1\}$. To obtain the EM path from $\{2, 1, \dots, 1\}$ to another vertex $\wp'$, note that $\wp'$ contains at least one row of length $k_i \geq 3$ or/and additional rows of length 2. Each of these rows

of length $k_i \geq 2$ can be obtained by successive EM arrows that start with combining two rows of $\{2, 1, \dots, 1\}$ and keep adding one box at a time to rows, using the remaining rows of length 1 of $\{2, 1, \dots, 1\}$. The aperiodicity is immediate because of the self-loop SM arrows at $\{2, 1, \dots, 1\}$ and at other partitions consisting of 2's and 1's. Ergodicity then follows from strong connectedness and aperiodicity. $\square$

Note that even if we restrict to the subgraph without the SM self-loops we still do have aperiodicity, for all $n \geq 4$, because any three $\wp, \wp', \wp''$ that contain the diagrams shown below have both a length 2 and a length 3 cycle coming from this subgraph of $\mathcal{G}_{\mathcal{P}(4)}^{\text{EM-SM}}$:



Remark 8.4. While strong connectedness holds for all the graphs $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$, direct inspection of the example of $n = 5$ above shows that it is not a “robust” property of the graph, in the sense that it is sufficient to remove a single edge (the SM arrow between the $\{2, 2, 1\}$ and the $\{2, 1, 1, 1\}$ partitions) to lose the strong connectivity property. Indeed, without that arrow, the graph would split out into two strongly connected components (the first recurrent, the second transient):



This “robustness” property can be measured in terms of *strong k -edge connectedness*. A graph is k -edge connected if it remains connected after the removal of up to $k - 1$ edges and it is strongly k -edge connected if it remains strongly connected after removal of up to $k - 1$ edges. The graph $\mathcal{G}_{\mathcal{P}(5)}^{\text{EM-SM}}$ is strongly 1-edge connected (namely it is strongly connected) but it is not strongly 2-edge connected.

Indeed, the strong edge connectedness never improves when n grows.

Proposition 8.5. *For all $n \geq 3$ the graphs $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ are strongly 1-edge-connected but never strongly 2-edge-connected.*

Proof. Strong 1-edge-connectedness (strong connectedness) is shown in Lemma 8.3. The case $n = 3$ just consists of two vertices with an arrow in each direction, so clearly removing one edge causes the strong connectedness property to fail. For any $n \geq 4$ consider the two partitions $\wp = \{2, 1, \dots, 1\}$ and $\wp' = \{3, 1, \dots, 1\}$. The partition $\wp'$ cannot be the target of any minimal SM arrow because it does not contain any $k_i = 2$, so it can only be reached via an EM arrow. There is only one such arrow, which is coming from the vertex labelled by $\wp$. Thus, removing a single edge given by this EM arrow $\wp \rightarrow \wp'$ will cause $\wp'$ to become a transient state, in fact a source vertex, so the remaining graph is no longer strongly connected. $\square$

This shows that “fragility” of strong connectedness in the $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ networks is coming from the fact that some vertices cannot be directly reached via minimal SM arrows. Those vertices are only reachable via EM arrows. This makes certain EM arrows particularly crucial to the strong connectivity of the graph. Note that the argument of Proposition 8.5 identifies a crucial kind of

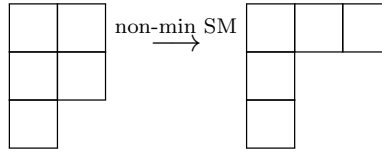
EM arrow as being, in the early stages of structure formation, the arrows that go from a cherry and a single leaf to a 3-leaves structure

$$\text{EM} : \alpha \sqcup \widehat{\beta \gamma} \mapsto \widehat{\alpha \beta} \gamma.$$

One can argue, from the linguistics perspective, that these EM arrows do indeed play a special role, as they are the ones that make it possible to transition from simple binary associations $\{\alpha, \beta\}$ of two elementary structures, to actual hierarchical structures like $\{\{\alpha, \beta\}, \gamma\}$.

This property also shows the fundamental asymmetry between EM and SM, with EM (and especially these particular EM arrows) playing a more crucial role in determining the properties of the dynamics.

Note, however, that if one includes all SM arrows, not just the minimal ones, then there would be SM arrows with target a partition that does not contain any $k_i = 2$, such as, for example



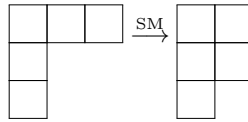
corresponding to non-minimal SM transformations like

$$\text{non-min SM} : \widehat{\alpha_1 \alpha_2} \sqcup \widehat{\alpha_3 \alpha_4} \sqcup \alpha_5 \mapsto \widehat{\alpha_2 \alpha_3 \alpha_4} \sqcup \alpha_1 \sqcup \alpha_5.$$

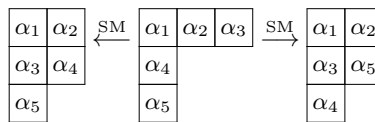
Thus, maintaining minimality of SM (as discussed in [26], see also §11 below) causes the “fragility” of strong connectedness and the special role of certain EM arrows.

8.3. Young tableaux diagram. As an intermediate step, before returning to discuss the full Merge action on workspaces and the graph $\mathcal{G}_{n,A}^{\text{EM-SM}}$, we refine the graph $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ by including the assignment of labels α_i at the leaves, partitioned according to the partition $\wp \in \mathcal{P}'(n)$. Thus, for $A = \{\alpha_1, \dots, \alpha_n\}$, we consider the graph $\mathcal{G}_{\mathcal{P}'(n),A}^{\text{EM-SM}}$ where the vertices are pairs $(\wp, \alpha)$ with $\alpha : A \rightarrow Y(\wp)$ a map that bijectively assigns the elements $\alpha_i \in A$ to the boxes of the Young diagram $Y(\wp)$ of the partition $\wp \in \mathcal{P}'(n)$, considered up to the action of the Young group $S_\wp$ of permutations within the rows, and up to permuting rows of equal length, so that the total number of such assignments $\alpha : A \rightarrow Y(\wp)$ is given by the generalized multinomial coefficient $\Upsilon_{\wp,n}$ of (2.8).

Again we do not include the partition $\{1, \dots, 1\}$ in $V(\mathcal{G}_{\mathcal{P}'(n),A}^{\text{EM-SM}})$. The maps are then the same maps EM and SM arrows as in $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ on the partitions, but now one also keeps track of the labeling, so that, for example an arrow like



splits into two arrows



and similarly for the other arrows of $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ that have different possible preimages under the projection map $\mathcal{G}_{\mathcal{P}'(n),A}^{\text{EM-SM}} \rightarrow \mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ that forgets the labels $\alpha_i \in A$.

Remark 8.6. Note that the self-looping SM maps present in the graph $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$, at vertices given by partitions $\wp$ containing 2's and 1's, are no longer self-loops in $\mathcal{G}_{\mathcal{P}'(n),A}^{\text{EM-SM}}$, since now they map a vertex $(\wp, \alpha)$ to a different vertex $(\wp, \alpha')$ with $\alpha \neq \alpha'$, as for example

$$\begin{array}{|c|c|} \hline \alpha_1 & \alpha_2 \\ \hline \alpha_3 & \\ \hline \end{array} \xrightarrow{\text{SM}} \begin{array}{|c|c|} \hline \alpha_1 & \alpha_3 \\ \hline & \alpha_2 \\ \hline \end{array}$$

Lemma 8.7. *The graph $\mathcal{G}_{\mathcal{P}'(n),A}^{\text{EM-SM}}$ is strongly connected and aperiodic, hence the Markov chain $\hat{\mathcal{K}}_{\mathcal{P}'(n),A}^{\text{EM-SM}}$ is ergodic.*

Proof. Given the strong connectedness of $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$, we can show that $\mathcal{G}_{\mathcal{P}'(n),A}^{\text{EM-SM}}$ is also strongly connected by showing that any permutation of the labels on a fixed partition $\wp$ can be achieved by a concatenation of arrows in $\mathcal{G}_{\mathcal{P}'(n),A}^{\text{EM-SM}}$. Consider the following case as illustration of the general method. Suppose in the following Young tableau we want to permute the labels α_3 and α_5 . This can be done with the following chain of SM and EM arrows:

$$\begin{array}{|c|c|c|c|} \hline \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 \\ \hline \alpha_5 & \alpha_6 & & \\ \hline \end{array} \xrightarrow{\text{SM}} \begin{array}{|c|c|c|} \hline \alpha_1 & \alpha_2 & \alpha_4 \\ \hline \alpha_5 & \alpha_3 & \\ \hline \alpha_6 & & \\ \hline \end{array} \xrightarrow{\text{SM}} \begin{array}{|c|c|} \hline \alpha_1 & \alpha_2 \\ \hline \alpha_5 & \alpha_4 \\ \hline \alpha_3 & \\ \hline \alpha_6 & \\ \hline \end{array} \xrightarrow{\text{EM}} \begin{array}{|c|c|c|c|} \hline \alpha_1 & \alpha_2 & \alpha_5 & \alpha_4 \\ \hline \alpha_3 & & & \\ \hline & \alpha_6 & & \\ \hline \end{array} \xrightarrow{\text{EM}} \begin{array}{|c|c|c|c|} \hline \alpha_1 & \alpha_2 & \alpha_5 & \alpha_4 \\ \hline \alpha_3 & & & \\ \hline \alpha_6 & & & \\ \hline \end{array}$$

This same procedure can be used to permute any two labels in different rows of the diagram. Since the assignments of labels to the diagrams is only up to permutations within rows, compositions of these operations suffice to achieve all possible permutations. The same argument used for $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$ shows aperiodicity for $\mathcal{G}_{\mathcal{P}'(n),A}^{\text{EM-SM}}$. $\square$

Thus, we obtain the following result.

Corollary 8.8. *The Markov chain $\hat{\mathcal{K}}_{\mathcal{P}'(n),A}^{\text{EM-SM}}$ induced on $\mathcal{G}_{\mathcal{P}'(n),A}^{\text{EM-SM}}$ by the Merge Hopf algebra markov chain $\hat{\mathcal{K}}^{(A,n)}$ on $\mathcal{G}_{n,A}$ has a unique asymptotic distribution*

$$\pi_{\mathcal{P}'(n),A} \hat{\mathcal{K}}_{\mathcal{P}'(n),A}^{\text{EM-SM}} = \pi_{\mathcal{P}'(n),A},$$

with $\pi_{\mathcal{P}'(n),A}(\wp, \alpha) > 0$ for all $(\wp, \alpha) \in V(\mathcal{G}_{\mathcal{P}'(n),A}^{\text{EM-SM}})$ and any initial distribution converges to $\pi_{\mathcal{P}'(n),A}$ under iterations of $\hat{\mathcal{K}}_{\mathcal{P}'(n),A}^{\text{EM-SM}}$.

Note that, unlike Internal Merge dynamics, the stationary distribution $\pi_{\mathcal{P}'(n),A}$ on $\mathcal{G}_{\mathcal{P}'(n),A}^{\text{EM-SM}}$ for External and Sideward Merge dynamics is usually not uniform.

9. FULL MERGE DYNAMICS WITH MINIMAL SIDEWARD MERGE

We now combine all the results that we have obtained by separately investigating the Internal Merge dynamics on the graphs $\mathcal{G}_{A,n,\wp,\sigma}$ and the combined Sideward Merge and External Merge induced dynamics on the graphs $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$, to understand the full Merge dynamics on $\mathcal{G}_{n,A}$. To this purpose, we will consider here an extension of the graph $\mathcal{G}_{\mathcal{P}'(n)}^{\text{EM-SM}}$, to a graph $\mathcal{G}_{\mathcal{P}'(n)}$ that also includes the self-loops that are images of the IM arrows of $\mathcal{G}_{n,A}$. To study the relation between the dynamics on this graph $\mathcal{G}_{\mathcal{P}'(n)}$ and on the full Merge graph $\mathcal{G}_{n,A}$ we first prove in §9.1 a general result on projections of Markov chains.

9.1. Markov Chain Projection. We show here that when a finite Markov chain has a particular symmetry with respect to a projection of the space of states to a smaller state, the stationary distribution can be computed on a smaller graph.

Definition 9.1. Let X be a finite set and $p : X \rightarrow Y$ a surjective map. Let G_X be a strongly connected aperiodic directed graph with $V(G_X) = X$, and let $\mathcal{K}^X$ be the adjacency matrix of G_X . We say that the graph G_X is p -symmetric if it has the property that the number of edges from a vertex x to vertices in a fiber $p^{-1}(y')$ is constant over $x \in p^{-1}(y)$

$$(9.1) \quad \sum_{x' \in p^{-1}(y')} \mathcal{K}_{x,x'}^X = \mathcal{K}_{y,y'}^{Y,R}, \quad \forall x \in p^{-1}(y)$$

and the number of edges to a vertex x' from vertices in a fiber $p^{-1}(y)$ is constant over $x' \in p^{-1}(y')$,

$$(9.2) \quad \sum_{x \in p^{-1}(y)} \mathcal{K}_{x,x'}^X = \mathcal{K}_{y,y'}^{Y,L}, \quad \forall x' \in p^{-1}(y'),$$

with

$$(9.3) \quad \#p^{-1}(y) \cdot \mathcal{K}_{y,y'}^{Y,R} = \#p^{-1}(y') \cdot \mathcal{K}_{y,y'}^{Y,L}.$$

counting the total number of directed edges in G_X from the fiber $p^{-1}(y)$ to the fiber $p^{-1}(y')$.

The number computed by (9.3) can be interpreted geometrically as follows. Let G_Y be the graph with vertex set Y and with a directed edge $y \rightarrow y'$ whenever there are some $F \in p^{-1}(y)$ and $F' \in p^{-1}(y')$ such that there is a directed edge $F \rightarrow F'$ in G_X between them. This way the map $p : X \rightarrow Y$ extends to a map $p : G_X \rightarrow G_Y$. We refer to G_Y as the graph induced by G_X and p . Then both sides of (9.3) compute the number of edges of G_X that are above an edge $y \rightarrow y'$ in G_Y with respect to the projection p .

Proposition 9.2. Let X be a finite set and $p : X \rightarrow Y$ a surjective map and G_X a strongly connected aperiodic p -symmetric graph on the vertex set X . Then the induced graph G_Y is strongly connected. Assume that it is also aperiodic. Let $c_y = \#p^{-1}(y)$ denote the size of the fiber over a point $y \in Y$. Let λ and η_X be the Perron-Frobenius eigenvalue and (right) eigenvector of the adjacency matrix $\mathcal{K}^X$ of G_X ,

$$(9.4) \quad \sum_{x' \in X} \mathcal{K}_{x,x'}^X \eta_X(x') = \lambda \eta_X(x),$$

and let $\hat{\mathcal{K}}^X$ be the associated Markov chain transition matrix

$$\hat{\mathcal{K}}_{x,x'}^X := \lambda^{-1} \frac{\eta_X(x')}{\eta_X(x)} \mathcal{K}_{x,x'}^X.$$

- (1) The right Perron-Frobenius eigenfunction $\eta_X(x) = \eta_Y(y)$ of $\mathcal{K}^X$ is constant on fibers $p^{-1}(y)$ for $y \in Y$ and its values $\eta_Y(y)$ are given by the right Perron-Frobenius eigenfunction of $\mathcal{K}^{Y,R}$

$$(9.5) \quad \sum_{y' \in Y} \mathcal{K}_{y,y'}^{Y,R} \eta_Y(y') = \lambda \eta_Y(y).$$

- (2) The left Perron-Frobenius eigenfunction $\xi_X(x) = \xi_Y(y)$ of $\mathcal{K}^X$ is also constant on fibers $p^{-1}(y)$ and given by the left Perron-Frobenius eigenfunction of $\mathcal{K}^{Y,L}$

$$(9.6) \quad \sum_{y \in Y} \xi_Y(y) \mathcal{K}_{y,y'}^{Y,L} = \lambda \xi_Y(y).$$

(3) The stationary distribution $\pi_X(x)$ of $\hat{\mathcal{K}}^X$,

$$(9.7) \quad \sum_{x \in X} \pi_X(x) \hat{\mathcal{K}}_{x,x'}^X = \pi_X(x')$$

is also constant, $\pi_X(x) = \pi_Y(y)$, on fibers $x \in p^{-1}(y)$, with $\pi_Y(y)$ the stationary distribution

$$(9.8) \quad \sum_{y \in Y} \pi_Y(y) \hat{\mathcal{K}}_{y,y'}^Y = \pi_Y(y'),$$

of the Markov chain on Y with transition matrix

$$\hat{\mathcal{K}}_{y,y'}^Y = \lambda^{-1} \frac{\eta_Y(y') \cdot c_y}{\eta_Y(y) \cdot c_{y'}} \mathcal{K}_{y,y'}^{Y,R}.$$

(4) Equivalently, the function

$$(9.9) \quad \psi_Y(y) := \frac{\pi(y) c_y}{\eta_Y(y)}$$

is the (left) Perron-Frobenius eigenvector of $\mathcal{K}^{Y,R}$, satisfying

$$(9.10) \quad \sum_y \psi_Y(y) \mathcal{K}_{y,y'}^{Y,R} = \lambda \psi_Y(y'),$$

(5) The stationary distribution of the Markov chain $\hat{\mathcal{K}}^X$ is given by

$$(9.11) \quad \pi_X(x) = \frac{\eta_Y(p(x)) \cdot \psi_Y(p(x))}{c_{p(x)}},$$

in terms of the left and right PF eigenvectors ψ_Y and η_Y of $\mathcal{K}^{Y,R}$ and the sizes c_y of the fibers of the projection $p : X \rightarrow Y$.

(6) the left Perron-Frobenius eigenvectors ξ_Y of $\mathcal{K}^{Y,L}$ and ψ_Y of $\mathcal{K}^{Y,R}$ are related (up to normalization) by

$$(9.12) \quad \xi_Y(y) = \frac{\psi_Y(y)}{c_y}.$$

Proof. The graph G_Y is strongly connected since G_X is. Indeed, given $y, y' \in Y$, two arbitrary vertices $F \in p^{-1}(y)$ and $F' \in p^{-1}(y')$ are connected by paths in G_X in both directions and subsequent edges of these paths in G_X project to subsequent edges in G_Y .

(1) The graph G_Y is also aperiodic by assumption, so the matrix $\mathcal{K}^{Y,R}$ has a unique (up to scaling) right Perron-Frobenius eigenfunction with $\eta_Y(y) > 0$ for all $y \in Y$, satisfying (9.6) with $\lambda > 0$ the PF eigenvalue of Y . We then have

$$\sum_{x' \in X} \mathcal{K}_{x,x'}^X \eta_Y(y') = \sum_{y'} \sum_{x' \in p^{-1}(y')} \mathcal{K}_{x,x'}^X \eta_Y(y') = \sum_{y'} \mathcal{K}_{y,y'}^{Y,R} \eta_Y(y') = \lambda \eta_Y(y),$$

so that $\eta_X(x) := \eta_Y(p(x))$ is a solution to (9.4) with the same λ . The Perron-Frobenius theorem for $\mathcal{K}^X$ ensures that the left/right PF eigenvectors are the only eigenvectors of $\mathcal{K}^X$ with all positive real entries. Thus, η_X has to be the right Perron-Frobenius eigenvector of $\mathcal{K}^X$, which is unique up to scale.

(2) The case of the left Perron-Frobenius eigenfunction of $\mathcal{K}^X$ is analogous with

$$\sum_{x \in X} \xi_X(x) \mathcal{K}_{x,x'}^X = \sum_y \sum_{x \in p^{-1}(y)} \xi_Y(y) \mathcal{K}_{x,x'}^X = \sum_y \xi_Y(y) \mathcal{K}_{y,y'}^{Y,L} = \lambda \xi_Y(y).$$

(3) In a similar way, given that η_X is constant on preimages of p , if $\pi_Y(y)$ is the stationary distribution of $\hat{\mathcal{K}}^Y$ satisfying (9.8), then $\pi_X(x) := \pi_Y(p(x))$ satisfies, for $x' \in p^{-1}(y')$,

$$\begin{aligned} \sum_x \pi_X(x) \hat{\mathcal{K}}_{x,x'} &= \sum_y \pi_Y(y) \lambda^{-1} \frac{\eta_Y(y')}{\eta_Y(y)} \sum_{x \in p^{-1}(y)} \mathcal{K}_{x,x'}^X = \sum_y \pi_Y(y) \lambda^{-1} \frac{\eta_Y(y')}{\eta_Y(y)} \mathcal{K}_{y,y'}^{Y,L} \\ &= \sum_y \pi_Y(y) \lambda^{-1} \frac{\eta_Y(y') \cdot c_y}{\eta_Y(y) \cdot c_{y'}} \mathcal{K}_{y,y'}^{Y,R} = \sum_y \pi_Y(y) \hat{\mathcal{K}}_{y,y'}^Y = \pi_Y(y') = \pi_X(x'). \end{aligned}$$

Since the graph G_X is strongly connected and aperiodic, the stationary distribution π_X is unique, so it has to be of this form.

(4) We can equivalently write the equation

$$\sum_{y \in y} \pi_Y(y) \lambda^{-1} \frac{\eta_Y(y') \cdot c_y}{\eta_Y(y) \cdot c_{y'}} \mathcal{K}_{y,y'}^{Y,R} = \pi_Y(y')$$

in the form

$$\sum_{y \in y} \frac{\pi_Y(y) c_y}{\eta_Y(y)} \mathcal{K}_{y,y'}^{Y,R} = \lambda \frac{\pi_Y(y') c_{y'}}{\eta_Y(y')},$$

which says that $\psi(y)$ as in (9.9) is the left Perron-Frobenius eigenvector equation for $\mathcal{K}^{Y,R}$ with PF eigenvalue λ .

(5) This follows directly from (4) and the form of the stationary distribution $\pi(x)$ as in (9.11).

(6) Since ψ_Y is the left PF eigenvector of $\mathcal{K}^{Y,R}$, we have

$$\sum_y \frac{\psi_Y(y)}{c_y} \frac{c_y}{c_{y'}} \mathcal{K}_{y,y'}^{Y,R} = \lambda \frac{\psi_Y(y')}{c_{y'}},$$

which by (9.3) and the uniqueness (up to normalization) of the left PF eigenvector of $\mathcal{K}^{Y,L}$ gives the relation (9.12). □

9.2. Stationary distribution of the Merge dynamics on workspaces. This result applies to the case of the Hopf algebra Markov chain of Merge we are considering here.

Proposition 9.3. *Let $X = V(\mathcal{G}_{n,A}) = \mathfrak{F}_{A,n} \subset \mathfrak{F}_{SC_0}$ be the set of nonplanar full binary forests with n leaves with set of labels $A = \{\alpha_1, \dots, \alpha_n\}$, and with non-empty set of edges. Let $Y = \mathcal{P}'(n)$ be the set of partitions of n (not including the partition $\{1, \dots, 1\}$). Let $p : \mathfrak{F}_{A,n} \rightarrow \mathcal{P}'(n)$ be the map that assigns to a forest F with n leaves the partition $\wp = p(F)$ of $n = k_1 + \dots + k_r$ into the numbers of leaves of each tree component $F = T_1 \sqcup \dots \sqcup T_r$ of the forest, $k_i = \#L(T_i)$. The graph $\mathcal{G}_{n,A}$ is p -symmetric with respect to this map. The projection $p : \mathfrak{F}_{A,n} \rightarrow \mathcal{P}'(n)$ induces a projection of graphs $\mathcal{G}_{n,A} \rightarrow \mathcal{G}_{\mathcal{P}'(n)}$, where the graph $\mathcal{G}_{\mathcal{P}'(n)}$ has vertex set $\mathcal{P}'(n)$ and EM, SM arrows as in Definition 8.2 and an additional IM self-loop at a vertex $\wp$ whenever there are IM arrows $F \rightarrow F'$ between some vertices $F, F' \in p^{-1}(\wp) = \mathcal{G}_{n,A,\wp}$.*

Proof. The key property of the graph $\mathcal{G}_{n,A}$ that is responsible for the p -symmetry with respect to the map $p : \mathfrak{F}_{A,n} \rightarrow \mathcal{P}'(n)$ is the fact that the number of edges from (or to) a vertex F in $\mathcal{G}_{n,A}$ depends only on $p(F)$ (assuming that we also include Sideward Merge arrows that extract the two leaves of a cherry when that cherry is part of a larger tree). This follows from Proposition 2.3. This property, in fact, can be further refined to show that the counting of the edges out (or in) a vertex F on $\mathcal{G}_{n,A}$ can be split into separate counting for the subsets of edges with target (respectively, source) vertex in each fiber $p^{-1}(\wp')$, and the counting over each of these sets also depends only on $\wp = p(F)$ and not on F itself. To see this, notice that for each type (EM/IM/SM) of arrows,

the counting of the outgoing (or incoming) edges of that type depends on counting choices of appropriate boxes in the partition $\wp = p(F)$. Any such choice of boxes, along with the type of operation (EM/IM/SM), suffice to determine the shape of the resulting partition $\wp'$ associated to the target vertex of the arrow. Thus each subset of arrows with source F that have the other end in a fiber $p^{-1}(\wp')$ is determined by knowing the partition $\wp$. This implies that both (9.2) and (9.1) hold. By construction, the map $\mathcal{G}_{n,A} \rightarrow \mathcal{G}_{\mathcal{P}'(n)}$ has fibers $p^{-1}(\wp) = \mathcal{G}_{n,A,\wp}$, hence the EM and SM arrows of $\mathcal{G}_{n,A}$ give rise to edges of $\mathcal{G}_{\mathcal{P}'(n)}$ as in Definition 8.2 and the IM arrows, which are always along components of fibers, give rise to looping edges in $\mathcal{G}_{\mathcal{P}'(n)}$. $\square$

Thus we obtain the form of the stationary distribution for the full Merge dynamics on the set of workspaces, given by the Hopf algebra Markov chain $\hat{\mathcal{K}}^{A,n}$ on the graph $\mathcal{G}_{n,A}$.

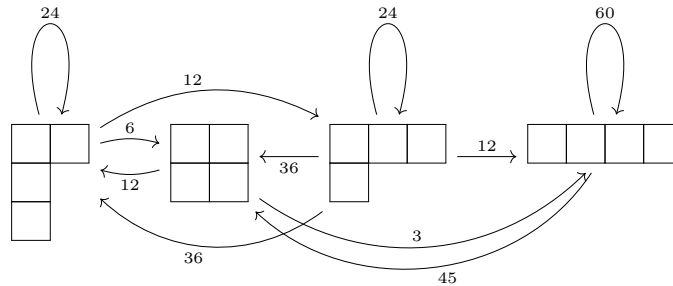
Theorem 9.1. *The Merge Hopf algebra Markov chain $\hat{\mathcal{K}}^{A,n}$ on the graph $\mathcal{G}_{n,A}$ has stationary distribution that is constant on the subsets $V_{n,A,\wp} = p^{-1}(\wp)$, that are the fibers of the projection $p : \mathcal{G}_{n,A} \rightarrow \mathcal{G}_{\mathcal{P}'(n)}$. Up to an overall multiplicative constant, the stationary distribution $\pi(F)$ of the Merge Hopf algebra Markov chain $\hat{\mathcal{K}}^{A,n}$ is obtained from the left and right Perron-Frobenius eigenvectors, $\eta_{\mathcal{P}'(n)}, \psi_{\mathcal{P}'(n)}$ of the adjacency matrix $\mathcal{K}^{\mathcal{P}'(n),R}$ of the graph $\mathcal{G}_{\mathcal{P}'(n)}$ (with the IM looping edges included) and the multiplicities of the fibers $\#V_{n,A,\wp} = \Lambda_{\wp,n}$ as in (2.9), in the form*

$$(9.13) \quad \pi(F) = \frac{\eta_{\mathcal{P}'(n)}(\wp) \psi_{\mathcal{P}'(n)}(\wp)}{\Lambda_{\wp,n}} \quad \text{for } \wp = p(F).$$

Proof. The form (9.13) of the distribution follows from Proposition 9.2 and the fact that the distribution is uniform on the $V_{n,A,\wp}$ follows from (5.1) and (7.2), as in Proposition 5.5. $\square$

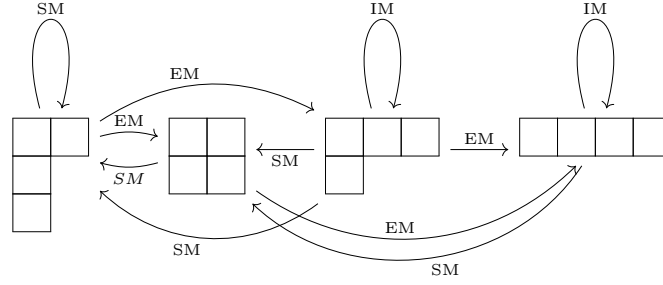
9.3. The example of $n = 4$. We show here an explicit example, which is simple enough that everything can be computed explicitly, but sufficient to illustrate all the steps of the procedure discussed above. The case with $n = 3$ leaves is too simple (and already fully discussed in [26]) so we look at the case of $n = 4$ leaves. We first describe the geometry of the projection $p : \mathcal{G}_{4,A} \rightarrow \mathcal{G}_{\mathcal{P}(4)}$.

Lemma 9.4. *In the case $n = 4$, the projection $p : \mathcal{G}_{4,A} \rightarrow \mathcal{G}_{\mathcal{P}(4)}$ has multiplicities above each edge given by*



while the multiplicities above the vertices are 6 for the first vertex, 3 for the second, 12 for the third and 15 for the fourth.

Proof. The graph $\mathcal{G}_{\mathcal{P}(4)}$ with the EM and SM arrows, and with the self-looping IM arrows, is of the form



To compute the weights of the arrows we need to compute the number of edges of $\mathcal{G}_{n,A}$ that are above each of the edges in this graph through the projections $\mathcal{G}_{n,A} \rightarrow \mathcal{G}_{\mathcal{P}'(n),A} \rightarrow \mathcal{G}_{\mathcal{P}'(n)}$. Since the number of incoming and outgoing edges at each vertex F in $\mathcal{G}_{n,A}$ only depends on the partition $\wp = p(F)$ (as we are here also allowing SM arrows that cut two leaves of a cherry that is inside a larger tree), the number of edges above an edge $\wp \rightarrow \wp'$ can be computed as the product of the number of vertices above $\wp$ in $\mathcal{G}_{n,A}$ times the number of edges from one of the vertices above $\wp$ to the fiber above $\wp'$. Thus, we obtain the following:

- There are 6 vertices of $\mathcal{G}_{4,A}$ above the vertex $\wp = \{2, 1, 1\}$ of $\mathcal{G}_{\mathcal{P}(4)}$, namely the choices of labels at the leaves

$$\begin{array}{c} \wedge \\ \alpha_i \quad \alpha_j \end{array} \sqcup \alpha_k \sqcup \alpha_\ell$$

namely all choices of two out of $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ to form the cherry leaves.

- There are 3 vertices of $\mathcal{G}_{4,A}$ above the vertex $\wp = \{2, 2\}$ of $\mathcal{G}_{\mathcal{P}(4)}$, again given by all choices of labels in

$$\begin{array}{c} \wedge \\ \alpha_i \quad \alpha_j \end{array} \sqcup \begin{array}{c} \wedge \\ \alpha_k \quad \alpha_\ell \end{array}$$

which can be seen as the 6 choices of two labels to form the first cherry, taken up to the symmetry between the two cherries that cuts the counting down to 3.

- There are 12 vertices of $\mathcal{G}_{4,A}$ above the vertex $\wp = \{3, 1\}$ of $\mathcal{G}_{\mathcal{P}(4)}$ given by the assignments of labels to

$$\begin{array}{c} \wedge \\ \alpha_i \quad \alpha_j \quad \alpha_k \end{array} \sqcup \alpha_\ell$$

so 4 choices of the label for the isolated vertex, and 3 remaining choices of the label at the leaf attached to the root in the tree.

- There are 15 vertices of $\mathcal{G}_{4,A}$ above the vertex $\wp = \{4\}$ of $\mathcal{G}_{\mathcal{P}(4)}$ given by the sum over the two tree topologies of the different possible labelings at the leaves,

$$\begin{array}{c} \wedge \\ \alpha_i \quad \alpha_j \quad \alpha_k \quad \alpha_\ell \end{array} \quad \text{and} \quad \begin{array}{c} \wedge \\ \alpha_i \quad \alpha_j \quad \alpha_k \quad \alpha_\ell \end{array}.$$

In the first case there are 3 possible assignments of leaves (as in the case of $\wp = \{2, 2\}$) while in the second case there are 12 (4 choices of label at the vertex attached to the root and 3 remaining at the second vertex).

We then just need to count the number of edges from one vertex above $\wp$ with target above $\wp'$ as this number is constant at all vertices above $\wp$. These are counted by our general counting of indegrees and outdegrees for the $\mathcal{G}_{n,A}$.

- There are 4 Sideward Merge arrows from F above $\wp = \{2, 1, 1\}$ to the same fiber, hence $24 = 6 \cdot 4$ edges above $\{2, 1, 1\} \rightarrow \{2, 1, 1\}$.

- There is 1 External Merge arrow from F above $\wp = \{2, 1, 1\}$ to vertices above $\wp' = \{2, 2\}$, hence 6 edges above $\{2, 1, 1\} \rightarrow \{2, 2\}$.
- There are 2 External Merge arrows from F above $\wp = \{2, 1, 1\}$ to vertices above $\wp' = \{3, 1\}$, hence 12 edges above $\{2, 1, 1\} \rightarrow \{3, 1\}$.
- There are 4 Sideward Merge arrows from F above $\wp = \{2, 2\}$ to vertices above $\wp' = \{2, 1, 1\}$, hence 12 edges above $\{2, 2\} \rightarrow \{2, 1, 1\}$.
- There is 1 External Merge arrow from F above $\wp = \{2, 2\}$ to vertices above $\wp' = \{4\}$, hence 3 edges above $\{2, 2\} \rightarrow \{4\}$.
- There are 2 Internal Merge arrows from a vertex F above $\wp = \{3, 1\}$ to the same fiber, hence $24 = 12 \cdot 2$ edges above $\{3, 1\} \rightarrow \{3, 1\}$.
- There are 3 Sideward Merge arrows from F above $\wp = \{3, 1\}$ to vertices above $\wp' = \{2, 2\}$, hence $36 = 12 \cdot 3$ edges above $\{3, 1\} \rightarrow \{2, 2\}$.
- There are 3 Sideward Merge arrows from F above $\wp = \{3, 1\}$ to vertices above $\wp' = \{2, 1, 1\}$, hence $36 = 12 \cdot 3$ edges above $\{3, 1\} \rightarrow \{2, 1, 1\}$.
- There is 1 External Merge arrows from F above $\wp = \{3, 1\}$ to vertices above $\wp' = \{4\}$, hence 12 edges above $\{3, 1\} \rightarrow \{4\}$.
- There are 3 Sideward Merge arrows from F above $\wp = \{4\}$ to vertices above $\wp' = \{2, 2\}$, hence $45 = 15 \cdot 3$ edges above $\{4\} \rightarrow \{2, 2\}$.
- There are 4 Internal Merge arrows from a vertex F above $\wp = \{4\}$ to the same fiber, hence $60 = 15 \cdot 4$ edges above $\{4\} \rightarrow \{4\}$.

□

We then compute the matrix $\mathcal{K}^{\mathcal{P}(4),R}$ of Proposition 9.2.

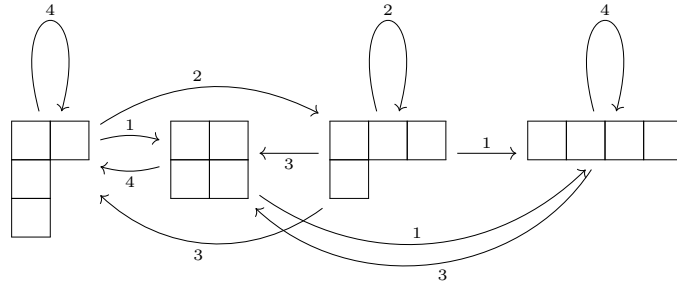
Lemma 9.5. *For $n = 4$, the matrix $\mathcal{K}^{\mathcal{P}(4),R}$ satisfying*

$$\mathcal{K}_{\wp, \wp'}^{\mathcal{P}(4),R} = \sum_{F' \in p^{-1}(\wp')} \mathcal{K}_{F, F'}^{(A,4)},$$

with $\mathcal{K}_{F, F'}^{(A,4)}$ the adjacency matrix of $\mathcal{G}_{4,A}$, is given by

$$(9.14) \quad \mathcal{K}^{\mathcal{P}(4),R} = \begin{pmatrix} 4 & 1 & 2 & 0 \\ 4 & 0 & 0 & 1 \\ 3 & 3 & 2 & 1 \\ 0 & 3 & 0 & 4 \end{pmatrix}$$

and corresponds to the weighted graph (with vertices listed from left to right)



The stationary distribution for $\hat{\mathcal{K}}^{A,4}$ on $\mathcal{G}_{4,A}$ takes the form

$$(9.15) \quad \pi_{4,A}(F) \sim \begin{cases} 0.5267 & F \in p^{-1}(\{2, 1, 1, 1\}) \\ 0.3109 & F \in p^{-1}(\{2, 2\}) \\ 0.1217 & F \in p^{-1}(\{3, 1\}) \\ 0.0406 & F \in p^{-1}(\{4\}). \end{cases}$$

Proof. The matrix $\mathcal{K}^{\mathcal{P}(4),R}$ of (9.14) has Perron–Frobenius eigenvalue $\lambda \sim 6.9656$, right Perron–Frobenius eigenvector with $\eta_{\mathcal{P}(4)}(\{2, 1, 1, 1\}) \sim 0.5565$, $\eta_{\mathcal{P}(4)}(\{2, 2\}) \sim 0.3739$, $\eta_{\mathcal{P}(4)}(\{3, 1\}) \sim 0.6383$, $\eta_{\mathcal{P}(4)}(\{4\}) \sim 0.3782$, and left Perron–Frobenius eigenvector $\psi_{\mathcal{P}(4)}(\{2, 1, 1, 1\}) \sim 0.8345$, $\psi_{\mathcal{P}(4)}(\{2, 2\}) \sim 0.3666$, $\psi_{\mathcal{P}(4)}(\{3, 1\}) \sim 0.3361$, $\psi_{\mathcal{P}(4)}(\{4\}) \sim 0.2369$. The size $c_\varphi = \#p^{-1}(\varphi)$ of the fibers is computed in Lemma 9.4 above, so using the expression (9.11) for the stationary distribution, and normalizing the result so that it is a probability distribution, we obtain (9.15). $\square$

9.4. The Sideward Merge problem. One sees easily, already from the explicit example of $n = 4$, the general problem produced by the presence of the Sideward Merge arrows, even while restricting only to the minimal ones: they tend to push the dynamics towards the more disconnected structures, entirely preventing the process of structure formation. We see this reflected in the fact that the fully connected structures (the trees in the preimage of $\varphi = \{4\}$) have a very low weight in the stationary distribution, compared to the most disconnected structures (in the preimage of $\varphi = \{2, 1, 1\}$) that receive the highest weight.

Notice how the behavior, already in the case $n = 4$, is different from the case $n = 3$ analyzed in [26]. In the case $n = 3$, when EM, SM, IM arrows are considered, the dynamics has a uniform stationary distribution where all workspaces carry the same weight. This already shows a problem, created by Sideward Merge: the dynamics does not converge to the fully formed sentences (the workspaces consisting of a single tree). This problem is corrected by the introduction of cost functions, see §11.

Indeed, this behavior clearly does not reflect the reality of linguistic structure formation and it means that the different types of arrows, External Merge, Internal Merge, and Sideward Merge, must have different weights in the graph $\mathcal{G}_{n,A}$, according to some cost function. As already discussed in [26], Sideward Merge is indeed non-optimal with respect to some measures of optimality used in the linguistics literature. Also there are only a few linguistic phenomena that may require Sideward Merge as an explanation, so it should occur very infrequently in Merge derivations, contrary to what one sees in the dynamics of the Markov chain $\hat{\mathcal{K}}^{A,n}$ we have been discussing here, where Sideward Merge arrows end up playing a dominant role in the dynamics. This means that a realistic model, as already discussed in [26], will have to replace $\hat{\mathcal{K}}^{A,n}$ with a weighted version, where different costs are accounted for. We will be discussing this in Section 11.

10. MERGE DYNAMICS WITH THE CONTRACTION COPRODUCT

Before introducing cost functions and analyzing their effect on the different parts (EM, IM, SM) of the dynamics, we comment in this section on possible variants of the dynamics discussed here. It is shown in [22] that there are different possible forms of the coproduct on workspaces that is used to define the Merge operations. All the discussion and results in the previous sections are based on the use of what is called in [22] the “deletion” coproduct, where the quotient $T/dF_{\underline{v}}$, for $F_{\underline{v}} = T_{v_1} \sqcup \cdots \sqcup T_{v_k}$ a collection of disjoint accessible terms in T , is defined as the maximal full binary rooted tree obtained (via edge contractions) from the non-full binary tree (including non-branching vertices) resulting from the admissible cut that removes the edges above each of the vertices v_i (the roots of the accessible terms T_{v_i}). The essential property of this coproduct Δ^d that we used in analyzing the dynamics is that the number of leaves adds over this decomposition:

$$\#L(T) = \#L(F_{\underline{v}}) + \#L(T/dF_{\underline{v}}).$$

In terms of the linguistics model, this coproduct is suitable for the interface of the core Merge dynamics with Externalization, where the trace of movement is not expressed.

The other coproduct used in [22] is the “contraction” coproduct, where the quotient $T/cF_{\underline{v}}$ is obtained by shrinking each component T_{v_i} of $F_{\underline{v}} = T_{v_1} \sqcup \cdots \sqcup T_{v_k} \subset T$ to its vertex v_i . Each of

these vertices v_i becomes in this way a new leaf of $T/^c F_v$, labelled by an element of $\mathcal{SO}_0$ that maintains a “trace” of the movement of the extracted accessible terms. One denotes such label by the symbol $\mathcal{T}_{\overline{v_i}}$, to mark that it corresponds to the cancellation of the lower copy of T_{v_i} under movement. This form of coproduct Δ^c now no longer satisfies the additivity of the number of leaves, as now we have

$$\#L(T) + k = \#L(F_v) + \#L(T/^c F_v),$$

for $F_v = T_{v_1} \sqcup \dots \sqcup T_{v_k}$.

10.1. Labels of traces of movement. Before discussing further the Merge operation constructed using this coproduct (as discussed in §1.3 of [22]), it is useful to clarify the meaning of the labeling system $\mathcal{T}_{\overline{v}}$ used to mark the new leaf v of the quotient $T/^c T_v$, when the subtree $T_v \subset T$ is contracted to its root vertex v . This section clarifies and expands in more detail the brief discussion in §1.3 of [22].

The form of the quotient $T/^c T_v$ requires an explanation: since the label of the leaf resulting from the contraction of T_v is usually written as $\mathcal{T}_{\overline{v}}$, this does not look like it can be compatible with having a fixed finite set of leaf labels. Indeed, if the label $\mathcal{T}_{\overline{v}}$ would really contain the full information about T_v , together with a “cancellation symbol” to mark it as a “trace” of movement, it would give rise to a countably infinite set of labels, since T_v can be an arbitrary syntactic object (since T itself can be arbitrary), and the set of all possible syntactic objects is countably infinite. However, the label $\mathcal{T}_{\overline{v}}$ in fact does not have to retain the full structure of T_v as a syntactic object (as a full nonplanar binary rooted tree). After all, the extracted term T_v is still present, on the other side of the coproduct, so that information is not lost. All that is needed is that $\mathcal{T}_{\overline{v}}$ remembers a feature of T_v sufficient for interpretation: namely, that it remembers the role that the moved element T_v played in its original position. We show here that this can be achieved through a labeling algorithm that uses a head function (in the sense of [22], that we recall below) and the structure of T_v to assign a label to the non-leaf vertices, including the root v .

To begin with, we make a refined choice of the initial set $\mathcal{SO}_0$ of lexical items and syntactic features. This is not needed in the formulation we gave in the previous sections, as it is only relevant when the coproduct Δ^c is involved.

Definition 10.1. We replace $\mathcal{SO}_0$ with a set of the form $\mathcal{SO}_0 \sqcup \neg\mathcal{SO}_0$, where $\neg\mathcal{SO}_0$ stands for the set of symbols of the form

$$(10.1) \quad \neg\mathcal{SO}_0 := \{\neg\alpha \mid \alpha \in \mathcal{SO}_0\}.$$

We refer to elements of $\neg\mathcal{SO}_0$ as “trace labels”.

We also recall from §1.13 of [22] the definition of a *head function*.

Definition 10.2. A head function on a full non-planar binary rooted tree T is a function $h_T : V^o(T) \rightarrow L(T)$, from the set of non-leaf vertices of T to the set of leaves, with the property that $h_T(v) \in L(T_v)$ and that if $T_w \subset T_v$ and $h_T(v) \in L(T_w)$ then $h_T(v) = h_T(w)$.

We can now define more precisely the meaning of the labels $\mathcal{T}_{\overline{v}}$ as follows.

Definition 10.3. Let $T \in \mathfrak{T}_{\mathcal{SO}_0}$ be a syntactic object endowed with a head function h_T , and let $T_v \subset T$ be an accessible term. If $T_v = \ell$ is a single leaf, decorated by an element $\alpha_\ell \in \mathcal{SO}_0$, then we simply define $\mathcal{T}_{\overline{v}}$ as the label $\alpha_\ell \in \neg\mathcal{SO}_0$. If T_v is a subtree containing more than a single leaf, the label $\mathcal{T}_{\overline{v}}$ is defined as the element $\alpha_{h_T(v)} \in \mathcal{SO}_0$, where $\ell = h_T(v)$ is the value of the head function h_T at the vertex $v \in V^o(T)$ and $\alpha_{h(T_v)}$ is the labeling of this leaf in $\mathcal{SO}_0$.

We need to verify that this definition of the labeling $\mathcal{T}_{\overline{v}}$ is compatible with the coassociativity of the coproduct Δ^c .

Lemma 10.4. *The coproduct Δ^c with the leaves-labels $\mathbb{T}_{v_i}$ of the contracted accessible terms T_{v_i} given by $\alpha_{h(T_{v_i})}$ with $h = h_T$ a head function is coassociative.*

Proof. As shown in [22] a head function partitions the vertices of T into disjoint paths γ_ℓ for $\ell \in L(T)$, where for all γ_ℓ consisting of more than the single vertex ℓ , the other end-vertex v_ℓ is called the “maximal projection” of ℓ . For each vertex v in γ_ℓ one has $h_T(v) = \ell$. The leaf ℓ with the property that its maximal projection is the root of T is called the head of T , and denoted by $\ell = h(T)$. The partitioning of vertices into paths γ_ℓ determines a labeling of all vertices of T , where vertices on the path γ_ℓ acquire the same label α_ℓ originally assigned to the leaf ℓ . Thus, the labeling $\alpha_{h(T_v)}$ assigns to the root vertex v of T_v the label $\alpha_{h_T(v)}$ with the additional cancellation symbol $\alpha_{\overline{h_T(v)}}$ that indicates the new leaf v is a “trace” of movement. What we need to check is that this assignment of labels maintains the coassociativity of the coproduct Δ^c . The coassociativity $(\Delta^c \otimes \text{id}) \circ \Delta^c = (\text{id} \otimes \Delta^c) \circ \Delta^c$ is an identity involving two successive admissible cuts C, C' , where the first cut C corresponds to the first application of the coproduct Δ^c and the second cut C' can take place in either the left or the right side ($\Delta^c \otimes \text{id}$ or $\text{id} \otimes \Delta^c$) of the output of the first Δ^c . Without leaf-labels, the coassociativity of this form of coproduct is well known, so the issue is only whether the labeling is consistent on the two sides of the coassociativity identity. Since the coproduct is multiplicative, we only need to check this on a tree T , rather than on an arbitrary forest F . For an admissible cut C we use the notation $\pi_C(T) = F_v$ for the forest extracted by the cut, which is in the left side of the corresponding terms of the coproduct, and we write $\rho_C^c(T) = T / {}^c F_v$ for the quotient term on the right side, so the term of the coproduct determined by the cut C is $\pi_C(T) \otimes \rho_C^c(T) = F_v \otimes T / {}^c F_v$, and the full coproduct is $\Delta^c(T) = \sum_C \pi_C(T) \otimes \rho_C^c(T)$, including the primitive part $T \otimes 1 + 1 \otimes T$ in the form of the two trivial cuts (above the root, below the leaves). The coassociativity relation compares terms of the form

$$\pi_{C'}(\pi_C(T)) \otimes \rho_{C'}^c(\pi_C(T)) \otimes \rho_C^c(T)$$

in $(\Delta^c \otimes \text{id})(T)$ with terms of the form

$$\pi_{C'}(T) \otimes \pi_C(\rho_{C'}^c(T)) \otimes \rho_C^c(\rho_{C'}^c(T))$$

in $(\text{id} \otimes \Delta^c)(T)$, where in this second term C' is the first cut and C the second. The terms $\pi_{C'}(\pi_C(T)) = \pi_{C'}(T)$ (since the cut C' is placed below C) do not involve any trace labeling so nothing needs to be checked. The middle term $\rho_{C'}^c(\pi_C(T)) = \pi_C(\rho_{C'}^c(T))$ has trace labels at the vertices w_j that are the roots of the trees in $\pi_{C'}(T) = \sqcup_j T_{w_j}$. These are labelled as $\alpha_{\overline{h_T(w_j)}}$ in $\rho_{C'}^c(T)$, and maintain the same label in $\pi_C(\rho_{C'}^c(T))$, while they are labelled as $\alpha_{\overline{h_{T_{v_i}}(w_j)}}$ in $\rho_C^c(\pi_C(T))$, where $\pi_C(T) = \sqcup_i T_{v_i}$. But the properties of the head function ensure that $h_{T_{v_i}}(w_j) = h_T(w_j)$ whenever $T_{w_j} \subset T_{v_i} \subset T$, so the labels match. Similarly, in the case of the terms $\rho_C^c(T) = \rho_C^c(\rho_{C'}^c(T))$, the trace labels are at the vertices v_i that were the root vertices of the T_{v_i} with $\pi_C(T) = \sqcup_i T_{v_i}$. In both $\rho_C^c(T)$ and $\rho_C^c(\rho_{C'}^c(T))$ these are labelled by these are labelled by $\alpha_{\overline{h_T(v_i)}}$ so again there is no labeling mismatch. $\square$

This labeling method for the traces of movement require that all the components T_i of all the workspaces $F = \sqcup_i T_i$ have an assigned head function h_{T_i} . On a tree T_i with $k_i = \#L(T_i)$ leaves there are 2^{k_i-1} possible choices of a head function. Here we simply assume that one such choice is made. More realistically, one wants a head function that correctly models the syntactic head and the structure of the Extended Projection, as discussed in [24]. We will return to discuss this in §15.

10.2. Merge operations with the contraction coproduct. It is natural then to ask what happens to the Merge dynamics $\mathcal{K} = \sum_{S, S'} \mathfrak{M}_{S, S'}^c$, acting on workspaces in $\mathfrak{F}_{\mathcal{SO}_0 \sqcup \mathcal{SO}_\infty}$, if we define

$\mathfrak{M}_{S,S'}^c$ as

$$(10.2) \quad \mathfrak{M}_{S,S'}^c = \sqcup \circ (\mathcal{B} \otimes \text{id}) \circ \delta_{S,S'}^c \circ \Delta^c$$

instead of the Merge operators considered in the previous sections, with

$$\mathfrak{M}_{S,S'} = \sqcup \circ (\mathcal{B} \otimes \text{id}) \circ \delta_{S,S'} \circ \Delta^d.$$

It is important to point out that, in passing from the coproduct Δ^d to Δ^c , one is also modifying the selection $\delta_{S,S'}^c$ of which terms of the coproduct are used by Merge. (This is reflected in [22] in the counting of the effect of Merge on the number of accessible terms, though it is not spelled out explicitly, so we make it more explicit here.) In linguistics, the trace of previous movement is no longer extracted by successive applications of Merge in a derivation for further movement: in other words, traces no longer move. To reflect this fact, the selection $\delta_{S,S'}^c$ of accessible terms (in the case of the coproduct Δ^d), that is defined in [22] simply as $\delta_{S,S'} = \gamma_{S,S'} \otimes \text{id}$ where

$$\gamma_{S,S'}(F) = \begin{cases} F & F = S \sqcup S' \\ 0 & \text{otherwise,} \end{cases}$$

can be defined in the slightly modified form

$$\delta_{S,S'}^c = \gamma_{S,S'}^c \otimes \tilde{\gamma}_{S,S'}^c,$$

where

$$(10.3) \quad \gamma_{S,S'}^c(F) = \begin{cases} F & F = S \sqcup S' \text{ and } S, S' \notin \mathfrak{T}_{-\mathcal{SO}_0-} \\ 0 & \text{otherwise,} \end{cases}$$

and, for $F = T_1 \sqcup \dots \sqcup T_r$,

$$(10.4) \quad \tilde{\gamma}_{S,S'}^c(F) = \begin{cases} F & T_i \notin \mathfrak{T}_{-\mathcal{SO}_0-}, \forall i = 1, \dots, r \\ 0 & \text{otherwise.} \end{cases}$$

Note that $\gamma_{S,S'}^c$ does not prevent the extraction of any term $S = T_w$, where $T_w \in \mathfrak{T}_{\mathcal{SO}_0 \sqcup \mathcal{SO}_0-}$ has some of the leaves decorated by elements in $-\mathcal{SO}_0-$: it only prevents extraction of accessible terms that have *all* leaves labelled by trace labels in $-\mathcal{SO}_0-$.

The operator $\tilde{\gamma}_{S,S'}^c$ correspondingly prevents the creation of components in the right-hand-side of the coproduct that have only trace-leaves. The reason for including here also the constraint $\tilde{\gamma}_{S,S'}^c$ is in order to better relate the action of Merge with the two forms of the coproduct. For example, an Internal Merge transformation involving a term $T/^cT_v \in \mathcal{T}_{-\mathcal{SO}_0-}$ would have no analog in terms of the coproduct Δ^d , hence in linguistic terms it would not be viable at Externalization.

The presence of both constraints $\gamma_{S,S'}^c$ and $\tilde{\gamma}_{S,S'}^c$ ensures that all the Merge transformation that can take place in $\mathcal{K}^c = \sum_{S,S'} \mathfrak{M}_{S,S'}^c$ also have an associated form in $\mathcal{K} = \sum_{S,S'} \mathfrak{M}_{S,S'}$, expressing the idea that the same mechanism of structure formation always simultaneously maps to the Externalization interface (where the traces are not expressed) and to the Conceptual-Intentional (also called Syntax-Semantics) interface where the traces are present for interpretation.

We address in the rest of this section the behavior of the dynamics of Merge in the form based on the coproduct Δ^c .

10.3. Edges of the Merge graph with the contraction coproduct. The arrows in the graph of the dynamical system obtained using the Merge operators as defined as in (10.2) again correspond to External Merge, Internal Merge, and Sideward Merge operations.

- (1) The External Merge arrows are exactly the same considered in the previous sections: they remain unchanged with respect to the choice of the Δ^d or Δ^c coproduct, since they only

involve the primitive part of the coproduct that partitions the workspace into components, without extracting accessible terms. Since in this case no quotient is involved, both coproducts have the same primitive part.

- (2) The Internal Merge arrows correspond to compositions $\mathfrak{M}_{T_v, T/^c T_v}^c \circ \mathfrak{M}_{T_v, 1}$, hence they differ from the case with the coproduct Δ^d , due to the presence of the additional leaf with label $\mathcal{T}_{\bar{v}}$ in $T/^c T_v$
- (3) The Sideward Merge arrows also similarly differ from the case of the coproduct Δ^d . Since again we are only considering the minimal SM arrows that extract only leaves, we have cases
 - $\mathfrak{M}(\beta, \alpha) \sqcup T/^c \alpha$ (when β and T are components of the workspace SM acts on), where the resulting component $T/^c \alpha$ now still have the leaf α of T but with label changed to α ,
 - $\mathfrak{M}(\beta, \alpha) \sqcup T/^c(\alpha \sqcup \beta)$, where the quotient $T/^c(\alpha \sqcup \beta)$ has the same leaves as T , but those that were respectively labelled α and β are now relabelled α and β ,
 - $\mathfrak{M}(\beta, \alpha) \sqcup T/^c \alpha \sqcup T' /^c \beta$ where similarly the resulting components $T/^c \alpha$ and $T' /^c \beta$ maintain the same number of leaves as T and T' , with α and β labels changed to α and β , respectively.

In all cases involving a quotient $T/^c T_v$ (or $T/^c F_v$), one should interpret the labels $\mathcal{T}_{\bar{v}}$ as in Definition 10.3.

Definition 10.5. We define the contraction Merge graph $\mathcal{G}_{\mathfrak{F}_{SO_0 \sqcup SO_0^-}}$ as the infinite, locally finite graph with countably infinite set of vertices $\mathfrak{F}_{SO_0 \sqcup SO_0^-}$ and with finitely many incoming and outgoing edges at each vertex given by all the possible EM, IM, SM transformations listed above.

10.4. Transient dynamics and projection. We show here that looking at the Merge dynamics in terms of the coproduct Δ^c does not provide any useful information, since the dynamics now involves a countably infinite set of states (it is no longer a finite Markov chain) where all states become transient, so in particular there is no stationary distribution that describes the long term behavior of the dynamics. However, we also show that the Merge dynamics in the Δ^d case provides complete information to also determine the dynamics of the Δ^c case. The key reason for this lies in the fact that, in linguistics, traces of previous movement are no longer extracted for further movement.

Proposition 10.6. Let $\mathcal{K}^c = \sum_{S, S'} \mathfrak{M}_{S, S'}^c$ be the action of Merge on the linear span of workspaces $\mathcal{V}(\mathfrak{F}_{SO_0 \sqcup SO_0^-})$, with $\mathfrak{M}_{S, S'}^c$ as in (10.2). This action does not preserve the subspaces spanned by the finite sets $\mathfrak{F}_{A, n}$, and all $F \in \mathfrak{F}_{SO_0 \sqcup SO_0^-}$ with $F \notin \mathfrak{F}_{SO_0^-}$ are transient states for the dynamics defined by $\mathcal{K}^c$.

Proof. As discussed in §10.3, the EM arrows in $\mathcal{G}_{\mathfrak{F}_{SO_0}}$ are the same as in $\sqcup_n \mathcal{G}_{n, A}$ so they do preserve the $\mathfrak{F}_{A, n} = V(\mathcal{G}_{n, A})$. However, this is not the case for IM and SM arrows. Indeed, in the case of IM arrows, if $F = T \sqcup \hat{F} \in \mathfrak{F}_{A, n}$, its image under an IM transformation that uses the form Δ^c of the coproduct and acts on the T component is a new workspace

$$F' = \mathfrak{M}(T_v, T/^c T_v) \sqcup \hat{F} \in \mathfrak{F}_{B, n+1}, \quad \text{with} \quad B = A \cup \{\mathcal{T}_{\bar{v}}\}$$

and with the label $\mathcal{T}_{\bar{v}}$ as in Definition 10.3. Similarly, in the case of minimal SM, as illustrated in §10.3, we can obtain from $F \in \mathfrak{F}_{A, n}$ a new workspace

$$F' = \begin{cases} \mathfrak{M}(\alpha, \beta) \sqcup T/^c \alpha \sqcup \hat{F} \in \mathfrak{F}_{A \cup \{\alpha\}, n+1} & \text{for } F = \beta \sqcup T \sqcup \hat{F} \\ \mathfrak{M}(\alpha, \beta) \sqcup T/^c(\alpha \sqcup \beta) \sqcup \hat{F} \in \mathfrak{F}_{A \cup \{\alpha, \beta\}, n+2} & \text{for } F = T \sqcup \hat{F} \\ \mathfrak{M}(\alpha, \beta) \sqcup T/^c \alpha \sqcup T' /^c \beta \sqcup \hat{F} \in \mathfrak{F}_{A \cup \{\alpha, \beta\}, n+2} & F = T \sqcup T' \sqcup \hat{F}. \end{cases}$$

The resulting graph $\mathcal{G}_{\mathfrak{F}_{SO_0} \sqcup \mathfrak{S}\Theta_0^-}$ cannot be strongly connected, as the IM and SM arrows always map any subset of vertices $\mathfrak{F}_{A,n}$ to a subset of vertices $\mathfrak{F}_{B,n+1}$ of $\mathfrak{F}_{B,n+2}$ while the EM arrows map $\mathfrak{F}_{A,n}$ to $\mathfrak{F}_{A,n}$, while there are no directed arrows that can decrease the number of leaves. Since every $F \in \mathfrak{F}_{A,n}$ has some outgoing IM or SM arrows, this implies that every $F \in \mathfrak{F}_{A,n}$ is a transient state for the dynamical system $\mathcal{K}^c$. $\square$

This shows that, when using the coproduct Δ^c instead of Δ^d , there is no analog here of the Hopf algebra Markov chain $\hat{\mathcal{K}}$ that we studied in the previous sections and its stationary distribution that governs the long term dynamics and provides information on the dynamical behavior.

On the other hand, this does not mean that we do not have any viable information about how $\mathcal{K}^c$ behaves. In fact, the following result shows that the dynamics of $\mathcal{K}^c$, acting on workspaces $F \in \mathfrak{F}_{SO_0}$ is completely determined by its image under the projection $\Pi_{d,c}$ on $\mathcal{V}(\mathfrak{T}_{SO_0})$ (as in §1.3 of [22]).

Definition 10.7. *Let $\Pi_{d,c} : \mathcal{V}(\mathfrak{F}_{SO_0} \sqcup \mathfrak{S}\Theta_0^-) \rightarrow \mathcal{V}(\mathfrak{F}_{SO_0})$ be the projection that cuts from each component T of F all the trace-labelled leaves and then contracts edges to obtain the maximal full binary rooted tree determined by what remains.*

The iterates of the dynamical systems $\mathcal{K} = \sum_{S,S'} \mathfrak{M}_{S,S'}$ and $\mathcal{K}^c = \sum_{S,S'} \mathfrak{M}_{S,S'}^c$, based on the two forms of the coproduct, Δ^d and Δ^c , satisfy the following relation.

Proposition 10.8. *For $F \in \mathfrak{F}_{A,n} \subset \mathfrak{F}_{SO_0}$, the orbit $(\mathcal{K}^c)^m(F)$ under iterations of $\mathcal{K}^c$ is contained in the fiber $\Pi_{d,c}^{-1}(\mathcal{K}^m(F)) \subset \mathcal{V}(\mathfrak{F}_{SO_0} \sqcup \mathfrak{S}\Theta_0^-)$ and is completely determined by the orbit of F under $\mathcal{K}$.*

Proof. We first show that, for $F \in \mathfrak{F}_{SO_0}$, we have

$$(10.5) \quad \Pi_{d,c} \circ \mathcal{K}^c = \mathcal{K}.$$

For a forest $F \in \mathfrak{F}_{SO_0}$, we have

$$(10.6) \quad (\text{id} \otimes \Pi_{d,c}) \circ \Delta^c(F) = \Delta^d(F),$$

or equivalently

$$\pi_C(T) \otimes \Pi_{d,c}(\rho_C^c(T)) = \pi_C(T) \otimes \rho_C(T).$$

Note that, in this case, since F has no trace labels, none of the components of the terms $F_v = \pi_C(F)$ extracted from $F \in \mathfrak{F}_{A,n}$ can be in $\mathfrak{T}_{\mathfrak{S}\Theta_0^-}$. Thus, we have $\delta_{S,S'}^c \Delta^c(T) = \delta_{S,S'} \Delta^d(F)$ and we obtain in this case

$$\begin{aligned} \mathfrak{M}_{S,S'}(F) &= \sqcup \circ (\mathcal{B} \otimes \text{id}) \circ (\gamma_{S,S'} \otimes \Pi_{d,c}) \Delta^c(F) = \sqcup \circ (\text{id} \otimes \Pi_{d,c}) \circ (\mathcal{B} \otimes \text{id}) \circ \delta_{S,S'}^c \Delta^c(F) = \\ &= \sqcup \circ (\Pi_{d,c} \otimes \Pi_{d,c}) \circ (\mathcal{B} \otimes \text{id}) \circ \delta_{S,S'}^c \circ \Delta^c(F) = \Pi_{d,c} \circ \mathfrak{M}_{S,S'}^c(F) \end{aligned}$$

so that we get (10.5). We now proceed inductively. Suppose that, starting with an $F \in \mathfrak{F}_{SO_0}$ and applying the m -th iterate of the action of $\mathcal{K}^c$ we obtain that $\Pi_{d,c} \circ (\mathcal{K}^c)^m(F) = \mathcal{K}^m(F)$. Let $\tilde{F}_i$ be any of the components of $(\mathcal{K}^c)^m(F) = \sum_i c_i \tilde{F}_i \in \mathcal{V}(\mathfrak{F}_{SO_0} \sqcup \mathfrak{S}\Theta_0^-)$. When we take $\mathcal{K}^c(\tilde{F}_i) = \sum_j c_{ij} \tilde{F}_{ij}$ each component $\tilde{F}_{ij}$ has trace-leaves that are either already trace-leaves of $\tilde{F}_i$ or that are added by the IM and SM arrows of $\mathcal{K}^c$, so that we have

$$\Pi_{d,c}(\mathcal{K}^c(\tilde{F}_i)) = \Pi_{d,c}(\mathcal{K}^c(\Pi_{d,c}(\tilde{F}_i))).$$

Using (10.5) we then see that

$$\Pi_{d,c}(\mathcal{K}^c(\Pi_{d,c}(\tilde{F}_i))) = \mathcal{K}(\Pi_{d,c}(\tilde{F}_i))$$

hence, by linearity

$$\Pi_{d,c}((\mathcal{K}^c)^{m+1}(F)) = \Pi_{d,c}(\mathcal{K}^c(\Pi_{d,c}((\mathcal{K}^c)^m(F)))) = \mathcal{K}(\Pi_{d,c}((\mathcal{K}^c)^m(F)))$$

and by induction hypothesis we then get

$$(10.7) \quad \Pi_{d,c}((\mathcal{K}^c)^{m+1}(F)) = \mathcal{K}^{m+1}(F).$$

This implies that the orbit $(\mathcal{K}^c)^m(F)$ of $F \in \mathfrak{F}_{A,n}$ is contained in $\Pi_{d,c}^{-1}(\mathcal{K}^m(F))$. One can also see that the orbit $(\mathcal{K}^c)^m(F)$ is completely determined by the orbit $\mathcal{K}^m(F)$, for $F \in \mathfrak{F}_{\mathcal{SO}_0}$, because at each application of $\mathcal{K}^c$ the only additional information that is needed to determine the image is the trace-leaves created by the coproduct Δ^c . These are determined by knowing the vertices v where the accessible terms T_v are extracted and the value of the head function $h_T(v)$ in the component $T \supset T_v$. This can also be shown inductively as above. At the first step, the extracted T_v are the same for $\mathcal{K}^c$ and $\mathcal{K}$ since the orbit starts at a point is $F \in \mathfrak{F}_{\mathcal{SO}_0}$. Thus, comparing F with the components of $\mathcal{K}(F)$ identifies the terms T_v involved in each arrow. We are assuming here that all the components of the workspaces involved have a head function (we will return to discuss this in §15). So for $T_v \subset T$, by the properties of the head function this satisfied $h_T(v) = h_{T_v}(v) = h_{T'}(v)$ where T' is the component of $\mathcal{K}(F)$ that contains the vertex v , so this is computable from $\mathcal{K}(F)$. Inductively, assume that in $(\mathcal{K}^c)^m(F)$ the trace-leaves and their labels computed through the head function are all determined from data computable from the orbit $\{\mathcal{K}^k(F)\}_{k \leq m}$. The next iterate $\mathcal{K}^c((\mathcal{K}^c)^m(F))$ does not change the previous trace-leaves and labels but adds new ones through the IM and SM arrows in $\mathcal{K}^c$. Again these depend on the extracted accessible terms and the value of the head function at their root vertices. These, as in the case of a single iteration only depend on data in $(\mathcal{K}^c)^m(F)$ that are determined by the orbit $\{\mathcal{K}^k(F)\}_{k \leq m}$. $\square$

Thus, the conclusion is that the Merge dynamics is fully described by the Hopf algebra Markov chains $\hat{\mathcal{K}}^{A,n}$ that we studied in the previous sections and the change of the coproduct from Δ^d to Δ^c does not provide any additional information.

In terms of the linguistic model, the result of Proposition 10.8 can be interpreted in the following way. The dynamics of $\mathcal{K}^c$ determines (via the projection $\Pi_{d,c}$) and is in turn determined by the dynamics of $\mathcal{K}$ (via the argument given in Proposition 10.8). This means that, while the presence of traces is considered necessary for semantic parsing, it is not necessary to the structure building dynamics itself, which can be fully determined without using trace-leaves. As we have argued, this is a consequence of the fact that, in linguistics, traces are no longer extracted for movement.

11. WEIGHTED MERGE DYNAMICS: COST FUNCTIONS

The conclusions we have drawn from the previous sections agree with what is stated in the linguistics literature, in the formulation of the action of Merge on workspaces of [7] and [8]. Namely, some natural cost function must weight the different forms of Merge differently, with EM and IM being optimal and SM being sub-optimal.

As proposed in [7] and [8], there are two main forms of optimality that Merge is expected to satisfy. One is what is called Minimal Search, which optimizes the cost of searching in the workspace for the accessible terms that Merge applies to. The other goes under the name of Resource Restrictions, and especially Minimal Yield and is formulated as a balance between the effect of a Merge transformation on the number of connected components of the workspace and on the number of accessible terms. With these two forms of optimality, one can include an additional one, which is referred to in [22] as “no complexity loss”, and which is usually bundled up, in the linguistics literature, together with the “Extension Condition”: it requires that structures are *grown* by Merge (with the second part of the Extension Condition also requiring that growth happens at the root, as discussed at length in [26]).

The Minimal Yield condition, in turn, has slightly different formulations in [22] (and in [26]) from the original formulation of [7] and [8]. We will recall briefly these cost functions here, in

§11.1, following the formulation as in [26]. Using the original form of the cost function does not affect the main results stated here, so we adopt the formulation of [22] and [26], since it is more immediately transparent in terms of the tree topology.

We then describe in §12 a technique for estimating the leading order in the weight parameter of the Perron–Frobenius eigenvalue and eigenvector of the cost-weighted Merge dynamics, and of the stationary distribution of the associated Hopf algebra Markov chain.

The algebra of orders of magnitude is governed by the min-plus (also known as tropical) semiring. Thus, the estimation of the leading order of the Perron-Frobenius data for the cost-weighted Merge is phrased in terms of a Perron-Frobenius problem in the tropical semiring.

A general approach to such tropical Perron-Frobenius problems reduces evaluating the order of the Perron-Frobenius eigenvalue to identifying critical circuits (cost-minimal circuits) in the Merge graph, and the construction of a basis of the Perron-Frobenius eigenspace to the construction of optimal arborescences for the Merge graph. We discuss the details of this approach and the result in §12.

The conclusion of this analysis is that, due to the non-uniqueness of critical circuits, we do not identify a unique Perron-Frobenius eigenvector, but a basis for the eigenspace. The form of this basis shows that, to leading order, the weighted dynamics moves away from the more disconnected structures that dominate in the unweighted one as an effect of SM. On the other hand, instead of convergence to the tree structures (i.e. the partition $\wp = \{n\}$), as observed in [26] in the weighted dynamics for $n = 3$, one finds that any of the partitions $\wp \in \mathcal{P}'(n)$ that have at least some $k_i \geq 3$ (hence that have a non-trivial IM self-loop) contribute to the Perron-Frobenius eigenspace.

This shows that, despite what the simpler case $n = 3$ described in [26] suggests, the total cost function that combines Minimal Search (in a bottom-up formulation as in [26]), Minimal Yield (as formulated in [22]) and Complexity-Loss, does not suffice to fully eliminate the unwanted effects of Sideward Merge on the dynamics (while maintaining the desirable strong connectedness properties) and ensure convergence to the tree structure.

We describe how to correct this problem by modifying the cost function and introducing in it a term that depends on the Shannon entropy of a distribution $\mathbb{P}_\wp$ associated to the partition $\wp \in \mathcal{P}'(n)$, which simply describes the leading order of the combinatorial multinomial coefficient describing the partitioning of the n leaves into the different tree components of the workspace. We show that the presence of this additional quantity $\text{Sh}(\mathbb{P}_\wp)$ in the cost function suffices to drive the dynamics toward the connected workspaces (consisting of a single tree). Namely, we show that the stationary distribution of the Merge Markov chain weighted by $t^\mathfrak{c}$ for this modified cost function $\mathfrak{c}$ converges, in the limit $t \rightarrow 0$ to the uniform distribution on the trees. In other words, the $\hat{\mathcal{K}}^{(A,n)}$ dynamics on $\mathcal{G}_{n,A}$, weighted by this modified cost function, converges to the IM dynamics on $\mathcal{G}_{n,A,\{n\}}^{\text{IM}}$. The leading order in t of all the components of the stationary distribution can be expressed explicitly in terms of costs of certain loops of edges in the graph $\mathcal{G}_{n,A}$ that lie along optimal arborescences in $\mathcal{G}_{n,A}$.

In fact we show an even stronger fact: that we can use *only* the cost function given by the Shannon entropy $\text{Sh}(\mathbb{P}_\wp)$ and obtain the same result, without including the Minimal Search, Minimal Yield, and Complexity-Loss cost functions at all. These cost functions, usually considered in linguistics, then play a role only in the initial selection of only the “minimal” Sideward Merge arrows that involve only atomic components, while the influence on the dynamics is entirely coming from the information-theoretic cost function.

11.1. Combined cost functions. We consider here the three main types of cost function as discussed in [26], namely a bottom-up version of Minimal Search, and two types of Resource Restriction costs: Minimal Yield and Complexity Loss.

11.1.1. Minimal Search costs. As in [26], we consider a “bottom-up” version of the Minimal Search cost (rather than the “top-down”) version used in [22]. In this approach to Minimal Search we assume that, instead of the usual description of search algorithms on rooted trees that start at the root and descend into the tree, the search on syntactic structures, which are built in a bottom up way, starts with the lexical items at the leaves and searches for subsets of leaves that form a constituency, namely that determine an accessible term that can be extracted for use by the Merge operations. Viewed in this way, the cost of the search depends on the (relative) size of the set of leaves that needs to be identified. This leads to a definition of cost function where

$$\mathbf{c}_{MS}(T_v) = \frac{\ell(T_v)}{\ell(T)},$$

where $\ell(T) = \#L(T)$ is the number of leaves (the grading in the Hopf algebra of workspaces). This gives a cost to the extraction of an accessible term. The cost of the corresponding quotient operation T/T_v is then set to be

$$\mathbf{c}_{MS}(T/T_v) = \frac{\ell(T/T_v)}{\ell(T)} = 1 - \frac{\ell(T_v)}{\ell(T)} = 1 - \mathbf{c}_{MS}(T_v).$$

As in [22] we also assign a resulting cost to the merging of two components or accessible terms A, B of the form

$$\mathbf{c}_{MS}(\mathfrak{M}(A, B)) = \mathfrak{b}(A, B) - \mathbf{c}(A) - \mathbf{c}(B),$$

where $\mathfrak{b}(A, B) = 1$ if A and B come from the same component of the workspace and $\mathfrak{b}(A, B) = 2$ if they come from different components. This choice encodes the idea that Minimal Search would first (lower cost) search for terms within a fixed components and then (higher cost) for terms across different components. With this cost function we see that we have for the different cases of Merge, EM $\mathfrak{M}(T, T')$, IM $\mathfrak{M}(T_v, T/T_v)$, SM(1) $\mathfrak{M}(T_v, T') \sqcup T/T_v$, SM(2) $\mathfrak{M}(T_w, T'_w) \sqcup T/T_v \sqcup T'/T'_w$, and SM(3) $\mathfrak{M}(T_v, T_w) \sqcup T/(T_v \sqcup T_w)$ the respective costs

	EM	IM	SM(1)	SM(2)	SM(3)
$\mathbf{c}_{MS}$	0	0	$2 - \mathbf{c}_{MS}(T_v)$	$2 - \mathbf{c}_{MS}(T_v) - \mathbf{c}_{MS}(T'_w)$	$1 - \mathbf{c}_{MS}(T_v) - \mathbf{c}_{MS}(T_w)$

In particular, in the case of a tree T with $k = \#L(T)$, if we only consider the minimal SM transformations, where the terms T_v, T_w, T'_w, T' involved in the operations of extracting and Merging are all atomic components consisting of a single leaf, we obtain the costs

	EM	IM	min SM(1)	min SM(2)	min SM(3)
$\mathbf{c}_{MS}$	0	0	$2 - 1/k_i$	$2 - 1/k_i - 1/k_j$	$1 - 2/k_i$

This means that, on a workspace F with $n = \#L(F)$ partitioned across components $F = T_1 \sqcup \dots \sqcup T_r$ with $n = k_1 + \dots + k_r$, $k_i = \#T_i$, an SM transformation that extracts from a component T_i (for SM(1) and SM(3)) or to two components T_i and T_j (for SM(2)), has cost, $2 - 1/k_i$, $2 - 1/k_i - 1/k_j$ and $1 - 2/k_i$, for SM(1), SM(2), and SM(3), respectively.

11.1.2. Minimal Yield costs. For the computation of Minimal Yield costs we proceed as in [22] and [26] and we evaluate the effect of the different Merge operations (EM, IM, SM(1), SM(2), SM(3)) on the number $b_0(F)$ of connected components and the number $\alpha(F)$ of accessible terms of the workspace, evaluating the change in the total number $\sigma(F) = b_0(F) + \alpha(F)$. We consider here only the case of the coproduct $\Delta = \Delta^d$ as we have argued that this suffices for studying the properties of Merge as a dynamical system. We count for each type of Merge operation the

difference between the number of components of the resulting workspace and of the initial one. This is equal to -1 for EM (which reduces the number of connected components), 0 for IM and for SM(1), and $+1$ for SM(2) and SM(3) that increase by one the components. We also count the resulting change on the number of accessible terms: EM increases it by 2, IM and SM(1) leave it unchanged and SM(2) and SM(3) reduce it, changing it by -2 (each quotient T/dT_v involves removal of one vertex to obtain a full binary tree). Thus the total count of the change in the sum of number of components and accessible terms gives a total Minimal Yield counting of

	EM	IM	SM(1)	SM(2)	SM(3)
$\mathbf{c}_{MY}$	1	0	0	-1	-1

This counting is independent of whether SM is minimal (extracting and merging only atomic components) or not.

11.1.3. Complexity Loss cost. Complexity loss is another possible measure of costs, discussed in [22]. It encodes one of the properties of the linguistic model. This is sometime grouped together with the “Extension Condition” in the linguistic literature, but it is better treated independently, because, as argued in [26], the Extension Condition is really an “algebraic” constraint (hard constraint) of the model, while this is an optimality condition (soft constraint). One can formulate it as the requirement that “syntactic composition always grows or expands structures”. More precisely, we can say that there is no complexity loss if in the transformed workspace, after the action of a Merge operation, each component of the original workspace ends up as part of a component that is at least as complex as the original one. A simple way to measure the complexity of a component of the workspace is by its size, measured by its degree in the Hopf algebra of workspaces (the number of leaves). Thus the complexity loss is measured as the largest drop in degree over the components of the workspace. Thus, for an EM of the form $\mathfrak{M}(T, T')$ the components T, T' end up as parts of a component $\mathfrak{M}(T, T')$ of largest degree (and any other components are unchanged), so there is no loss. For an IM of the form $\mathfrak{M}(T_v, T/T_v)$ the original component T and the resulting $\mathfrak{M}(T_v, T/T_v)$ have the same degree and the other components are unchanged, so also there is no complexity loss. For an SM(1) of the form $\mathfrak{M}(T', T_v)$, the component T' ends up in a larger component (so it does not incur any complexity loss) but the component T from which T_v is extracted results in a component T/T_v of lower degree, with a complexity loss of $\ell(T_v)$. For SM(2) there are two components T and T' that result, respectively, in components T/T_v and T'/T'_w for a loss of $\ell(T_v) + \ell(T'_w)$ and for SM(3) there is a component T resulting in $T/(T_v \sqcup T_w)$ for a loss of $\ell(T_v) + \ell(T_w)$. So the CL cost function is given by

	EM	IM	SM(1)	SM(2)	SM(3)
$\mathbf{c}_{CL}$	0	0	$\ell(T_v)$	$\ell(T_v) + \ell(T'_w)$	$\ell(T_v) + \ell(T_w)$

For the case of minimal SM operations involving only atomic (single leaf) elements, this gives

	EM	IM	min SM(1)	min SM(2)	min SM(3)
$\mathbf{c}_{CL}$	0	0	1	2	2

11.1.4. Combined cost. Thus, following [26], the costs according to these different cost functions for the Merge operations are summarized as follows. For Merge acting on components T_i , or T_i and T_j , of a forest with $n = k_1 + \dots + k_r$ leaves, one has weights $\mathbf{c}_{\text{tot}}$ given by:

	MS	MY	CL	total weight $\mathbf{c}_{\text{tot}}$
EM	0	1	0	1
IM	0	0	0	0
SM(1)	$2 - 1/k_i$	0	1	$3 - 1/k_i$
SM(2)	$2 - 1/k_i - 1/k_j$	-1	2	$3 - 1/k_i - 1/k_j$
SM(3)	$1 - 2/k_i$	-1	2	$2 - 2/k_i$

With this counting of the total cost function of Minimal Search, Minimal Yield, and Complexity Loss, we consider a weighted version of the Merge dynamics. We think of the unweighted Merge dynamics discussed so far as the value $t = 1$ of a deformed dynamics, dependent on a deformation parameter $0 < t \leq 1$, which recovers the unweighted case at $t = 1$ and, for $t \rightarrow 0$ has leading order given by the least costly part of the dynamics. This is achieved by weighting each component of the Merge graph adjacency matrix by a cost function $t^{\mathbf{c}_{\text{tot}}}$. We analyze in the next section the effect of this deformation on the properties of the resulting dynamical system.

12. TROPICAL PERRON-FROBENIUS AND EIGENVECTOR ESTIMATES

We have seen in Theorem 9.1 that the stationary distribution for the Merge Hopf algebra Markov chain is given by

$$\pi(F) = \frac{\eta_{\mathcal{P}'(n)}(\wp) \psi_{\mathcal{P}'(n)}(\wp)}{\Lambda_{\wp,n}} \quad \text{for } \wp = p(F)$$

where $\eta_{\mathcal{P}'(n)}$ and $\psi_{\mathcal{P}'(n)}$ are, respectively, the left and right Perron–Frobenius eigenvectors of the adjacency matrix of the graph $\mathcal{G}_{\mathcal{P}'(n)}$. In the case where we consider the weighting by the cost functions, the same result holds for the weighted Merge Hopf algebra Markov chain dynamics. Namely, we have the following.

Proposition 12.1. *Consider the directed Merge graph $\mathcal{G}_{n,A}$, where the arrows are weighted by a weight $t^{\mathbf{c}_{\text{tot}}}$, according to the total cost $\mathbf{c}_{\text{tot}}$ (combining MS, MY, CL) as computed in §11.1.4, and with $t \in \mathbb{R}_+^*$ a parameter. We write $\mathcal{K}^{A,n,w}(t)$ for the resulting non-negative matrix with these weights,*

$$(12.1) \quad \mathcal{K}^{A,n,w}(t)_{F,F'} = t^{\mathbf{c}_{\text{tot},F,F'}} \mathcal{K}_{F,F'}^{A,n}$$

seen as a function of the parameter t . Consider again the projection map $p : \mathcal{G}_{n,A} \rightarrow \mathcal{G}_{\mathcal{P}'(n)}$, where in the graph $\mathcal{G}_{\mathcal{P}'(n)}$ we consider the EM and SM arrows weighted by $t^{\mathbf{c}_{\text{tot}}}$, and the IM loops weighted by $t^{\mathbf{c}_{\text{tot}}} = 1$. We write $\mathcal{K}^{\mathcal{P}'(n),R,w}(t)$ for this resulting weighted adjacency matrix of $\mathcal{G}_{\mathcal{P}'(n)}$. Then the stationary distribution of the weighted Merge Hopf algebra Markov chain $\hat{\mathcal{K}}^{A,n,w}(t)$ determined by this weighted $\mathcal{K}^{A,n,w}(t)$ has stationary distribution

$$(12.2) \quad \pi^w(F, t) = \frac{\eta_{\mathcal{P}'(n),w}(\wp, t) \psi_{\mathcal{P}'(n),w}(\wp, t)}{\Lambda_{\wp,n}} \quad \text{for } \wp = p(F),$$

where $\eta_{\mathcal{P}'(n),w}$ and $\psi_{\mathcal{P}'(n),w}$ are the left and right Perron–Frobenius eigenfunctions of $\mathcal{K}^{\mathcal{P}'(n),R,w}(t)$.

Proof. The argument is as in Proposition 9.2, but taking into account the weights at the edges. All the IM edges in the $\mathcal{G}_{n,A}$ have $\mathbf{c}_{\text{tot}} = 0$, so all the edges in the fibers of the projection p do not carry any weight. We have the same numerical factors $\Lambda_{\wp,n} = \#V_{n,A,\wp}$ in the fiber multiplicities at the vertices. $\square$

Note that the form (12.1) is analogous to the weighted adjacency matrices that we discussed in §3.4, after setting $t = e^{-\beta}$ for the thermodynamic parameter of the Boltzmann distribution.

The form (12.2) of the stationary distribution reduces, as in the unweighted case, the problem to the computation of the left and right Perron-Frobenius eigenfunctions of the matrix $\hat{\mathcal{K}}^{\mathcal{P}'(n),R,w}(t)$ for the weighted graph $\mathcal{G}_{\mathcal{P}'(n)}$.

While the graph $\mathcal{G}_{\mathcal{P}'(n)}$ is much smaller than the Merge graph $\mathcal{G}_{n,A}$, it is still computationally difficult to obtain a closed expression (as a function of the parameter t) for the Perron-Frobenius eigenvalue and eigenvector, as done in the simple case of $n = 3$ in [26]. This means that, in the general case of arbitrary $n \geq 4$, one needs to obtain an estimate of the leading order behavior in t for the Perron-Frobenius eigenvalue and left/right eigenvector.

There are general methods (see for instance [1]) for evaluating the leading order of the Perron-Frobenius eigenvalue and eigenvector of a non-negative matrix that depends on a real parameter, which is based on solving an associated Perron-Frobenius problem in a semiring that provides the appropriate “algebra of orders of magnitudes”.

We follow the setting of [1], adapted to our case. In particular, the large real parameter $p > 0$ of [1] will be here the parameter $-\log t$, for $0 < t < 1$. Our parameter t is designed to give back the unweighted case discussed in the previous sections when $t = 1$ and give the limiting form of the dynamics, which we will be discussing in this section, when $t \rightarrow 0$, as in the $n = 3$ example analyzed in [26]. Thus, our $t \rightarrow 0$ limit will indeed correspond to the $p = -\log t \rightarrow \infty$ limit of [1]. The matrix denoted by $\mathcal{A}_p$ in [1] will be our t -dependent matrix $\mathcal{K}^{\mathcal{P}'(n),R,w}(t)$, which has, by construction, the following form.

Lemma 12.2. *The entries of the matrix $\mathcal{K}^{\mathcal{P}'(n),R,w}(t)$ are of the form*

$$(12.3) \quad \mathcal{K}_{\wp,\wp'}^{\mathcal{P}'(n),R,w}(t) = \mathcal{K}_{\wp,\wp'}^{\mathcal{P}'(n),R} t^{\mathbf{c}_{\wp,\wp'}} ,$$

where $\mathcal{K}_{\wp,\wp'}^{\mathcal{P}'(n),R}$ are the entries of the unweighted matrix $\mathcal{K}^{\mathcal{P}'(n),R}$, and where

$$\mathbf{c}_{\wp,\wp'} := \mathbf{c}_{\text{tot}}(v_{\wp\wp'}) ,$$

with the label $v_{\wp,\wp'}$ indicating whether the corresponding entry $\mathcal{K}_{\wp,\wp'}^{\mathcal{P}'(n),R}$ is an EM, SM(1), SM(2), or SM(3) arrow and $\mathbf{c}_{\text{tot}}(v_{\wp\wp'})$ the associated cost according to the table of §11.1.4.

12.1. Tropical semiring and Perron-Frobenius problem. We want to extract the dominant term in t of the Perron-Frobenius eigenvalue and eigenvector of $\mathcal{K}^{\mathcal{P}'(n),R,w}(t)$, in the limit where $t \rightarrow 0$, as in [26]. To this purpose, we turn the Perron-Frobenius problem for $\mathcal{K}^{\mathcal{P}'(n),R,w}(t)$ into an associated Perron-Frobenius problem in the min-plus semiring $S = (\mathbb{R} \sqcup \{\infty\}, \min, +, \infty, 0)$ that evaluates orders of magnitudes as $t \rightarrow 0$.

Lemma 12.3. *For $\mathcal{K}^{\mathcal{P}'(n),R,w}(t)$ as in (12.3), the Perron-Frobenius problems*

$$(12.4) \quad \sum_{\wp'} \mathcal{K}_{\wp,\wp'}^{\mathcal{P}'(n),R,w}(t) \eta_{\wp'}(t) = \lambda_t \eta_{\wp}(t)$$

$$(12.5) \quad \sum_{\wp} \psi_{\wp}(t) \mathcal{K}_{\wp,\wp'}^{\mathcal{P}'(n),R,w}(t) = \lambda_t \psi_{\wp'}(t)$$

with leading orders $\lambda_t = \lambda t^{\ell} + o(t^{\ell})$, $\eta_{\wp}(t) = \eta_{\wp} t^{u_{\wp}} + o(t^{u_{\wp}})$ and $\psi_{\wp}(t) = \psi_{\wp} t^{q_{\wp}} + o(t^{q_{\wp}})$ determines Perron-Frobenius problems for the orders of magnitudes $u_{\wp}$, $q_{\wp}$, ℓ in the min-plus semiring of the form

$$(12.6) \quad \min_{\wp'} \{\mathbf{c}_{\wp,\wp'} + u_{\wp'}\} = \ell + u_{\wp}$$

$$(12.7) \quad \min_{\wp} \{\mathbf{c}_{\wp,\wp'} + q_{\wp}\} = \ell + q_{\wp'} .$$

Proof. We write (12.4) as

$$\sum_{\wp'} (\mathcal{K}_{\wp, \wp'}^{\mathcal{P}'(n), R} t^{\mathbf{c}_{\wp, \wp'}} \eta_{\wp'} t^{u_{\wp'}} + o(t^{u_{\wp'} + \mathbf{c}_{\wp, \wp'}})) = \lambda \eta_{\wp} t^{\ell + u_{\wp}} + O(t^{\ell + u_{\wp}}).$$

Due to the positivity of the matrix and the PF data, there are no cancellations. The leading term when $t \rightarrow 0$ in the sum on the left-hand-side is then equal to $\min\{\mathbf{c}_{\wp, \wp'} + u_{\wp'}\}$ and this has to match the leading term $\ell + u_{\wp}$ on the right-hand-side, giving (12.6). The case of (12.5) and (12.7) is analogous. $\square$

12.2. Optimal arborescences and Perron–Frobenius eigenvalue. To give an explicit description of the Perron–Frobenius eigenvectors of the min-plus Perron–Frobenius problem (12.6) and (12.7), we first recall the notion of *optimal arborescence* in a weighted directed graph.

Definition 12.4. Let G be a directed graph with edges weighted by a cost function $\mathbf{c} : E(G) \rightarrow \mathbb{R}$. The weight of a subgraph $G' \subset G$ is given by the total weight of its edges,

$$\mathbf{c}(G') = \sum_{e \in E(G')} \mathbf{c}(e).$$

For a choice of a vertex $v_r \in V(G)$, an arborescence T with sink at v_r is a subgraph $T \subset G$ that satisfies

- (1) T is a spanning tree of the underlying undirected graph of G ,
- (2) for every $v \in V(G)$ there is a directed path of edges in $E(T)$ from v to the root vertex v_r .

An arborescence $\tilde{T}$ with source at v_r satisfies condition (1) above and

- (2') for every $v \in V(G)$ there is a directed path of edges in $E(T)$ from the root vertex v_r to v .

An arborescence T or $\tilde{T}$ is an optimal arborescence if it also satisfies the third property:

- (3) T (or $\tilde{T}$) has the minimal weight among all arborescences with sink (or with source) at v_r .

If the graph G is strongly connected, an arborescence with sink at v_r exists for any choice of the vertex v_r . When this is the case, an optimal arborescence can be constructed through the Chu–Liu/Edmonds/Bock Algorithm, which first constructs a subgraph G' of G obtained by retaining at each vertex of G the edges of minimal cost among all the incoming edges, and then proceeding with a recipe for eliminating loops in G' to obtain T . The case with source at v_r is analogous but selecting outgoing edges with minimal cost. (For a discussion of algorithms to find minimal arborescences, see [16].)

We have the following description of the asymptotic orders of magnitude as $t \rightarrow 0$ for the Perron–Frobenius eigenvalue and eigenspace, in the case of the Merge dynamics weighted with the total cost function $\mathbf{c}_{\text{tot}}$.

Theorem 12.1. Consider the $\mathbf{c}_{\text{tot}}$ -weighted Merge dynamics with matrix $\mathcal{K}^{\mathcal{P}'(n), R, w}(t)$ as in (12.3), and with asymptotics given by the solutions to the min-plus Perron–Frobenius problem (12.6) and (12.7). The min-plus Perron–Frobenius eigenvalue is $\ell = 0$.

Proof. We compute what the order of magnitude ℓ for the Perron–Frobenius eigenvalue should be, so that the PF systems (12.6) and (12.7) become equations for the orders of magnitudes $u_{\wp}$ and $q_{\wp}$, with $\mathbf{c}_{\wp, \wp'}$ and ℓ as known quantities.

There is a general method for computing the Perron–Frobenius eigenvalue over semirings, which is used for instance in [1], [2], [10], [15]. For a semiring $(S, \oplus, \odot, e, \epsilon)$ (in particular the min-plus

semiring we are interested in here), one defines the *permanent* of an S -valued $n \times n$ matrix as

$$\text{per}(K) = \bigoplus_{\sigma \in S_n} \bigodot_{i=1}^n K_{i\sigma(i)}.$$

In the case of the min-plus semiring this gives

$$\text{per}(K) = \min_{\sigma \in S_n} (K_{1,\sigma(1)} + \cdots + K_{n,\sigma(n)}).$$

One can use this to define an analog of the characteristic polynomial

$$\text{per}(Y \text{Id} \oplus K) = \bigoplus_{\sigma \in S_n} \bigodot_{i=1}^n (Y \delta_{i,\sigma(i)} \oplus K_{i,\sigma(i)}) \in S[Y],$$

whose coefficients are the semiring k -th traces

$$\text{Tr}_k^S(K) := \bigoplus_{J \subset \{1, \dots, n\}, \#J=k} \left(\bigoplus_{\sigma \in S_J} \bigodot_{j \in J} A_{j,\sigma(j)} \right)$$

with $\sigma \in S_J$ the permutations of the subset J . The roots of this characteristic polynomial are called “algebraic eigenvalues” of K (see [2]). Note that in general not all of these algebraic eigenvalues will have eigenvectors in this semiring setting: those that do are referred to as “geometric eigenvalues”. For the min-plus semiring, in particular, the minimal algebraic eigenvalue is given by

$$(12.8) \quad \rho_{\min}(K) = \bigoplus_{k=1}^n \bigoplus_{i_1, \dots, i_k} (K_{i_1 i_2} \odot \cdots \odot K_{i_k i_1})^{1/k} = \min_{k=1, \dots, n} \min_{c=(i_1, \dots, i_k)} \frac{1}{k} (K_{i_1 i_2} + \cdots + K_{i_k i_1}).$$

where $c = (i_1, \dots, i_k)$ runs over all circuits of length k in the graph $G(K)$ with n vertices and an edge e_{ij} for each nontrivial entry K_{ij} . A circuit $c = (i_1, \dots, i_k)$ that realizes the minimum is a “critical circuit”. The critical graph $G_{\text{crit}}(K) \subset G(K)$ consists of vertices and edges that belong to critical circuits. If $G(K)$ is strongly connected, then $\rho_{\min}(K)$ is the unique geometric eigenvalue and its multiplicity is equal to the number of strongly connected components of $G_{\text{crit}}(K)$.

The graph $\mathcal{G}_{\mathcal{P}'(n)}$ with the edges $e_{\wp, \wp'}$ weighted by the entries $\mathcal{K}_{\wp, \wp'}^{\mathcal{P}'(n), R} t^{\mathbf{c}_{\wp, \wp'}}$ of the matrix $\mathcal{K}^{\mathcal{P}'(n), R}(t)$ is strongly connected and has IM self-loops at all the partitions $\wp$ that contain at least one row of size $k_i \geq 3$. Since the cost $\mathbf{c}_{\text{tot}}(\text{IM}) = 0$, these IM diagonal entries of $\mathcal{K}^{\mathcal{P}'(n), R}(t)$ are independent of t and equal to the corresponding entry $\mathcal{K}_{\wp, \wp}^{\mathcal{P}'(n), R}$.

Let K be the min-plus valued matrix with entries the costs $K_{\wp, \wp'} = \mathbf{c}_{\wp, \wp'}$ of the Merge transformations associated to the edges, as described in §11.1.4. Let $G(K)$ be the graph associated to the min-plus matrix K as above. The graph $G(K)$ has the same vertices and edges of $\mathcal{G}_{\mathcal{P}'(n)}$, with the edges weighted by $\mathbf{c}_{\wp, \wp'}$, hence it is also strongly connected, since none of the $\mathbf{c}_{\wp, \wp'}$ equals ∞ (the additive unit of the min-plus semiring). The IM self-loops of $\mathcal{G}_{\mathcal{P}'(n)}$ have weight $\mathbf{c}_{\wp, \wp'} = 0$ (the multiplicative unit of the max-plus semiring). Since all other costs coming from non-IM (EM or SM) arrows are larger, $\mathbf{c}_{\wp, \wp'} > 0$, the circuits $c = (\wp_1, \dots, \wp_k)$ that realize the minima of $(\mathbf{c}_{\wp_1, \wp_2} + \cdots + \mathbf{c}_{\wp_k, \wp_1})/k$ are the single-edge IM loops $\mathbf{c}_{\wp, \wp} = 0$ (with $k = 1$). Thus, there are many critical circuits, each consisting of a single edge, and a single vertex (any partition $\wp$ with at least one row of length at least three is such a vertex). Thus, the critical graph $G_{\text{crit}}(K)$ consists of a disjoint union of single-edge loops, one for each such partition $\wp$.

Thus, we have $\ell = 0$ in (12.6) and (12.7), as this ℓ corresponds to the unique geometric eigenvalue (12.8). $\square$

12.3. Tropical Perron-Frobenius eigenspace. We then analyze the Perron-Frobenius eigenspace of the tropical Perron-Frobenius problem (12.6) and (12.7) with $\ell = 0$.

Since there are multiple critical circuits, we cannot use the eigenvalue asymptotics of [1], [2] that relies on the uniqueness of the critical circuits to relate the min-plus (or max- $\times$) Perron-Frobenius eigenvector to the Perron-Frobenius eigenvector of the original problem (12.4), (12.5). However, we can describe the Perron-Frobenius eigenspace in terms of its generators, as in [3].

Theorem 12.2. *Consider the cost function $\mathbf{c}_{\text{tot}}$ of the dynamics $\mathcal{K}^{\mathcal{P}'(n),R,w}(t)$ as in (12.3), with asymptotics determined by the min-plus Perron-Frobenius problem (12.6) and (12.7). The right Perron-Frobenius eigenspace is generated by a family $\{u^{(\wp)}\}$ of min-plus eigenvectors as in (12.6), where $\wp$ ranges over those partitions $\wp \in \mathcal{P}'(n)$ whose Young diagram has at least one row of length at least 3. The entry $u_{\wp'}^{(\wp)}$ of the basis eigenvector $u^{(\wp)} = (u_{\wp'}^{(\wp)})_{\wp' \in \mathcal{P}'(n)}$ is given by the cost $\mathbf{c}(\gamma_{\wp'\wp})$ of the directed path in an optimal arborescence $T_{\wp}$ in $\mathcal{G}_{\mathcal{P}'(n)}$ with sink at $\wp$. Similarly, the left Perron-Frobenius eigenspace is generated by a family $\{q^{(\wp)}\}$ of solutions of (12.7), for $\wp$ in the same set of partitions $\wp$ with at least one row of length at least 3, where for a $\wp' \in \mathcal{P}'(n)$, the entry $q_{\wp'}^{(\wp)}$ is the cost $\mathbf{c}(\tilde{\gamma}_{\wp\wp'})$ of the directed path $\tilde{\gamma}_{\wp\wp'}$ in an optimal arborescence $\tilde{T}_{\wp}$ in $\mathcal{G}_{\mathcal{P}'(n)}$ with source at $\wp$.*

Proof. One defines the Kleene star K^* of a min-plus valued matrix K in the form

$$(12.9) \quad K^* := \mathcal{I} \oplus K \oplus K^2 \oplus \cdots \oplus K^{n-1},$$

where $\mathcal{I}$ is the identity matrix (in the min-plus sense). The entries are given by

$$(K^*)_{ij} = K_{i=i_0, i_1} \odot \cdots \odot K_{i_{k-1}, i_k=j}$$

along the directed path of minimal weight from i to j in $G(K)$, which is of some length $1 \leq k \leq n-1$. As shown in Proposition 2.6 of [3] (specialized to our case with $\rho_{\min}(K) = 0$ and with each critical cycle C_j consisting of a single vertex j and a looping edge), there is an eigenvector u_j of K for each critical cycle C_j , proportional to the j -th column of the matrix K^* . These eigenvectors are the generators of the eigenspace of $\rho_{\min}(K)$.

In our case, this means that, for each partition $\wp$ with at least one row of length $k_i \geq 3$ there is an eigenvector $u^{(\wp)} = (u_{\wp'}^{(\wp)})_{\wp' \in \mathcal{P}'(n)}$ that satisfies (12.6) with $\ell = 0$. Correspondingly, the same argument applied to the transpose matrix shows that for each such $\wp$ there is an eigenvector $q^{(\wp)} = (q_{\wp'}^{(\wp)})_{\wp' \in \mathcal{P}'(n)}$ that satisfies (12.7) with $\ell = 0$. The set $\{u^{(\wp)}\}$ (respectively, $\{q^{(\wp)}\}$) of these eigenvectors spans the eigenspace.

Moreover, the description of these generators in terms of columns of the Kleene star matrix can be made geometrically more explicit in our case, in terms of the graph $\mathcal{G}_{\mathcal{P}'(n)}$. To see this, we use the notion of optimal arborescence in a weighted directed graph, recalled in Definition 12.4 above.

Let $\wp$ be one of the partitions in $\mathcal{P}'(n)$ that have at least one row with $k_i \geq 3$ so that they are a non-trivial IM self-loop edge in $\mathcal{G}_{\mathcal{P}'(n)}$. And let $T_{\wp}$ denote an optimal arborescence with sink at $\wp$, with respect to the cost function $\mathbf{c}_{\text{tot}}$. This is a directed spanning tree of $\mathcal{G}_{\mathcal{P}'(n)}$, where each vertex $\wp' \neq \wp$ has a single outgoing edge in $T_{\wp}$, with minimal cost among all the outgoing edges at $\wp'$ in $\mathcal{G}_{\mathcal{P}'(n)}$, while the vertex $\wp$ has no outgoing edges in $T_{\wp}$.

In the construction of the Kleene star matrix, we have

$$(K^*)_{\wp', \wp} = \mathbf{c}(e_{\wp', \wp_1}) + \cdots + \mathbf{c}(e_{\wp_{k-1}, \wp})$$

where $e_{\wp', \wp_1} \cdots e_{\wp_{k-1}, \wp}$ is a minimal cost path from $\wp'$ to $\wp$. In particular, this means (Bellman's optimality principle) that for any $0 \leq i < j \leq k$ the subpath $e_{\wp_i, \wp_{i+1}} \cdots e_{\wp_{j-1}, \wp_j}$ is also a cost minimizing path between $\wp_i$ and $\wp_j$, so in particular at each vertex $\wp_i$ the next outgoing edge in

the path can be taken to be an edge in $T_\wp$ and we can identify $(K^\star)_{\wp',\wp}$ with the $\mathbf{c}$ -cost of the unique path $\gamma_{\wp',\wp}$ in $T_\wp$ from $\wp'$ to $\wp$. Thus, the eigenvectors $u^{(\wp)}$ satisfying (12.6) that span the eigenspace are associated to the optimal arborescences $T_\wp$ and have components $u^{(\wp)} = (u_{\wp'}^{(\wp)})_{\wp' \in \mathcal{P}'(n)}$ given by the costs $u_{\wp'}^{(\wp)} = \mathbf{c}(\gamma_{\wp',\wp})$ of the paths in the arborescence.

In the case of the eigenvalues $q^{(\wp)}$ of the (12.7) min-plus Perron-Frobenius problem, we are considering the same setting but with the transpose K^τ of the K matrix. Thus, the vector $q^{(\wp)}$ is the $\wp$ -column of the Kleene star matrix $(K^\tau)^\star$ of the transpose K^τ . The entries of this Kleene star matrix are given by

$$(K^\tau)_{\wp',\wp}^\star = \mathbf{c}(e_{\wp,\wp_1}) + \cdots + \mathbf{c}(e_{\wp_{k-1},\wp'}) = \mathbf{c}(\tilde{\gamma}_{\wp,\wp'}),$$

where $\tilde{\gamma}_{\wp,\wp'} = e_{\wp,\wp_1} \dots e_{\wp_{k-1},\wp'}$ is a minimal cost path from $\wp$ to $\wp'$. Thus, in this case the eigenvectors $q^{(\wp)}$ correspond to optimal arborescences $\tilde{T}_\wp$ with source at $\wp$, with the components $q_{\wp'}^{(\wp)} = \mathbf{c}(\tilde{\gamma}_{\wp',\wp})$ given by the costs of the paths in the arborescence. $\square$

Moreover, the result of [3] shows that the left/right Perron-Frobenius eigenvector of $\mathcal{K}^{\mathcal{P}'(n),R,w}(t)$ has asymptotics $u = (u_\wp)_{\wp \in \mathcal{P}'(n)}$ (respectively, $q = (q_\wp)_{\wp \in \mathcal{P}'(n)}$) in the eigenspace, although not necessarily equal to one of the above generators of the eigenspace. In fact, explicit examples computed in [18] show that it is in general not the case that the asymptotics of the Perron-Frobenius eigenvector is one of the generators of the eigenspace of the min-plus Perron-Frobenius problem.

12.4. Optimization with respect to cost functions. The introduction of optimality with respect to Minimal Search, Minimal Yield, and Complexity Loss in the linguistics model of Minimalism, [7], [8], [22], formulated in the setting we are considering here, of the Merge Hopf algebra Markov chain dynamical system, becomes the minimization property of the Markov chain

$$\hat{\mathcal{K}}^{(A,n,(\mathbf{c}_{\text{tot}},\beta))} = \hat{\mathcal{K}}^{(A,n,\mathbf{c}_{\text{tot}})}(t) \quad \text{for } t = e^{-\beta},$$

with $\mathbf{c}_{\text{tot}}$ the cost function of §11.1.4, with respect to the free energy function

$$\mathbb{F}(\hat{\mathcal{K}}^{(A,n,(\mathbf{c}_{\text{tot}},\beta))}) = \bar{\mathbb{E}}_{\mathbb{P}_{(\mathbf{c}_{\text{tot}},\beta)}} - \beta^{-1} \text{Sh}(\mathbb{P}_{(\mathbf{c}_{\text{tot}},\beta)})$$

where $\mathbb{P}_{(\mathbf{c}_{\text{tot}},\beta)}$ is the Boltzmann distribution with $\mathbb{E} = \mathbf{c}_{\text{tot}}$, as explained in §3.4.

We discuss in §12.5 to what extent the use of the cost function $\mathbf{c}_{\text{tot}}$ achieves the goal of eliminating the unwanted properties of the Sideward Merge arrows, allowing for convergence via External Merge to fully formed sentences (tree structures). We will then present in §13 an alternative choice of a simple cost function which corrects for the remaining problems and formulates optimization entirely in information theoretic terms.

12.5. The Sideward Merge problem in the weighted dynamics. The result of our discussion of the weighted dynamics with cost function $\mathbf{c}_{\text{tot}}$ and weights $t^{\mathbf{c}_{\text{tot}}}$ on the adjacency matrix of the Merge graph $\mathcal{G}_{n,A}$ shows that the cost function corrects for some of the unwanted properties of the SM arrows in the unweighted dynamics. However, the surprise with respect to the simple case $n = 3$ discussed in [26] is that the cost counting $\mathbf{c}_{\text{tot}}$ alone is not sufficient to completely correct for the SM pull on the dynamics in the direction of less connected structures.

More precisely, we found that, to leading order when $t \rightarrow 0$, the stationary distribution for the weighted dynamics lies in the Perron-Frobenius eigenspace described above. Since the basis of this eigenspace only involves partitions $\wp$ with at least one of the $k_i \geq 3$, we no longer see the phenomenon typical of the unweighted dynamics, where the stationary distribution assigns highest probability to the most disconnected structure $\wp_{1^n}$ due to the dominance of SM. However,

the leading order of the stationary distribution can now have contributions from all the $\wp$ with at least one of the $k_i \geq 3$ (all those with IM dynamics), and is not supported only on $\wp = \{n\}$ (convergence to the connected structures, workspaces consisting of a single tree, with the IM dynamics), as one expects when EM dominates the SM–EM tension in the dynamics, and as happens in the case $n = 3$ of [26] with the same weights and cost function.

This reveals that Minimal Search, Minimal Yield, and Complexity Loss as usually considered in the linguistics literature (the slightly different MY formulation of [7], [8] makes no difference) are not sufficient cost functions to eliminate the unwanted effects of Sideward Merge.

As we discuss in the next section, what is missing is the role of *information optimization*, which as we have already observed in a different form in §3, and especially §3.3, that the Merge Hopf algebra Markov chain satisfies. We show here that adding a Shannon information count to the cost function achieves the result of concentrating the dynamics, in the leading order part of the stationary distribution, on just the workspaces consisting of a single tree with the IM dynamics.

13. SHANNON ENTROPY OPTIMIZATION AND THE MERGE DYNAMICS

Definition 13.1. *To a partition $\wp \in \mathcal{P}'(n)$, of the form $n = a_1 k_1 + \dots + a_r k_r$, we associate the probability distribution*

$$(13.1) \quad \mathbb{P}_\wp := \left(\underbrace{\frac{k_1}{n}, \dots, \frac{k_1}{n}}_{a_1\text{-times}}, \dots, \underbrace{\frac{k_r}{n}, \dots, \frac{k_r}{n}}_{a_r\text{-times}} \right).$$

We extend the cost function $\mathbf{c}_{\text{tot}}$ considered in the previous section, by including a term that depends on the Shannon entropy $\text{Sh}(\mathbb{P}_\wp)$ of the distribution (13.1).

Definition 13.2. *On the graph $\mathcal{G}_{\mathcal{P}'(n)}$ we consider a cost function $\mathbf{c} : E(\mathcal{G}_{\mathcal{P}'(n)}) \rightarrow \mathbb{R}$ of the form*

$$(13.2) \quad \tilde{\mathbf{c}}_{\text{tot}}(e) = \mathbf{c}_{\text{tot}}(e) + \text{Sh}(\mathbb{P}_{s(e)}),$$

where $\mathbf{c}_{\text{tot}}$ is the total cost function of MS, MY, CL, as described in §11.1.4 and $\mathbb{P}_{s(e)}$ is the distribution (13.1) at the partition $\wp = s(e) \in \mathcal{P}'(n)$, the source vertex of the edge e in the directed graph $\mathcal{G}_{\mathcal{P}'(n)}$.

The same argument used in Theorem 12.1 and Theorem 12.2, applied to the modified cost function (13.2), leads to the following result.

Proposition 13.3. *With the modified cost function $\tilde{\mathbf{c}}_{\text{tot}}$ of (13.2), the dynamics $\mathcal{K}^{\mathcal{P}'(n), R, w}(t)$ as in (12.3) has asymptotics given by the solution of the min-plus Perron-Frobenius problem (12.6) and (12.7) with $\ell = 0$ and with PF L/R-eigenvectors $v = v^{(\wp=\{n\})}$, with $v_{\wp'} = \tilde{\mathbf{c}}(\gamma_{\wp'\{n\}})$, the cost of the directed path in an optimal arborescence $T_{\{n\}}$ with sink at $\wp = \{n\}$, and $q = q^{(\wp=\{n\})}$, with $q_{\wp'} = \tilde{\mathbf{c}}(\tilde{\gamma}_{\{n\}\wp'})$, the cost of the directed path in an optimal arborescence $\tilde{T}_{\{n\}}$ with source at $\wp = \{n\}$.*

Proof. The main difference with respect to Theorem 12.1 and Theorem 12.2 is that, with the modified cost function (13.2), there is only one critical circuit, consisting of the IM self-loop at the partition $\wp = \{n\}$. Indeed, the added term $\text{Sh}(\mathbb{P}_{s(e)})$ in the cost function is minimal, $\text{Sh}(\mathbb{P}_\wp) = 0$ iff $\wp = \{n\}$ and is strictly positive at all other $\wp \in \mathcal{P}'(n)$. Thus, the only minimal cost loop, with $\tilde{\mathbf{c}}_{\text{tot}}(\gamma) = 0$ is the single IM edge with source and target $\wp = \{n\}$. This again gives $\ell = 0$ for the min-plus PF eigenvalue as in Theorem 12.1. Since in this case we do have a unique critical circuit, we can obtain not only a basis of the PF eigenspace of the min-plus Perron-Frobenius problem, but also directly the PF eigenvector which gives the asymptotic behavior of the original Perron-Frobenius problem. In particular, this gives the form $v_{\wp'} = \tilde{\mathbf{c}}(\gamma_{\wp'\{n\}})$ and $q_{\wp'} = \tilde{\mathbf{c}}(\tilde{\gamma}_{\{n\}\wp'})$ or the R/L-eigenvector of the min-plus problem. $\square$

Consequently we obtain the asymptotic behavior of the stationary distribution of the weighted Merge dynamics on $\mathcal{G}_{n,A}$.

Theorem 13.1. *The asymptotic behavior for $t \rightarrow 0$ of the stationary distribution of the weighted form $\hat{\mathcal{K}}^{A,n,w}(t)$ of the Hopf algebra Markov chain of Merge acting on workspaces, with weights $t^{\tilde{\mathbf{c}}}$ and cost function $\tilde{\mathbf{c}}$ of (13.2) is given by the total cost of the loop along the optimal arborescences $T_{\{n\}}$ and $\tilde{T}_{\{n\}}$,*

$$(13.3) \quad \text{order}_{t \rightarrow 0}(\pi^w(F, t)) = \tilde{\mathbf{c}}(\gamma_{p(F)\{n\}} \cup \tilde{\gamma}_{\{n\}p(F)}).$$

This implies that in the limit $t \rightarrow 0$ the stationary distribution of $\hat{\mathcal{K}}^{A,n,w}(t)$ is the uniform distribution supported on the workspaces that consist of a single tree, $F = T \in p^{-1}(\{n\})$. Thus, the Merge dynamics weighted by the cost function $\tilde{\mathbf{c}}$ converges to fully formed trees with the remaining action of Internal Merge.

Proof. We know by Proposition 12.1 that the Hopf algebra Markov chain $\hat{\mathcal{K}}^{A,n,w}(t)$ for the dynamics of Merge on $\mathcal{G}_{n,A}$, with the weighted adjacency matrix

$$\mathcal{K}_{F,F'}^{A,n,w}(t) = t^{\tilde{\mathbf{c}}_{\text{tot},F,F'}} \mathcal{K}_{F,F'}^{A,n},$$

has stationary distribution as in (12.2), given by

$$\pi^w(F, t) = \frac{\eta_{\mathcal{P}'(n),w}(\varnothing, t) \psi_{\mathcal{P}'(n),w}(\varnothing, t)}{\Lambda_{\varnothing,n}},$$

with $\varnothing = p(F)$ under the projection map $p : \mathcal{G}_{n,A} \rightarrow \mathcal{G}_{\mathcal{P}'(n)}$, where $\eta_{\mathcal{P}'(n),w}$ and $\psi_{\mathcal{P}'(n),w}$ are the left and right Perron–Frobenius eigenfunctions of $\mathcal{K}^{\mathcal{P}'(n),R,w}(t)$ on $\mathcal{G}_{\mathcal{P}'(n)}$. Thus, when looking at the asymptotic behavior for $t \rightarrow 0$ of $\pi^w(F, t)$, we see from Proposition 13.3 that this is given by

$$\text{order}_{t \rightarrow 0}(\pi^w(F, t)) = v_{p(F)} + q_{p(F)}$$

with $v = v_{(\varnothing=\{n\})}$ and $q = q_{(\varnothing=\{n\})}$ as in Proposition 13.3. This gives (13.3) as $v_{p(F)} + q_{p(F)}$ is the $\tilde{\mathbf{c}}$ -cost of the loop $\gamma_{p(F)\{n\}} \cup \tilde{\gamma}_{\{n\}p(F)}$. All the costs of these loops are strictly positive, except for the case where $p(F) = \{n\}$ where the cost is zero. Thus, in the limit $t \rightarrow 0$ the $\pi^w(F, t)$ will converge to the uniform distribution on $p^{-1}(\{n\})$, the workspaces with a single connected component. $\square$

In this form, the weighted dynamics has exactly the expected effect, namely it compensates for the fact that the presence of Sideward Merge tends to push the dynamics to less connected structures preventing convergence of the structure formation process to a single tree. At the same time, one maintains the good properties of Sideward Merge, the strong connectedness of the graph (irreducibility of the dynamics).

13.1. Shannon entropy cost function. In fact, while we maintained in Proposition 13.3 and Theorem 13.1 a combination, as in Definition 13.2 of the cost function $\mathbf{c}_{\text{tot}}$ coming from the usual notions of Minimal Search, Minimal Yield, and Resource Restriction developed in the linguistics literature (see [7], [8]), in our weighted dynamics it is really the Shannon entropy part of the cost function that has the desired role of forcing the dynamics to converge to the workspaces consisting of a single tree as one wants in the structure formation process of syntax. The following result is proved exactly as Theorem 13.1.

Theorem 13.2. *Consider the Merge dynamics $\hat{\mathcal{K}}^{A,n,\text{Sh}}(t)$ weighted only by the Shannon entropy cost, namely using the weighted adjacency matrix on $\mathcal{G}_{n,A}$ of the form*

$$(13.4) \quad \mathcal{K}_{F,F'}^{A,n,\text{Sh}}(t) := t^{\text{Sh}(\mathbb{P}_{p(F)})} \mathcal{K}_{F,F'}^{A,n}.$$

Then the stationary distribution has asymptotic behavior

$$(13.5) \quad \text{order}_{t \rightarrow 0}(\pi^{\text{Sh}}(F, t)) = \sum_{e \in \gamma_{p(F)}\{n\}} \text{Sh}(\mathbb{P}_{s(e)}) + \sum_{e' \in \tilde{\gamma}_{\{n\}} p(F)} \text{Sh}(\mathbb{P}_{s(e')}).$$

In particular, $\text{order}_{t \rightarrow 0}(\pi^{\text{Sh}}(F, t)) > 0$ for all $F \in p^{-1}(\wp)$ with $\wp \neq \{n\}$, while $\text{order}_{t \rightarrow 0}(\pi^{\text{Sh}}(F, t)) = 0$ for $F = T \in p^{-1}(\{n\})$. Thus, in the limit $t \rightarrow 0$ the stationary distribution converges to the uniform distribution on the trees $T \in p^{-1}(\{n\})$, where only Internal Merge is acting.

Remark 13.4. The result of Theorem 13.2 shows that the Minimal Search and Resource Restrictions cost functions play a role in simplifying the graph $\mathcal{G}_{n,A}$ by identifying the “minimal” Sideward Merge arrows that suffice for strong connectedness while significantly reducing the size of the graph, making it sparse. Optimization of information-theoretic data then drives the dynamics towards the connected workspaces, compensating for the presence of Sideward Merge.

This observation leads to the natural question on the meaning of the added cost function $\mathfrak{c}_{\text{Sh}}(e) = \text{Sh}(\mathbb{P}_{s(e)})$ in the context of the geometry of the projection map $p : \mathcal{G}_{n,A} \rightarrow \mathcal{G}_{\mathcal{P}'(n)}$.

We have seen in §3.3 how the Merge Hopf algebra Markov chain $\hat{\mathcal{K}}^{A,n}$ has an entropy maximization property for the Shannon entropy of the probability on paths determined by the probability on edges defined by the transition probabilities of a Markov chain. We have also seen in §3.4 how, introducing a weight by a cost function on edges, one can reformulate this optimization as a minimization of the free energy functional $\bar{\mathbb{F}} = \bar{\mathbb{E}} - \beta^{-1} \text{Sh}$, with the energy $\bar{\mathbb{E}}$ determined by the cost function and the probability distribution on edges determined by the transition probabilities of the (weighted) Markov chain.

We are considering here an additional datum of a probability distribution $\mathbb{P}_{\wp}$ associated to each vertex $\wp \in \mathcal{P}'(n)$. The meaning of the quantity $\text{Sh}(\mathbb{P}_{\wp})$ can be described as follows.

Remark 13.5. By definition, the Shannon entropy $\text{Sh}(\mathbb{P}_{\wp})$ measures the expected amount of information (or the expected surprisal) in observing the outcome of an event distributed according to $\mathbb{P}_{\wp}$. If we fix a workspace $F = T_1 \sqcup \cdots \sqcup T_r$ in $p^{-1}(\wp) \subset V(\mathcal{G}_{n,A})$, we can observe how the lexical material (the elements of the set A) at the leaves is distributed among the components T_i of the workspace F . Since we are here dealing only with *free* structure formation (the free symmetric Merge), there are no constraints on how the elements of A can be distributed among the leaves of the T_i , and this implies that the only law governing the possible assignments is the probability $\mathbb{P}_{\wp}$. (We will see in §15 and §16 that filtering for theta roles assignments, phase structure, and syntactic parameters will impose constraints that change this probability distribution.)

Remark 13.6. The probability distribution $\mathbb{P}_{\wp}$ and the Shannon entropy $\text{Sh}(\mathbb{P}_{\wp})$ are directly related to the combinatorial counting of the assignments of the elements of A to the components of the workspace, via the well known relation of $\text{Sh}(\mathbb{P}_{\wp})$ to the multinomial coefficient $\mu_{\wp,n}$ of (2.4), (2.6), namely the asymptotic behavior

$$(13.6) \quad \mu_{n,\wp} \sim e^{n \text{Sh}(\mathbb{P}_{\wp}) + o(n)}.$$

This can be derived directly from the Stirling approximation of factorials, which also shows that the lower order term behaves like $o(n) = O(\log(n))$. The behavior in (13.6) can be interpreted in terms of the fact that $\mu_{n,\wp}$ counts words of length n in an alphabet of r symbols, where the i -th symbol occurs k_i times.

Remark 13.7. In the counting of the multiplicities of the projection map $p : \mathcal{G}_{n,A} \rightarrow \mathcal{G}_{\mathcal{P}'(n)}$, namely the sizes of the fibers $\#p^{-1}(\wp) = \Lambda_{\wp,n}$ we have $\Lambda_{\wp,n} = \mu_{n,\wp} \cdot \Gamma_{\wp,n}$ with $\Gamma_{\wp,n}$ as in (2.11), where the factor $\Gamma_{\wp,n}$ accounts for permuting rows of equal length in the partition $\wp$ and for

counting all the different possible tree topologies of a tree T_i with k_i leaves. However, unlike in the counting of Proposition 5.3, when we have fixed a workspace $F = T_1 \sqcup \cdots \sqcup T_r$ in $p^{-1}(\wp)$, and we look at the distribution of the lexical material A among the T_i , only the $\mu_{n,\wp}$ part of the multiplicity $\Lambda_{\wp,n}$ remains relevant, as the other choices encoded in the $\Gamma_{\wp,n}$ factor have been fixed.

The effect on the partition of various Merge transformations has a corresponding effect on the probability distribution $\mathbb{P}_\wp$ and its Shannon entropy that can be phrased in terms of natural properties of the entropy functional. For example, we have the following observation.

Proposition 13.8. *External Merge, acting on partitions by*

$$\wp = \{k_1, \dots, k_r\} \mapsto \wp' = \{k_1, \dots, k_i + k_j, \dots, k_r\}$$

acts on $\text{Sh}(\mathbb{P}_\wp)$ as the chain rule of the Shannon entropy as

$$(13.7) \quad \text{EM} : \text{Sh}(\mathbb{P}_\wp) \mapsto \text{Sh}(\mathbb{P}_{\wp'}) = \text{Sh}(\mathbb{P}_\wp) - \frac{k_i + k_j}{n} \text{Sh}\left(\frac{k_i}{k_i + k_j}, \frac{k_j}{k_i + k_j}\right)$$

Proof. By the symmetry property of the Shannon entropy, we can assume without loss of generality that the pair k_i, k_j is k_1, k_2 . The chain rule for the Shannon entropy (also known as extensivity condition) gives

$$(13.8) \quad \text{Sh}\left(\frac{k_1}{n}, \dots, \frac{k_r}{n}\right) = \text{Sh}\left(\frac{k_1 + k_2}{n}, \frac{k_3}{n}, \dots, \frac{k_r}{n}\right) + \frac{k_1 + k_2}{n} \text{Sh}\left(\frac{k_1}{k_1 + k_2}, \frac{k_2}{k_1 + k_2}\right),$$

which translates into the action of an EM transformation that merges the components T_i and T_j of the workspace to $\mathfrak{M}(T_i, T_j)$ on the Shannon entropy by (13.7). $\square$

Remark 13.9. The identity of multinomial coefficients

$$\binom{n}{m, k_3, \dots, k_r} \cdot \binom{m}{k_1, k_2} = \binom{n}{k_1, k_2, k_3, \dots, k_r}$$

implies, in the asymptotic behavior the chain rule identity (13.8) for the Shannon entropy.

Remark 13.10. Note how this singles out External Merge as being associated to the chain rule of the Shannon entropy, which is the key property in the axiomatic characterization (Khinchin axioms) of information. It seems interesting to investigate whether Internal Merge and Sideward Merge also have a meaning in terms of information theoretic properties.

When we use $\text{Sh}(\mathbb{P}_\wp)$ as an energy/cost function, by setting $\mathfrak{c}_{\text{Sh}}(e) = \text{Sh}(\mathbb{P}_{p(s(e))})$, we weight the adjacency matrix $\mathcal{K}^{A,n}$ of $\mathcal{G}_{n,A}$ as

$$\mathcal{K}_{F,F'}^{A,n,\text{Sh}} = t^{\text{Sh}_p(F)} \mathcal{K}_{F,F'}^{A,n} = e^{-\beta \text{Sh}_p(F)} \mathcal{K}_{F,F'}^{A,n},$$

with $t = e^{-\beta}$, with β the inverse temperature thermodynamic parameter of the Boltzmann distribution, so that the range $0 < t < 1$ corresponds to the usual range $0 < \beta < \infty$ for the inverse temperature, with $t \rightarrow 0$ corresponding to the zero-temperature limit $\beta \rightarrow \infty$. Using this notation, we write here $\mathcal{K}_{F,F'}^{A,n,\text{Sh},\beta}$ rather than $\mathcal{K}_{F,F'}^{A,n,\text{Sh}}(t)$ as before, for consistency with §3.4.

Proposition 13.11. *In the thermodynamic limit of zero temperature, $\beta \rightarrow \infty$ the free energy optimization for $\hat{\mathcal{K}}_{F,F'}^{A,n,\text{Sh}}$ is achieved by the uniform distribution on edges of the Internal Merge graph $\mathcal{G}_{n,A,\{n\}}^{\text{IM}}$ that maximizes the Shannon entropy.*

Proof. As shown in §3.4, the corresponding Markov chain $\hat{\mathcal{K}}^{A,n,\text{Sh},\beta}$ minimizes the free energy function

$$\mathbb{F}(\hat{\mathcal{K}}^{A,n,\text{Sh},\beta}) = \bar{\mathbb{E}}(\hat{\mathcal{K}}^{A,n,\text{Sh},\beta}) - \beta^{-1} \text{Sh}(\hat{\mathcal{K}}^{A,n,\text{Sh},\beta}).$$

In order to look at the $\beta \rightarrow \infty$ behavior, it is more convenient to consider the maximization of the form of the energy function as in (3.19),

$$\tilde{\mathbb{F}}(\mathcal{S}) = \text{Sh}(\mathcal{S}) - \beta \bar{\mathbb{E}}(\mathcal{S}).$$

As shown in Lemma 3.7, this free energy is given as in (3.24) by

$$\tilde{\mathbb{F}}(\hat{\mathcal{K}}^{A,n,\text{Sh},\beta}) = -\beta \bar{\mathbb{E}}_{\mathbb{P}_{(\text{Sh},\beta)}} + \text{Sh}(\mathbb{P}_{(\text{Sh},\beta)}),$$

where

$$\mathbb{P}_{(\text{Sh},\beta)}(e) = \frac{1}{\lambda} \psi(s(e)) e^{-\beta \text{Sh}(\mathbb{P}_{p(s(e))})} \eta(t(e))$$

with λ and ψ , η the PF eigenvalue and L/R PF eigenvector of $\mathcal{K}^{A,n,\text{Sh},\beta}$. Thus, we have

$$\bar{\mathbb{E}}(\hat{\mathcal{K}}^{A,n,\text{Sh},\beta}) = \sum_F \pi(F) \text{Sh}(\mathbb{P}_{p(F)})$$

with π the stationary distribution of $\hat{\mathcal{K}}^{A,n,\text{Sh},\beta}$. As discussed in §3.4, we have $\mathbb{F}(\hat{\mathcal{K}}^{A,n,\text{Sh},\beta}) = -\beta^{-1}(\log(\lambda) + \text{Sh}(\pi))$ as in (3.27), for $\lambda = \lambda_{(\text{Sh},\beta)}$ the PF eigenvalue of $\mathcal{K}^{A,n,\text{Sh},\beta}$.

Theorem 13.2 further shows that, in the zero-temperature limit $\beta \rightarrow \infty$ ($t \rightarrow 0$), the stationary distribution $\pi(F) = \pi^{(\text{Sh},\beta)}(F)$ converges to the uniform distribution on the fiber $p^{-1}(\{n\})$. Thus, the terms in the sum

$$\sum_F \pi^{(\text{Sh},\beta)}(F) \text{Sh}(\mathbb{P}_{p(F)})$$

have $\pi^{(\text{Sh},\beta)}(F) \rightarrow 0$ for $\beta \rightarrow \infty$, for all $F \notin p^{-1}(\{n\})$, where $\text{Sh}(\mathbb{P}_{p(F)}) \neq 0$ and $\pi^{(\text{Sh},\beta)}(F) \rightarrow 1$ for $F \in p^{-1}(\{n\})$, where $\text{Sh}(\mathbb{P}_{p(F)}) = 0$ so that the sum vanishes in the limit.

Moreover, the distribution $\mathbb{P}_{(\text{Sh},\beta)}(e)$ on $\mathcal{G}_{n,A}$ reduces to the distribution $\mathbb{P}_{\text{IM}}(e) = \hat{\mathcal{K}}_{T,T'}^{\text{IM}}$ on the set of edges of the Internal-Merge graph $\mathcal{G}_{n,A,\{n\}}^{\text{IM}}$, with fixed $T = s(e)$ and with $T' = t(e)$. By Proposition 5.5 this distribution is the uniform distribution

$$\mathbb{P}_{\text{IM}}(e) = \frac{1}{\lambda_{\text{IM}}} \frac{\eta^{\text{IM}}(t(e))}{\eta^{\text{IM}}(s(e))} = \frac{1}{2n-4}$$

where $\eta^{\text{IM}}(T) = 1$ for all $T \in V_{n,A,\{n\}}$ and $\lambda_{\text{IM}} = d_{\{n\}} = 2n-4$. Thus, in the $\beta \rightarrow \infty$ limit, where the maximization reduces to the maximization of the Shannon entropy of the probability distribution on edges of $\mathcal{G}_{n,A,\{n\}}^{\text{IM}}$, achieved at the uniform distribution. $\square$

14. PARAMETERIZING CUBE AND DYNAMICS

In this and the following sections we mention some further directions in which it may be interesting to explore the dynamical properties of the action of Merge on workspaces. One of the questions posed in [22] concerns the extent to which the structure formation process of Merge can be encoded in the image of syntactic objects inside some semantic spaces, in the syntax-semantics interface.

The model of syntax-semantics interface developed in [22] is based on the idea that the computational structure of language resides entirely on the side of syntax (syntax-first model), while the semantics side can be reduced to a minimal kind of topological structure, capturing notions of proximity in some ambient “semantic space” (a topological or metric space, such as a manifold or stratified space). Since this model of semantics is not, in itself, endowed with a computational structure-formation process like Merge, it is natural to ask whether Merge would still be somehow encoded in, and at least partially reconstructible from, the image of syntactic objects in a semantic space. This question also has relevance to the current simulations of human language in machine architectures like large language models, where one can ask whether there are circuits

(in the sense of mechanistic interpretability) involving the weights of attention modules of transformer architectures, that would correspond to the computation of syntax via the Merge action on workspaces.

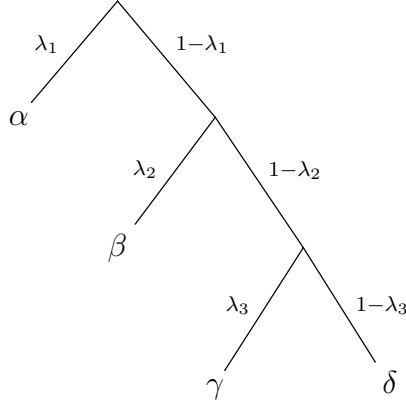
Another related question is about the realization of the process of structure formation via Merge in neurocomputational architectures: the broad question of neuroscience correlates of structure formation in syntax. A recent proposal [21] on possible ways of encoding the Merge operation in function spaces (such as wavelets) shows that a good encoding of the Merge operation is possible in a deformation of the tropical semiring determined by the Rényi entropy.

In both this setting and in the model of the syntax-semantics interface of [22], one considers a parameterizing space of non-planar full binary rooted trees with labelled leaves, where continuous parameters are introduced, reflecting a metric embedding for such trees inside some ambient semantic space. In the model of the syntax-semantics interface of [22] one uses the BHV moduli spaces [4] of non-planar trees to obtain this parameterization, following [13]. While the BHV moduli spaces appear adequate to describe the embeddings of syntactic objects in semantic spaces and the interaction with the planarization of trees in externalization (via the relation to the associahedra and the moduli space of real curves of genus zero as in [13]), this geometric setting does not seem ideal to capture a possible image of the Merge operation. The BHV moduli spaces have an interesting associated operad structure, the mosaic operad discussed in [12]. However, there is no immediate direct relation between the boundary strata of the BHV moduli spaces, which correspond to operations of shrinking edges to zero length, and the coproduct operation that gives rise to the Merge action, which corresponds to extracting accessible terms via admissible cuts.

On the other hand, in [21], continuous parameters are also associated to syntactic objects, in the form of parameters $\lambda_v \in [0, 1]$ associated to the non-leaf vertices of the tree, that determine a probability distribution at the leaves of the tree, as a polynomial function in these variables, which encodes the tree structure and is optimized over in the Merge operation represented as a sum operation in the deformed semiring. Since in that setting the Merge operation becomes expressed in terms of an optimization over continuous parameters, this also raises the question of how the discrete Hopf algebra Markov chain describing the action of Merge on workspaces can be mapped to a setting where continuous parameterizations (metric trees) are involved.

We do not address this question in the present paper, but we make some general considerations in this section on continuous parameterizations of metric trees, as a way of illustrating the problem and some possible approaches.

It was shown in [21] that, given a syntactic object $T \in \mathfrak{T}_{\mathcal{SO}_0}$, endowed with a head function h_T as in Definition 10.2 (see also [22], §1.13.3), one can assign a metric structure to the edges of T by a set of parameters $\lambda = (\lambda_v)_{v \in V^o(T)}$, where each variable λ_v is valued in $[0, 1]$, so that $\lambda \in [0, 1]^{n-1}$, with $n = \#L(T)$ and $V^o(T)$ the set of non-leaf vertices of T . The edge lengths are assigned so that the two edges below a given $v \in V^o(T)$ get lengths λ_v and $1 - \lambda_v$, as for example:



Lemma 14.1. *The datum of a head function is equivalent to a choice of which of the two edges gets weight λ_v and which $1 - \lambda_v$.*

Proof. As recalled in Definition 10.2, a head function $h_T : V^o(T) \rightarrow L(T)$ assigns a leaf to each internal (non-leaf) vertex, with the rule that if $T_w \subset T_v$ and $h_T(v) \in L(T_w)$ then $h_T(v) = h_T(w)$. Thus, if $h_T(v)$ is the leaf assigned to $v \in V^o(T)$ by the head function h_T and $\gamma_{h_T(v)}$ is the path in T from v to the leaf $h_T(v)$, we can assign the weights λ_v and $1 - \lambda_v$ according to the following rule: to e_v one of the two edges below a vertex $v \in V^o(T)$ we assign weight

$$(14.1) \quad \epsilon_{e_v} = \begin{cases} \lambda_v & e_v \in \gamma_{h_T(v)} \\ 1 - \lambda_v & e_v \notin \gamma_{h_T(v)} \end{cases}$$

Conversely, given an assignment of λ_v and $1 - \lambda_v$ to the pair of edges below v , for each $v \in V^o(T)$, the function $h_T : V^o(T) \rightarrow L(T)$ that assigns to v the leaf obtained by following the path from v consisting of edges marked by the variable λ_v (as opposed to $1 - \lambda_v$) is a head function. $\square$

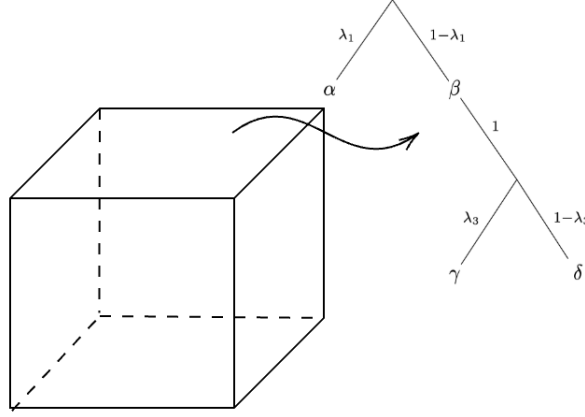
The labeling of the edges by these continuous variables induces a probability distribution $A^T = (a_\ell^T)_{\ell \in L(T)}$ on the leaves of T , as shown in [22], where each a_ℓ^T is a polynomial function $a_\ell^T = P_\ell^T(\lambda)$ of the variables λ_v , of the form

$$(14.2) \quad a_\ell^T = \prod_{v \in \gamma_{v_0, \ell}} \epsilon_{e_v},$$

where $\gamma_{v_0, \ell}$ is the path in T from the root vertex v_0 to the leaf ℓ .

The probability A is used in [21] to combine representations of lexical data $\alpha_\ell \in \mathcal{SO}_0$ in an appropriate function space, to obtain a representation of the syntactic object T .

We now consider the operation of restricting the point $\lambda = (\lambda_v)$ to one of the faces in the boundary of the cube $[0, 1]^{n-1}$, namely restricting one of the variables λ_v to 0 or 1, as shown in the figure below.



Lemma 14.2. *Taking one of the variables λ_v to 0 or 1 corresponds to the contraction of T/dT_w , for w the vertex below v connected by the edge e_v with $\epsilon_{e_v} \rightarrow 0$.*

Proof. Consider the probability distribution $A^T = (a_\ell^T)_{\ell \in L(T)}$ and the effect of taking one of the λ_v to be 0 or 1. If $\lambda_v = 0$, then the edge $e_{v,1}$ with $e_{v,1} \in \gamma_{h_T(v)}$ has length $\lambda_v = 0$ and the other edge $e_{v,2}$ below v has length $1 - \lambda_v = 1$. Thus the probability distribution A^T changes in the following way

$$(14.3) \quad a_\ell^T|_{\lambda_v=0} = \prod_{w \neq v \in \gamma_{v_0,\ell}} \epsilon_{e_w},$$

if the path $\gamma_{v_0,\ell}$ contains the edge $e_{v,2}$, and

$$(14.4) \quad a_\ell^T|_{\lambda_v=0} = 0,$$

if the path $\gamma_{v_0,\ell}$ contains the edge $e_{v,1}$, and is unchanged,

$$(14.5) \quad a_\ell^T|_{\lambda_v=0} = a_\ell,$$

if $\gamma_{v_0,\ell}$ contains neither $e_{v,1}$ or $e_{v,2}$. Observe then that (14.3), together with (14.5) (and (14.4) since those leaves are no longer present) gives exactly the resulting probability distribution A^{T/dT_w} , on the quotient tree T/dT_w , for $w = t(e_{v,1})$ the vertex below v along $e_{v,1}$, as shown in the example $\square$

This result extends from a single edge to an admissible cut as follows.

Corollary 14.3. *Let C be an admissible cut on T and let $F_{\underline{v}} = \pi_C(T)$ be the forest extracted by the admissible cut, with $\rho_C^{(d)}(T) = T/dF_{\underline{v}}$ the corresponding quotient, in the coproduct term $\pi_C(T) \otimes \rho_C^{(d)}(T)$. Then setting $\epsilon_e = 0$ for each edge $e \in C$ of the cut gives*

$$(14.6) \quad A^T|_{\epsilon_e=0, e \in C} = A^{T/dF_{\underline{v}}}.$$

Proof. As in Lemma 14.2 we have that $A^T|_{\epsilon_e=0, e \in C}$ is zero on all leaves in $F_{\underline{v}}$; is unchanged on all leaves where the path to the root does not contain any vertex above an edge of the cut C ; and, for leaves where the path contains an edge sister to an edge of C , is the product of the ϵ_e of all the other edges of the path that are not sister to a cut edge. A path from a leaf to the root cannot contain more than one edge of C because C is an admissible cut. Thus the resulting $A^T|_{\epsilon_e=0, e \in C}$ is exactly $A^{T/dF_{\underline{v}}}$. $\square$

We can also obtain the probability distribution of the extracted accessible terms T_v from A^T by specialization to some of the deeper strata of the boundary of the parameterizing cube $[0, 1]^{n-1}$.

Lemma 14.4. *Let $T_v \subset T$ be an accessible term of a syntactic object T , and let $\gamma_{v_0, v}$ be the path in T from the root vertex v_0 to v . Then specialization of the variables ϵ_e for all edges $e \in \gamma_{v_0, v}$ to $\epsilon_e = 1$ gives*

$$(14.7) \quad A^T|_{\epsilon_e=1, e \in \gamma_{v_0, v}} = A^{T_v}.$$

Proof. We have, for $\ell \in L(T_v)$

$$(14.8) \quad a_\ell^T|_{\epsilon_e=1, e \in \gamma_{v_0, v}} = \prod_{w \in \gamma_{v, \ell}} \epsilon_w,$$

while for $\ell \in L(T) \setminus L(T_v)$, the path $\gamma_{v_0, \ell}$ will contain a sister edge of one of the edges in $\gamma_{v_0, v}$. Since the weights ϵ_e of the edges in $\gamma_{v_0, v}$ are specialized to $\epsilon_e = 1$, then their sister edges are specialized to 0. This implies that for all $\ell \in L(T) \setminus L(T_v)$

$$(14.9) \quad a_\ell^T|_{\epsilon_e=1, e \in \gamma_{v_0, v}} = 0.$$

This then gives exactly the probability distribution A^{T_v} . $\square$

We can then associate the following geometric locus in the parameterizing cube, to the extraction of an accessible term $T_v \subset T$ and the corresponding quotient T/dT_v .

Proposition 14.5. *Let $T_v \subset T$ be an accessible term. To the term $T_v \otimes T/dT_v$ in the coproduct $\Delta^d(T)$ we can associate the pair $(A^{T_v}, A^{T/dT_v})$ of probability distributions on $L(T_v)$ and $L(T/dT_v) = L(T) \setminus L(T_v)$, respectively. When expressed in terms of the variables $\lambda = (\lambda_v)_{v \in V^o(T)}$, this pair of probabilities corresponds to a point $(\lambda^{T_v}, \lambda^{T/dT_v})$ in the product $\mathcal{Q}_{T_v} \times \mathcal{Q}_{T/dT_v}$ of two sub-cubes of the parameterizing cube $[0, 1]^{n-1}$, with dimensions $\dim \mathcal{Q}_{T_v} + \dim \mathcal{Q}_{T/dT_v} = n - 2$, with $\mathcal{Q}_{T_v} \simeq [0, 1]^{V^o(T_v)} \subset [0, 1]^{V^o(T)}$ and $\mathcal{Q}_{T/dT_v} \simeq [0, 1]^{V^o(T/dT_v)} \subset [0, 1]^{V^o(T)}$.*

Proof. In Lemma 14.2, the probability A^{T/dT_v} is a point in a stratum $\Delta_{L(T/dT_v)}$ of dimension $\#L(T/dT_v) - 1$ inside the $(n - 1)$ -simplex $\Delta_{L(T)}$. On the other hand, the locus inside the cube $[0, 1]^{n-1}$ cut out by the condition $\lambda_v = 0$ (or $\lambda_v = 1$) is a face $[0, 1]^{n-2}$ of codimension 1. The difference lies in the fact that the weights λ_w of vertices $w \in V^0(T_v)$ do not contribute to A^{T/dT_v} , because of (14.4), while they are coordinates in $[0, 1]^{n-2}$. Thus one can obtain a sub-locus of the face $[0, 1]^{n-2}$ that corresponds to points A^{T/dT_v} by fixing the value of the variables λ_w with $w \in V^0(T_v)$ (because of (14.4), it does not matter to which value: we will use $\lambda_w = 0$). We denote by $\mathcal{Q}_{T/dT_v} \subset [0, 1]^{n-1}$ for the resulting $(\#L(T/dT_v) - 1)$ -dimensional sub-cube, which we can identify with $\mathcal{Q}_{T/dT_v} \simeq [0, 1]^{V^o(T/dT_v)} \subset [0, 1]^{V^o(T)}$.

Similarly, in the case of Lemma 14.4, the probability A^{T_v} corresponds to a point in a stratum $\Delta_{L(T_v)}$ of dimension $\#L(T_v) - 1$ inside the $n - 1$ dimensional simplex $\Delta_{L(T)}$. On the other hand, the locus $\{\epsilon_e = 1, e \in \gamma_{v_0, v}\}$ determines a sub-cube $[0, 1]^{n-r-1} \subset \partial[0, 1]^{n-1}$ in the boundary of the parameterizing cube, where $r = \text{length}(\gamma_{v_0, v})$, which is of larger codimension, since in $A^{T_v} \in \Delta_{L(T_v)}$ the variables ϵ_w of all vertices w lying on paths from $L(T) \setminus L(T_v)$ to the path $\gamma_{v_0, v}$ no longer contribute. Thus, we can restrict to a cube $[0, 1]^{\#L(T_v)-1} \subset [0, 1]^{n-r-1}$ obtained by setting all the λ_w of these vertices to a fixed value. Again, it does not matter which one, since these weights do not contribute to A^{T_v} by (14.9). For later convenience we will set these values to 0. We denote by $\mathcal{Q}_{T_v} \subset [0, 1]^{n-1}$ the resulting $(\#L(T_v) - 1)$ -dimensional sub-cube, which we identify with $\mathcal{Q}_{T_v} \simeq [0, 1]^{V^o(T_v)} \subset [0, 1]^{V^o(T)}$.

Thus, for any accessible term $T_v \subset T$, we have obtained, by specialization of the variables in the parameterizing cube $[0, 1]^{n-1}$, a pair of lower dimensional cubes $\mathcal{Q}_{T_v}$ and $\mathcal{Q}_{T/dT_v}$ inside the boundary $\partial[0, 1]^{n-1}$, with $\dim \mathcal{Q}_{T_v} + \dim \mathcal{Q}_{T/dT_v} = n - 2$. $\square$

Corollary 14.6. *Under the Internal Merge operation $\mathfrak{M}(T_v, T/dT_v)$, the probability distribution $A^{\mathfrak{M}(T_v, T/dT_v)}$ at the leaves is a point in the join*

$$(14.10) \quad \Delta_{L(T_v)} \star \Delta_{L(T/dT_v)}$$

of the simplexes $\Delta_{L(T_v)}$ and $\Delta_{L(T/dT_v)}$, with $A^{\mathfrak{M}(T_v, T/dT_v)}$ a polynomial function as in (14.2) of $\lambda \in [0, 1] \times \mathcal{Q}_{T_v} \times \mathcal{Q}_{T/dT_v}$.

Proof. When we form the Internal Merge $\mathfrak{M}(T_v, T/dT_v)$, a new root vertex u is added, and correspondingly a new variable $\lambda_u \in [0, 1]$. The probability distribution at the leaves then satisfies

$$(14.11) \quad A^{\mathfrak{M}(T_v, T/dT_v)} = \begin{cases} \lambda_u A^{T_v} + (1 - \lambda_u) A^{T/dT_v} & h_T(v_0) \in L(T_v) \\ (1 - \lambda_u) A^{T_v} + \lambda_u A^{T/dT_v} & h_T(v_0) \notin L(T_v), \end{cases}$$

with v_0 the root of T . Namely $A^{\mathfrak{M}(T_v, T/dT_v)}$ is a point in the join (14.10), which can again be identified with the simplex $\Delta_{L(T)}$. In terms of the parameterizing cube, by Proposition 14.5, the probabilities $A^{\mathfrak{M}(T_v, T/dT_v)} \in \Delta_{L(T_v)} \star \Delta_{L(T/dT_v)}$ can be written as polynomial functions $a_\ell^{\mathfrak{M}(T_v, T/dT_v)} = \mathbb{P}_\ell(\lambda)$ of the form (14.2), for $\lambda \in [0, 1] \times \mathcal{Q}_{T_v} \times \mathcal{Q}_{T/dT_v}$. $\square$

We have obtained above a description of the Internal Merge operation in terms of a parameterizing cube $[0, 1]^n = [0, 1] \times [0, 1]^{n-1}$. Starting with a point $\lambda^T \in \{0\} \times [0, 1]^{n-1}$, with $A^T = P(\lambda^T) \in \Delta_{L(T)}$, one first assigns to it the pair of points $(\lambda^{T_v}, \lambda^{T/dT_v}) \in \mathcal{Q}_{T_v} \times \mathcal{Q}_{T/dT_v}$, with $(A^{T_v} = P(\lambda^{T_v}), A^{T/dT_v} = P(\lambda^{T/dT_v})) \in \Delta_{L(T_v)} \times \Delta_{L(T/dT_v)}$, by the specialization of variables as above. One then combines these to obtain a point as in (14.11) with $A^{\mathfrak{M}(T_v, T/dT_v)} = P(\lambda_u, \lambda^{T_v}, \lambda^{T/dT_v}) \in \Delta_{L(T_v)} \star \Delta_{L(T/dT_v)}$, for $(\lambda_u, \lambda^{T_v}, \lambda^{T/dT_v}) \in [0, 1] \times [0, 1]^{V^o(T_v)} \times [0, 1]^{V^o(T/dT_v)} \subset [0, 1]^n$. Thus, we can think of Internal Merge as the result of a continuous flow inside $[0, 1]^n$ that goes from a point in the sub-cube $\{0\} \times [0, 1]^{V^o(T)}$ to a point in the sub-cube $[0, 1] \times [0, 1]^{V^o(T_v)} \times [0, 1]^{V^o(T/dT_v)}$. One can then ask whether these Internal Merge transformations can be realized as part of a continuous flow on the parameterizing cube $[0, 1]^n$. Similarly, one can look at the EM and minimal SM transformations in terms of their effects on the probabilities $A^T = P(\lambda^T)$ and the parameters $\lambda^T \in [0, 1]^{V^o(T)}$, as a possible way to embed the Merge dynamics into a realization of syntactic objects as metric trees inside some semantic space (as in Chapter 3 of [22]) or in some function space representation as in [21]. Any such continuous flow would have to satisfy some compatibility condition with the discrete Markov chain that we described in the previous sections for the action of Merge on workspaces, in order to represent a realization of the Merge dynamics.

A related question arises when one considers the syntactic objects, endowed with an assignment of continuous parameters along the edges, as points in the BHV moduli spaces of [4]. The walls between the open cells of the BHV moduli space correspond to the trees where one (or more in the higher codimensional strata) of the edge variables are equal to 0. A wall crossing between two adjacent open cells separated by a codimension one boundary stratum corresponds to a transformation (at one of the internal vertices of T) of the form

$$\begin{array}{c} \diagup \quad \diagdown \\ T_{v,1} \quad T_{v,2} \quad T_{v'} \end{array} \mapsto \begin{array}{c} \diagup \quad \diagdown \\ T_{v,1} \quad T_{v,2} \quad T_{v'} \end{array}.$$

Since wall crossings in BHV happen in this way, by moving an edge across one of the non-leaf vertices, operations like IM that extract an accessible term, possibly from deep inside the tree,

and merges it back at the root, do not seem directly compatible with the form of the BHV boundary strata, in the sense that the terms $T_v \otimes T/dT_v$ do not seem to naturally correspond to codimension-one strata (or $F_v \otimes T/dF_v$ to higher codimensional strata).

Question 14.7. *Can the discrete Merge Hopf algebra Markov chain $\hat{\mathcal{K}}^{(A,n)}$ (and its weighted version) be embedded as a continuous Markov chain inside either the parameterizing cube $[0, 1]^n$ or the moduli space BHV_n , or in topological models of semantic spaces?*

We leave these considerations about continuous parameters, parameterizing spaces, and flows compatible with the discrete Markov chain as questions for future work.

15. I-LANGUAGE CONSTRAINTS AND COLORED MERGE DYNAMICS

The Hopf algebra Markov chain of the action of Merge on workspaces is the dynamical model of *free* structure formation in syntax through the free symmetric Merge.

In the Minimalist Model of syntax, freely formed structures produced by the free symmetric Merge are then filtered for compliance with theta role assignments (theta criterion) and for well formed phases (head and complement structure, extended projection). Both of these systems of filtering can be modeled mathematically in terms of bud generating systems of colored operads, as explained in [25] and [24].

It is also shown in [25] and [24] that filtering freely formed structures by these coloring rules is equivalent to imposing constraints on the Merge action and form only those structures that are compatible with the constraints. The reason for this equivalence is that all the bud generators of the relevant colored operads (see §15.1 for the definitions) are corollas with a single node, two input leaves, and one output root. One refers to this constrained form of Merge as “colored Merge”.

When interpreted in this way, one sees that the effect of selecting only those syntactic objects with well formed phases and theta roles corresponds to pruning the free Merge dynamics given by the Hopf algebra Markov chain $\hat{\mathcal{K}}^{A,n}$ on the Merge graph $\mathcal{G}_{n,A}$, so that the only arrows remaining are those that have coloring corresponding to one of the bud generators and the only vertices remaining are those workspaces that are obtainable through these colored Merge operations, with the resulting coloring, so that all the component trees of the workspace are elements of the given colored operad.

In this colored Merge formulation, instead of the free Merge operations described, as we have seen above, in the form

$$\mathfrak{M}_{S,S'} = \sqcup \circ (\mathcal{B} \otimes \text{id}) \circ \delta_{S,S'} \circ \Delta,$$

one considers a version of the form

$$(15.1) \quad \mathfrak{M}_{S,S'}^c = \sqcup \circ (\mathcal{B}^c \otimes \text{id}) \circ \delta_{c_S, c_{S'}}^c \delta_{S,S'} \circ \Delta,$$

where $c, c_S, c_{S'} \in \Omega$ are colors and

$$(15.2) \quad \begin{array}{c} c \\ \swarrow \quad \searrow \\ c_S \quad c_{S'} \end{array}$$

belong to the set $\mathbb{B}$ of bud generators of the colored operad $\mathfrak{O}_{\Omega, \mathbb{B}}$ (see §15.1 for more details), and S, S' are syntactic objects that belong to the colored operad, with c_S and $c_{S'}$ the colors assigned to the root vertex of S and S' respectively. The resulting $\mathfrak{M}^c(S, S')$ has root vertex colored by the color c and the remaining vertices colored according to the coloring of S and S' .

Let Ω be the set of colors ($\Omega_\Theta, \Omega_\Phi$ for theta roles and phases, or $\Omega_{\Theta, \Phi}$ for the combined case as discussed in [24]).

The colored Merge graph $\mathcal{G}_{n,A}^\Omega$ (in particular $\mathcal{G}_{n,A}^\Theta$ or $\mathcal{G}_{n,A}^\Phi$, or $\mathcal{G}_{n,A}^{\Theta,\Phi}$), can be obtained from $\mathcal{G}_{n,A}$ in two possible ways, reflecting the equivalent ways of thinking of coloring as filtering of already formed structures or as building via colored Merge. We describe these two constructions in §15.2 and §15.3 after recalling briefly the notions of colored operad and bud generating systems in §15.1.

15.1. Colored operads and bud generating systems. A colored operad (in the category of sets) is a collection of sets

$$\mathfrak{D} = \{\mathfrak{D}(c, \underline{c}) \mid \underline{c} = (c_1, \dots, c_n), n \geq 1, c, c_i \in \Omega\}$$

with Ω a (finite) set of colors, with composition operations

$$(15.3) \quad \begin{aligned} \gamma : \mathfrak{D}(c, (c_1, \dots, c_n)) \times \mathfrak{D}(c_1, (c_{1,1}, \dots, c_{1,k_1})) \times \dots \times \mathfrak{D}(c_n, (c_{n,1}, \dots, c_{n,k_n})) \\ \rightarrow \mathfrak{D}(c, (c_{1,1}, \dots, c_{1,k_1}, \dots, c_{n,1}, \dots, c_{n,k_n})) \end{aligned}$$

subject to an associativity property (which we do not write out explicitly here). An algebra over a colored operad is a collection of sets $A = \{A_c\}_{c \in \Omega}$ with an action

$$\gamma_A : \mathfrak{D}(c, (c_1, \dots, c_n)) \times A_{c_1} \times \dots \times A_{c_n} \rightarrow A_c$$

satisfying a compatibility condition with the composition γ (which we also do not write explicitly here).

A convenient way of constructing colored operads, which is the basis for the formalization of theta roles and phase structure in [25] and [24], is through *bud generating systems*, as in [17].

One starts with an ordinary operad $\mathfrak{D}$ (non-colored) and a finite set of colors Ω . First one forms a colored operad by all possible assignments of colors,

$$(15.4) \quad \mathbb{B}_\Omega(\mathfrak{D})(n) := \Omega \times \mathfrak{D}(n) \times \Omega^n.$$

Then one selects in $\mathbb{B}_\Omega(\mathfrak{D})(n)$ a finite set $\mathcal{R} \subset \mathbb{B}_\Omega(\mathfrak{D})$ of generators for the actual (more restrictive) colored operad one wishes to obtain.

One defines a *bud generating system* $\mathbb{B} = (\mathfrak{D}, \Omega, \mathcal{R}, \mathcal{I}, \mathcal{T})$ as the data of the (non-colored) operad $\mathfrak{D}$, the finite set of colors Ω , separated into a set $\mathcal{I}$ of initial colors and a set $\mathcal{T}$ of terminal colors, and the choice of the set of generators $\mathcal{R} \subset \mathbb{B}_\Omega(\mathfrak{D})$ (the local coloring rules).

One then obtains a colored operad $\mathfrak{D}_{\Omega, \mathbb{B}}$ generated by the bud system $\mathbb{B} = (\mathfrak{D}, \Omega, \mathcal{R}, \mathcal{I}, \mathcal{T})$ as

$$(15.5) \quad \mathfrak{D}_{\Omega, \mathbb{B}}(n) := \{x = (c, T, \underline{c}) \in \mathbb{B}_\Omega(\mathfrak{D})(n) \mid \mathbf{1}_c \rightarrow_{\mathbb{B}} x, c \in \Omega \setminus \mathcal{T}, \underline{c} \in (\Omega \setminus \mathcal{T})^n\},$$

where $\mathbf{1}_c \rightarrow_{\mathbb{B}} x$ means that x is obtained from the colored unit $\mathbf{1}_c$ through a sequence of operad compositions of generators $r_1, \dots, r_N \in \mathcal{R}$.

The language $\mathcal{L}(\mathbb{B})$ generated by the bud system $\mathbb{B} = (\mathfrak{D}, \Omega, \mathcal{R}, \mathcal{I}, \mathcal{T})$ is an algebra over the colored operad $\mathfrak{D}_{\Omega, \mathbb{B}}$ given by

$$(15.6) \quad \mathcal{L}(\mathbb{B}) = \{x = (c, T, \underline{c}) \in \mathbb{B}_\Omega(\mathfrak{D}) \mid \mathbf{1}_c \rightarrow_{\mathbb{B}} x, c \in \mathcal{I}, \underline{c} \in \mathcal{T}^n\}.$$

15.2. Colored Merge graph via filtering. For each $\alpha \in \mathcal{SO}_0$ there will be a subset $\Omega_\alpha \subset \Omega$ of colors that are compatible with α . Thus, for $A = \{\alpha_1, \dots, \alpha_n\}$ the lexical items at the leaves, we have corresponding choices of coloring

$$(15.7) \quad \Omega_A := \Omega_{\alpha_1} \times \dots \times \Omega_{\alpha_n}.$$

Given a choice of coloring $(c_1, \dots, c_n) \in \Omega_A$ of the leaves $\alpha_1, \dots, \alpha_n$ of F , for each vertex $F = T_1 \sqcup \dots \sqcup T_r \in \mathfrak{F}_{A,n}$ of the graph $\mathcal{G}_{n,A}$, and for each component T_i of F , consider all the possible ways in which T_i can be colored as an element of $\mathcal{L}(\mathbb{B})(k_i) \subset \mathfrak{D}_{\Omega, \mathbb{B}}(k_i)$, with $k_i = \#L(T_i)$.

If at least one of the components does not have any possible coloring in $\mathcal{L}(\mathbb{B})$ (for any available choice of coloring of the leaves), then the vertex F does not contribute any vertices to $\mathcal{G}_{n,A}^\Omega$. If there are colorings, then F contributes a number of vertices of $\mathcal{G}_{n,A}^\Omega$ “above F ”, equal to the number of different possible colorings.

For each of the coloring of an $F \in \mathfrak{F}_{A,n}$, and each Merge arrow (namely each $\mathcal{K}_{F,F'}^{A,n} = 1$ entry of the adjacency matrix of $\mathcal{G}_{n,A}$), there is an arrow in $\mathcal{G}_{n,A}^\Omega$ to one of the colorings of F' , since we know that the colorings can be described via colored Merge operations, and each coloring of F is the target of an arrow from one of the coloring of some F'' with $\mathcal{K}_{F'',F}^{A,n} = 1$.

The graph consisting of these vertices and arrows is the Colored Merge graph $\mathcal{G}_{n,A}^\Omega$. Correspondingly this gives a Colored Merge Markov chain $\hat{\mathcal{K}}^{A,n,\Omega}$, where $\mathcal{K}^{A,n,\Omega}$ is the adjacency matrix of $\mathcal{G}_{n,A}^\Omega$, and $\hat{\mathcal{K}}^{A,n,\Omega}$ is obtained as in (3.6).

Note that the relation between $\mathcal{G}_{n,A}^\Omega$ and $\mathcal{G}_{n,A}$ is not straightforward, because there is no projection map between them, to which we could apply our previously developed technique. There is rather a projection $p^\Omega : \mathcal{G}_{n,A}^\Omega \rightarrow \mathcal{G}_{n,A}^{\text{pruned}}$ to a pruned version of $\mathcal{G}_{n,A}$, where one removes all the vertices F that admit no coloring in $\mathcal{L}(\mathbb{B})$ compatible with the available colorings of the leaves and of the vertices adjacent to them.

The pruning that gives $\mathcal{G}_{n,A}^{\text{pruned}}$ will in general break the symmetry of $\mathcal{G}_{n,A}$ with respect to the projection $p : \mathcal{G}_{n,A} \rightarrow \mathcal{G}_{\mathcal{P}'(n)}$, making the dynamics of the Colored Merge Markov chain $\hat{\mathcal{K}}^{A,n,\Omega}$ more difficult to analyze than the Hopf algebra Markov chain $\hat{\mathcal{K}}^{A,n}$ of the free Merge action. Moreover, while the behavior of $\hat{\mathcal{K}}^{A,n}$ is independent of the specific choice of the lexical data $A = \{\alpha_1, \dots, \alpha_n\}$ at the leaves, the dynamics of $\hat{\mathcal{K}}^{A,n,\Omega}$ is sensitive to these data, and will be different for different choices of A , since the set $\Omega_A \subset \Omega^n$ of available colorings at the leaves changes with A and this set Ω_A determines which vertices are pruned in $\mathcal{G}_{n,A}^{\text{pruned}}$ and what the multiplicities (the sizes of the fibers) are in $p^\Omega : \mathcal{G}_{n,A}^\Omega \rightarrow \mathcal{G}_{n,A}^{\text{pruned}}$.

This has a significant implication for the linguistic model. We know it is in principle equivalent to think of structure formation in syntax as being performed by the free symmetric Merge (the dynamical system $\hat{\mathcal{K}}^{A,n}$, followed by filtering by coloring for phases and theta roles of the resulting fully formed structures $T \in p^{-1}(\{n\})$, or else only form structures via the Colored Merge operations, so that filtering by colorability happens at each step and the dynamics is governed by $\hat{\mathcal{K}}^{A,n,\Omega}$. These processes are equivalent in the sense that they give rise to the same set of well-colored objects in $\mathcal{L}(\mathbb{B})$ with underlying trees $T \in p^{-1}(\{n\})$.

However, $\hat{\mathcal{K}}^{A,n}$ and $\hat{\mathcal{K}}^{A,n,\Omega}$ are not equivalent as dynamical systems. This difference can be viewed as an analog of the difference between having the same weak generative capacity or the same strong generative capacity in the setting of formal languages. Both $\hat{\mathcal{K}}^{A,n,\Omega}$ or $\hat{\mathcal{K}}^{A,n}$ followed by colorability-filtering of the $T \in p^{-1}(\{n\})$ produce the same elements in $\mathcal{L}(\mathbb{B})$, and this can be seen as analogous to having the same weak generative capacity for formal languages. However, the dynamical processes of structure building involved are not in themselves equivalent.

From the dynamical systems point of view, it is more natural to think of structure formation as achieved by the free Merge via $\hat{\mathcal{K}}^{(A,n)}$ (and the weighted version) followed by filtering, so that filtering can be understood as conditioning on the stationary distribution π of the weighted free Merge dynamics.

15.3. Colored Merge graph via dynamical building. We also describe here how one can equivalently think of the construction of $\mathcal{G}_{n,A}^\Omega$ as a recursive process of replacing Merge arrows $\mathfrak{M}_{S,S'}$ with colored Merge arrows $\mathfrak{M}_{S,S'}^c$. One first needs to assume that S and S' have been colored, and in particular one needs to know the colors c_S and $c_{S'}$ assigned to their root vertices, and then this determines that the original arrow $\mathfrak{M}_{S,S'}$ is replaced by a collection arrows $\mathfrak{M}_{S,S'}^c$, one for each c with the property that (15.2) is in the set of colored operad generators. When new arrows out of the vertices $\mathfrak{M}_{S,S'}^c$ are similarly analyzed, the SM and IM arrows will add new coloring to the original S and S' , and so on until all allowed coloring and all allowed colored Merge operations are obtained.

One can start with the workspaces $F \in p^{-1}(\wp = \{2, \underline{1}^{n-2}\})$, namely the workspaces with a single cherry component and all the other components that are single leaves. Without loss of generality, say that $F = \mathfrak{M}(\alpha_1, \alpha_2) \sqcup \alpha_3 \sqcup \cdots \sqcup \alpha_n$.

For the single leaf components $\alpha_3 \sqcup \cdots \sqcup \alpha_n$ we have $\#(\Omega_{\alpha_3} \times \cdots \times \Omega_{\alpha_n})$ possible coloring, while for $\mathfrak{M}(\alpha_1, \alpha_2)$ the set of possible coloring

$$\Omega_{\alpha_1, \alpha_2} \subset \Omega \times \Omega_{\alpha_1} \times \Omega_{\alpha_2}$$

is the subset

$$\Omega_{\alpha_1, \alpha_2} = \{(c, c_1, c_2) \mid \begin{array}{c} c \\ \diagup \quad \diagdown \\ c_1 \quad c_2 \end{array} \in \mathbb{B}\}.$$

Thus, over each vertex $F \in p^{-1}(\wp = \{2, \underline{1}^{n-2}\})$ in $\mathcal{G}_{n,A}$ there will be $N_{\alpha_1, \alpha_2} \cdot \prod_{i=3}^n N_{\alpha_i}$ vertices in $\mathcal{G}_{n,A}^\Omega$, where $N_{\alpha_i} = \#\Omega_{\alpha_i}$ and $N_{\alpha_i, \alpha_j} = \#\Omega_{\alpha_i, \alpha_j}$.

One can then consider the EM arrows from the vertices of $\mathcal{G}_{n,A}^\Omega$ above the vertices in $p^{-1}(\wp = \{2, \underline{1}^{n-2}\})$ in $\mathcal{G}_{n,A}$ and color the resulting component $\mathfrak{M}(\mathfrak{M}(\alpha_i, \alpha_j), \alpha_k)$ by assigning color c' to the root vertex of $\mathfrak{M}(\mathfrak{M}(\alpha_i, \alpha_j), \alpha_k)$ if the colors c and c_k of the root of $\mathfrak{M}(\alpha_i, \alpha_j)$ and of α_k satisfy

$$\begin{array}{c} c' \\ \diagup \quad \diagdown \\ c \quad c_k \end{array} \in \mathbb{B},$$

while no EM arrow exists if there is no $c' \in \Omega$ with this property.

Iterating this process, for a given $\wp \in \mathcal{P}'(n)$, one considers all the workspaces $F \in p^{-1}(\wp)$ as vertices in $\mathcal{G}_{n,A}$. Since each such F is the target of some EM map, we obtain a collection of vertices in $\mathcal{G}_{n,A}^\Omega$ above F , obtained by the coloring of targets of EM maps as above, assuming that the source of these EM arrows have already been colored by the same method.

After obtaining all the possible coloring by colored EM, above each $F \in p^{-1}(\wp)$ for all $\wp \in \mathcal{P}'(n)$, we consider the IM arrows at each $F \in p^{-1}(\wp)$ where $\wp$ has at least one $k_i \geq 3$ (so that there are IM arrows). For each of the colored vertices above a given $F \in p^{-1}(\wp)$, we consider all the possible colorings of IM arrows.

As discussed in [25] and [24], in this case the coloring of the tree $\mathfrak{M}(T_v, T/dT_v)$ is obtained as two colored Merge operations: $\mathfrak{M}_{T_v, 1}^c$ with coloring

$$(15.8) \quad \begin{array}{c} c_1 \\ \diagup \quad \diagdown \\ c_v \quad c_0 \end{array} \in \mathbb{B}$$

with c_v the color of the vertex v in T , c_0 the coloring assigned to the unit element 1, and c_1 is another specified color for this kind of generator (like the edge-of-phase color $\mathfrak{s}^\downarrow$ in the case of phases or the non-theta position color for theta roles). This is then followed by a colored Merge $\mathfrak{M}^c(\mathfrak{M}_{T_v, 1}^c, T/dT_v)$, with

$$(15.9) \quad \begin{array}{c} c' \\ \diagup \quad \diagdown \\ c_1 \quad c \end{array} \in \mathbb{B}$$

for c the color of the root of T . Since the leaf marked with 1 is not counted as a leaf in $\mathfrak{M}(T_v, T/dT_v)$, this coloring is equivalent to just the condition (15.9) with c the color of the root of T and c_1 the color newly assigned to v (as replacement of its original color c_v).

Whenever a triple of colors satisfying the required condition does not exist, then the corresponding IM arrow of $\mathcal{G}_{n,A}$ is pruned and does not exist as an edge of $\mathcal{G}_{n,A}^\Omega$. These IM arrows will add

new colored vertices above $F \in p^{-1}(\varphi)$, as they may produce colorings not yet achieved by EM arrows. These vertices are added to $\mathcal{G}_{n,A}^\Omega$.

Similarly, one can consider from all the obtained vertices of $\mathcal{G}_{n,A}^\Omega$ all the possible minimal SM arrows and consider the corresponding coloring conditions. As shown in [26], these will use additional generators in $\mathbb{B}$ not used by EM and IM, hence they will create new possible colorings, hence adding new vertices to $\mathcal{G}_{n,A}^\Omega$.

One can then take again all the EM, IM, SM arrows from all the colored structures obtained, and see if they give rise to new colorings. The process stabilizes because the underlying tree structures are unchanged hence there is a maximal number of possible colorings. This produces a colored Merge graph $\mathcal{G}_{n,A}^\Omega$ from the Merge graph $\mathcal{G}_{n,A}$.

16. PARAMETER SETTING AND EXTERNALIZATION

There is another very important form of filtering on the syntactic objects formed by the free symmetric Merge, in addition to the “universal” (I-language) filtering for phases and theta roles, namely the filtering by language-specific syntactic parameters. The free symmetric Merge is seen as the core computational structure of the Universal Grammar, while the parametric variation of individual languages is modeled (since the original Principles and Parameters formulation) through a procedure of parameter-setting, which happens during language acquisition, and specializes the universal computational mechanism to the specifics of one particular language.

This process can be seen as further filtering the structures in $\mathcal{L}(\mathbb{B})$ obtained through the $\hat{\mathcal{K}}^{A,n,\Omega}$ dynamics (or the $\hat{\mathcal{K}}^{A,n}$ dynamics followed by colorability-filtering) according to language-specific filters, as well as a simultaneous assignment of a planar structure to the trees (also dependent on language-specific word order parameters). This process is referred to as Externalization.

Some discussion of Externalization in the context of the mathematical model of Minimalism is included in §1.12 of [22], and a more extensive discussion will appear elsewhere ([23], [32]), so we will not discuss here a general model of Externalization and syntactic parameters. We will, however, make some comments on how the process of parameter setting can be looked at from the perspective of the Markov chains $\hat{\mathcal{K}}^{A,n}$ and $\hat{\mathcal{K}}^{A,n,\Omega}$.

We give only a very impressionistic view here of how the process of parameter setting can be seen from this dynamical system viewpoint. A full investigation will have to incorporate a more detailed model of syntactic parameters and the process of parameter setting, some of which is being developed in [23], [32], so it will be left to future work.

We work here within the model of Externalization proposed in §1.12 of [22], where we regard Externalization as a two-step process: a section $\sigma_{\mathcal{L}}$ that assigns planar structure to trees (in a language-dependent way) and a projection $\Pi_{\mathcal{L}}$ that filters out structures (also in a language dependent way), thus resulting in a diagram of the form

$$\begin{array}{ccc} & \sigma_{\mathcal{L}} \nearrow & \mathfrak{T}_{SO_0}^{pl} \\ & \searrow & \downarrow \Pi_{\mathcal{L}} \\ \mathfrak{T}_{SO_0} & & \mathfrak{T}_{SO_0,\mathcal{L}}^{pl} \end{array}$$

with $\mathfrak{T}_{SO_0}^{pl}$ the set of planar binary rooted trees, and $\mathfrak{T}_{SO_0,\mathcal{L}}^{pl}$ the set of those planar binary rooted trees that are not filtered out by the syntactic parameters of the language $\mathcal{L}$.

Behind this picture we assume that there is a space of syntactic parameters (which we take here to be binary valued for simplicity, though it is often useful to allow a third “undefined” value). We identify the space of all possible parameter values with the binary cube $\mathbb{F}_2^N = \{0,1\}^N$, where

N is the total number of parameters. We assume here, also for simplicity of discussion, that these N parameters are independent binary variables, although in fact the problem of relations between syntactic parameters is one of the crucial questions in the modeling of Externalization, but for our purposes here we can temporarily ignore it, and assume that an actual sufficient set of independent parameters can be identified. Among these N parameters, a certain number $M < N$ deal with word order properties and have the effect of determining the section $\sigma_{\mathcal{L}}$. The remaining parameters determine the projection $\Pi_{\mathcal{L}}$. (For our purposes here it does not matter how exactly $\sigma_{\mathcal{L}}$ and $\Pi_{\mathcal{L}}$ are obtained from the parameter values, except for a few considerations that we make below.)

As part of the model, we assume that the lexical items $\alpha_{\ell} \in \mathcal{SO}_0$ at the leaves of the syntactic objects carry associated data φ_{ℓ} from a finite set Φ of syntactic features. The syntactic parameters $\mathbf{p} = \mathbf{p}(\varphi \in \Phi)$, are functions of the features $\varphi \in \Phi$, where typically each syntactic parameter depends only on a certain subset of features. Thus, a syntactic parameter $\mathbf{p}$ affects (as planarization or as filtering) a certain structure F^{Ω} (a vertex of $\mathcal{G}_{n,A}^{\Omega}$) iff those features $\varphi \in \Phi$ that $\mathbf{p}$ depends on are present at the leaves of the components of F^{Ω} . We write $\Phi_{\mathbf{p}} \subset \Phi$ for the subset of features that the parameter $\mathbf{p}$ depends on.

Parameter values are set in the process of language acquisition. As noted in [20], these two kinds of parameters behave differently in that respect. For syntactic parameters of the type that we associate to determining the projection $\Pi_{\mathcal{L}}$, there is empirical evidence for a default value of $-$ (or $0 \in \mathbb{F}_2$), and positive evidence in language acquisition is required to flip the value to $+$ ($1 \in \mathbb{F}_2$). Thus, we can assume that the subspace $\mathbb{F}_2^{N-M}$ is initialized at the $\underline{0}$ vector. The case of the word-order parameters, on the other hand, does not appear to have a default state. So it is suggested in [20] that these parameters should have an “unspecified” initial state, out of which the two possible $-/+$ states emerge as a symmetry breaking phenomenon. In the model we present here, constructing the section $\sigma_{\mathcal{L}}$ can be framed as a setting where word-order parameters are initialized in an undefined state and positive evidence sets them (equivalently, recursively constructs $\sigma_{\mathcal{L}}$). On the other hand, the analysis in [23] of the Final-Over-Final-Condition suggests that one may assume a default harmonic head-final planar embedding determined by the head function, with respect to which other embeddings are obtained via a group action, with the FOFC representing a minimality property with respect to generators of that group action. Note that all the syntactic objects that are colorable for phases have a head function, so the default assumption on the embedding is possible after filtering by well-structured phases. For simplicity, we just assume here that there is no default value for word order parameters (hence no default choice of planar embedding).

Assume that we start with the graph $\mathcal{G}_{n,A}^{\Omega}$, so that the filtering by theta roles and phases have already been taken care of.

As a first step, we need to perform a language-specific choice of planar embeddings (the section $\sigma_{\mathcal{L}}$) on the component trees T_i^{Ω} in the workspaces F^{Ω} (the vertices of $\mathcal{G}_{n,A}^{\Omega}$, constructed as described in the previous section). The Ω superscript is used here to remember that these trees are colored, $T_i^{\Omega} \in \mathcal{L}(\mathbb{B})$.

To that purpose, we can take, above each vertex $F^{\Omega} = T_1^{\Omega} \sqcup \cdots \sqcup T_r^{\Omega}$ of $\mathcal{G}_{n,A}^{\Omega}$, a collection of

$$2^{n-r} = \prod_{i=1}^r 2^{k_i-1}$$

vertices, given by all the possible choice of planar embeddings for each of the components T_i^{Ω} . (A full binary rooted tree T with labelled leaves has $2^{\#L(T)-1}$ planar embeddings.)

A priori, none of these vertices is selected. However, exposure to positive examples in language acquisition will select some $\sigma_{\mathcal{L}}(T^{\Omega})$, namely selects a particular planar embedding for some syntactic objects $T^{\Omega} \in \mathcal{G}_{n,A}^{\Omega}$. This selection automatically causes further selection of the same planar embedding wherever $\sigma_{\mathcal{L}}(T^{\Omega})$ occurs: as a component of some F^{Ω} in some other $\mathcal{G}_{m,A'}^{\Omega}$ with A' another set of lexical data, sharing the same features that the word-order parameters depend on as A . It also selects $\sigma_{\mathcal{L}}(T^{\Omega})$ when T^{Ω} occurs as an accessible term inside some larger $\tilde{T}^{\Omega}$, unless some other word-order parameter causes a change of planar embedding of T^{Ω} caused by other data present in the rest of $\tilde{T}^{\Omega}$. A locality and finiteness assumption on syntactic parameters (see [23], [32] for further discussion) implies that all $\sigma_{\mathcal{L}}(T^{\Omega})$ of arbitrary size, are determined by knowing the embeddings $\sigma_{\mathcal{L}}$ on the graphs $\mathcal{G}_{n,A}^{\Omega}$ with $n \leq n_0$ small, by identifying the planarization of accessible terms and quotient terms of small size, as discussed below. (See [23], [32] for specific bounds on the size n_0 needed.)

This process selects a set of vertices $\sigma_{\mathcal{L}}(F^{\Omega})$, which has the same cardinality as the set of vertices F^{Ω} of $\mathcal{G}_{n,A}^{\Omega}$, since $\sigma_{\mathcal{L}}$ assigns a unique planar structure to each F^{Ω} (we are not discussing here scrambling phenomena and how to account for languages with a relatively free word order). We can equivalently write these vertices as pairs $(F^{\Omega}, \pi_{\mathcal{L}})$ where $\pi_{\mathcal{L}}$ is a choice of one among the 2^{n-r} planar embeddings.

The remaining $N - M$ parameters that determine the projection $\Pi_{\mathcal{L}}$ are initialized as $\underline{0} \in \mathbb{F}_2^{N-M}$. Unless positive examples during language acquisition flip the values of some of these parameters to 1, they will remain in the 0 default value. A syntactic parameter $\mathbf{p}$ having default value 0 means that all structures $F^{\Omega} \in \mathcal{G}_{n,A}^{\Omega}$ where the syntactic features φ_{ℓ} associated to lexical data α_{ℓ} at the leaves $\ell \in L(T)$ are involved in determining the parameter $\mathbf{p}$ are filtered out in the projection $\Pi_{\mathcal{L}}$.

We assume two basic properties of the projection $\Pi_{\mathcal{L}}$.

- The projection $\Pi_{\mathcal{L}}$ decomposes into a product $\Pi_{\mathcal{L}} = \prod_{\mathbf{p}} \Pi_{\mathcal{L},\mathbf{p}}$. Thus, a structure F^{Ω} passes the filter iff it passes it with respect to each parameter, $\Pi_{\mathcal{L},\mathbf{p}}(F^{\Omega}) = F^{\Omega}$ for all $\mathbf{p}$.
- For $F^{\Omega} = T_1^{\Omega} \sqcup \dots \sqcup T_r^{\Omega}$, we have $\Pi_{\mathcal{L}}(F^{\Omega}) = 0$ iff $\Pi_{\mathcal{L}}(T_i^{\Omega}) = 0$ for some $1 \leq i \leq r$.

Initializing all the $N - M$ parameters to 0 means that at the start of the process *all* syntactic objects are filtered out, so nothing passes the Externalization filter. Suppose then that positive examples are encountered in the language acquisition process that cause one of the parameters $\mathbf{p}$ to switch to value 1. This change will affect only those structures $F^{\Omega} \in \mathcal{G}_{n,A}^{\Omega}$ where the features at the leaves satisfy $\{\varphi_{\ell}\}_{\ell \in L(F^{\Omega})} \supset \Phi_{\mathbf{p}}$.

By the considerations above on the projection $\Pi_{\mathcal{L}}$, we can assume that $F^{\Omega} = T^{\Omega}$ has a single component, and that the set of features $\{\varphi_{\ell}\}_{\ell \in L(T^{\Omega})}$ contains the set $\Phi_{\mathbf{p}}$.

To model the way in which changing the value of $\mathbf{p}$ to 1 changes $\Pi_{\mathcal{L},\mathbf{p}}$, we can first assume that positive examples suffice to determine which among the tree T^{Ω} with sufficiently small size $\#L(T^{\Omega}) \leq n_0$ and with $\{\varphi_{\ell}\}_{\ell \in L(T^{\Omega})} \supset \Phi_{\mathbf{p}}$ are now in the range of $\Pi_{\mathcal{L},\mathbf{p}}$ and no longer in the kernel. In other words, the set

$$(16.1) \quad \mathcal{T}_{\mathbf{p}} := \{T^{\Omega} \mid \#L(T^{\Omega}) \leq n_0 \text{ and } \{\varphi_{\ell}\}_{\ell \in L(T^{\Omega})} \supset \Phi_{\mathbf{p}}\}$$

is separated out into two subsets

$$(16.2) \quad \mathcal{T}_{\mathbf{p}} = \mathcal{T}_{\mathbf{p}}^{-} \sqcup \mathcal{T}_{\mathbf{p}}^{+}$$

where $\Pi_{\mathcal{L},\mathbf{p}}|_{\mathcal{T}_{\mathbf{p}}^{+}} = \text{id}$ and $\Pi_{\mathcal{L},\mathbf{p}}|_{\mathcal{T}_{\mathbf{p}}^{-}} = 0$.

For trees T^{Ω} of larger size, a locality condition for the parameter $\mathbf{p}$ (see [23]) would ensure that, if all the accessible terms $T_v^{\Omega} \subset T^{\Omega}$ and respective quotients $T_v^{\Omega}/^d T_v^{\Omega}$ of size at most n_0 , whose leaves contain the relevant features, are in the range of the projection $\Pi_{\mathcal{L},\mathbf{p}}$, then so is T^{Ω} . (Note

that T^Ω could still be in the kernel of other projections $\Pi_{\mathcal{L}, \mathbf{p}'}$ for other parameters $\mathbf{p}' \neq \mathbf{p}$ hence in the kernel of $\Pi_{\mathcal{L}}$.)

With this procedure of parameter setting, small positive examples suffice to “activate” structures F^Ω of arbitrary size through the locality condition. If all the accessible terms and quotients of components of F^Ω that are of size at most n_0 are in $\mathcal{T}_{\mathbf{p}}^+$ then $F^\Omega \in \mathcal{T}_{\mathbf{p}}^+$, so that $\Pi_{\mathcal{L}, \mathbf{p}}(F^\Omega) = F^\Omega$ and $\Pi_{\mathcal{L}, \mathbf{p}}(F^\Omega) = 0$ otherwise.

When all the parameters have been set in the language learning process, to a new vector $\mathbf{p} \in \mathbb{F}_2^N$, the process describes above results in activating a certain subset of the vertices $(F^\Omega, \pi_{\mathcal{L}})$ of the graphs $\mathcal{G}_{n,A}^\Omega$ for any choice of n and A . We denote this set of vertices as $V(\mathcal{G}_{n,A}^{\Omega, \mathcal{L}})$. The graph $\mathcal{G}_{n,A}^{\Omega, \mathcal{L}}$ is then the graph induced by $\mathcal{G}_{n,A}^\Omega$ on the set of vertices $V(\mathcal{G}_{n,A}^{\Omega, \mathcal{L}})$. The selection of $\mathcal{T}_{\mathbf{p}}^+$ and $\mathcal{T}_{\mathbf{p}}^-$ and the resulting construction of $\mathcal{G}_{n,A}^{\Omega, \mathcal{L}}$ completes the process of parameter setting.

For a given $\mathcal{L}$, the collection of the graphs $\mathcal{G}_{n,A}^{\Omega, \mathcal{L}}$, each with their associated MERW Markov chain, can be viewed as a model of the generative grammar of the language $\mathcal{L}$.

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